

Brain Before the Swipe: Can Consumer-Grade EEG Predict Viewer Engagement during Naturalistic TikTok Browsing?

Master's Thesis

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A thesis presented for the degree of
Master of Science

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Acknowledgements

I am deeply grateful to my three supervisors, each of whom shaped this work in a distinct and necessary way.

Dr. Irina-Catrinel Craciun has been part of this thesis from long before it had a topic. Through the Digital Story Lab, which we founded together, she shaped my development as a researcher well beyond the scope of this document, introduced me to teaching workshops on this subject, gave me the opportunity to present at conferences and in classes at HPI, and involved me in academic publication for the first time. Her critical eye consistently sharpened this work when it needed sharpening, and I could not have asked for a more generous or more exacting introduction to scientific practice.

Prof. Dr. Bert Arnrich gave me a humble and realistic view of what consumer-grade EEG hardware can and cannot do, connected me with the right people in his group to understand the technological opportunities and limits of this space, and consistently pushed me to include perspectives I had not yet considered myself. His encouragement to look beyond my own assumptions is reflected throughout this thesis.

Prof. Dr. Dr. h.c. mult. Lothar H. Wieler helped me stay focused and on topic, and to maintain a rigorously scientific approach to reporting these results. He kindly took the time to answer questions at every stage, and his guidance kept this thesis grounded when its scope threatened to grow beyond what could be responsibly claimed.

I am also sincerely grateful to HPI PhD Student Sidratul Moontaha, whose extensive expertise in EEG-based engagement and brain-activity measurement was invaluable to this work. She generously gave her time to help me understand the field, and lent me two EEG devices for the study, which I returned considerably later than promised. I hope the results below go some way toward justifying the delay.

My thanks go to every participant in this study: a genuinely curious and academically engaged group of people whose interest in the research question, and whose thoughtful input during and after their sessions, motivated me throughout the data collection phase and improved this thesis in ways a

dataset alone could not have.

I also thank the Ethics Committee of the University of Potsdam for their efficient, constructive, and collegial handling of the ethics application that made this study possible in the first place.

Finally, and most personally, I thank my partner and my family, who supported and pushed me to keep writing, keep researching, and keep going, again and again, with new energy each time it was needed.

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Abstract

Short-form video platforms such as TikTok structure user attention as a rapid sequence of binary micro-decisions: continue watching, or swipe away. Billions of such decisions are made every day, shaping what a population sees, knows, and acts on, though the data behind them rarely leaves the companies that collect it. Platform metrics and much of the surrounding public discourse treat continuous, uninterrupted viewing time as a proxy for user engagement, yet recent behavioral evidence suggests that prolonged, uninterrupted scrolling is at least as consistent with passive, dissociative consumption as with effortful, goal-directed attention. This thesis investigates whether electroencephalography (EEG), recorded unobtrusively during naturalistic short-form video browsing, can discriminate the neural state preceding a voluntary skip decision from the neural state during sustained, non-skip viewing, and whether any such discrimination exceeds what is achieved by the conventional Engagement Index ($\beta/(\alpha + \theta)$), a composite band-power ratio drawn from human-factors neuroscience.

Twenty-five participants ($n = 25$; 13 female, 12 male; age $M = 24.6$, $SD = 2.9$) were recorded with a consumer-grade, four-channel Muse S headband (AF7, AF8, TP9, TP10) while freely browsing a personal TikTok “For You” feed for 25 minutes in a private, distraction-free setting, following a two-minute resting baseline. Manual swipe events were synchronized to the EEG stream via keypress markers. A literature-motivated labeling scheme operationalized two classes: **SKIP**, drawn from a 3.0-second imminent-skip period ending 2.0 seconds before each swipe (a gap exceeding both the typical reported onset of the readiness potential (Bereitschaftspotential) and directly measured pre-movement EMG onset in a self-paced reaching task), and **STAY**, comprising all remaining continuous, non-boundary viewing. From this labeling, 236 univariate features spanning seven frequency bands, four electrodes, and the raw broadband signal were engineered, including band power statistics, Hjorth parameters, and two features defined in the present thesis that count waves per second (Peak Frequency, Macro Frequency).

Class separability, quantified via Cohen’s d , was consistently higher for

individual frequency bands, electrodes, and statistical descriptors than for the Engagement Index; the single strongest feature, raw-signal standard deviation at TP10, exceeded the Engagement Index’s separability by nearly tenfold. In logistic regression benchmarking against a coin-flip baseline, the Engagement Index achieved above-chance but class-asymmetric performance, detecting **STAY** states substantially better than **SKIP** states. Three candidate features identified through the Cohen’s d ranking, namely TP10 raw standard deviation, AF7 high-gamma mean and AF8 delta peak frequency, each yielded statistically significant improvements over the Engagement Index in **SKIP** detection at both the intra-subject and inter-subject level, though absolute performance remained modest and, in the cross-participant setting, did not exceed the chance-equivalent reference. A secondary behavioral analysis further showed that swipes cluster into rapid, individually variable “burst” sequences rather than occurring as uniformly spaced discrete events.

These findings indicate that the conventional Engagement Index captures only a limited, asymmetric component of the neural signal associated with short-form video browsing, and that more granular, region- and frequency-specific EEG features carry additional, currently underexploited discriminative information about imminent disengagement. The thesis further contributes a fully reproducible, open-source recording and analysis pipeline for consumer-grade EEG in naturalistic digital-media settings, and discusses the neuroscientific, psychological, behavioral, and public-health implications of these results, while explicitly flagging electromyographic and powerline-noise contamination as unresolved confounds requiring artifact-controlled replication.

Chapter 1

Introduction

1.1 Motivation

Short-form video platforms have become one of the dominant modes of digital media consumption worldwide, and TikTok in particular has popularized an interaction paradigm in which content is delivered as an effectively infinite, algorithmically curated stream, navigated by a single recurring behavioral unit: the swipe. Each swipe is the outward expression of a micro-decision, to continue watching or to move on, repeated hundreds of times within a single session. Understanding what happens, neurally and behaviorally, in the seconds preceding this decision is of interest not only to platform design and human–computer interaction research, but increasingly to public health communication, which must compete for exactly this attentional currency when delivering health messages, educational content, or behavior-change interventions through the same channels [1].

The default proxy for engagement used both by platforms and by much of the surrounding research and policy discourse is duration: how long a user continues watching before swiping away. This proxy is intuitive and trivially measurable from platform telemetry, but its behavioral validity is contested. Attention research using cued-recall paradigms has shown that sustained attention fluctuates from trial to trial as an internal state that can be decoded from EEG recorded before each stimulus [2]. Two recent studies of social media use suggest that long use does not reliably indicate productive attention: participants rarely attributed their spontaneously reported flow experiences to social media, and periods with more social media use than a participant’s usual went with fewer flow reports [3], while users of a custom Twitter app described getting “lost” in the feed, with one interviewee unable to remember what they had read [4]. Taken together, this literature suggests that con-

tinuous non-skip viewing time is an ambiguous behavioral signal, consistent both with effortful engagement and with passive, inattentive consumption.

This ambiguity motivates a different empirical question, one this thesis takes as its central object of study: rather than assuming that duration indexes engagement, can the moment of disengagement itself, the decision to skip, be characterized directly at the level of the EEG signal, independent of any prior assumption about what continuous viewing means? If a decodable neural signature precedes the voluntary skip action, it would offer a behaviorally grounded, physiologically anchored complement to duration-based metrics, and a candidate target for evaluating whether existing composite engagement measures, such as the Engagement Index ($\beta/(\alpha + \theta)$) widely used in educational and human-factors neuroscience [5], actually capture the process relevant to short-form video interaction. Neuroimaging evidence already indicates that neural signals recorded during video viewing predict individual engagement decisions and aggregate viewing behavior above and beyond self-report and behavioral measures [6]; the controlled laboratory constraints inherent to fMRI, however, preclude naturalistic paradigms, motivating the consumer-grade EEG approach adopted here at the acknowledged cost of spatial resolution and subcortical access.

1.2 Defining Engagement

“Engagement” is used across public health, media research, human–computer interaction, and neuroscience to mean different things, measured by different instruments, and the term is not interchangeable between them (Section 2.4). Because the construct is the object of this thesis, it is stated here, before anything is measured with it.

Working definition (this thesis). *Engagement is a viewer’s effortful, voluntary attention to the content currently on screen, held against the constant availability of the next video, and, if present, ending at the latest at the swipe.*

In the health-communication case that motivates this work, the content on screen is a health message, and the swipe is the moment that message stops being received. Three properties of this definition matter for what follows. It is *effortful and voluntary*, which excludes dissociated viewing, in which users report using a feed without really paying attention to what they are doing [4]. It is defined *against an alternative*, the next video, always one thumb movement away, so that holding attention is a real cost rather than

the absence of an exit. And it *ends at an observable event*, the swipe, rather than trailing off into a low score, which is what makes it measurable in this paradigm at all: each swipe is a time-stamped moment at which viewing of that content ended, whatever the viewer’s state had been before it.

This definition is deliberately narrow. It names a screen and a swipe, and so it holds only where both exist. Whether anything survives when those are removed, that is, whether the same construct can be stated for a learner, a patient using a digital intervention, or a colleague in distributed work, is taken up in Section 4.6, once there are results to constrain the answer. Throughout this thesis, **engagement** is the primary construct label; *attention*, *absorption*, *watch time*, *sustained viewing*, and the **STAY/SKIP** contrast introduced in Chapter 3 name indicators or operationalisations of it, never the construct itself. The **STAY/SKIP** labels hang on the swipe and are a partial operationalisation of engagement.

1.3 Engagement in Health Communication

In health communication, engagement is a precondition for any effect: a health intervention that nobody engages with cannot change behaviour [7]. The field measures engagement mostly as quantity of use, such as logins, views, and likes, and has not yet settled how much use counts as enough [8, 9]. Short-form video already carries health messages. Public-health research has analysed COVID-19 messages on TikTok [10], compared institutional and personal health accounts on TikTok [1], and proposed a public-health research agenda for TikTok [11]. Section 2.5 reviews how the health communication literature measures engagement.

Theories of how a health message persuades, such as narrative transportation [12] and entertainment-education [13], start from an audience that is still watching. The present thesis looks one step earlier, at the moment a viewer stops watching, and tests EEG as a candidate measure of that moment.

1.4 Research Gap

Four gaps follow from the literature above and in Chapter 2, and together they produce the research questions in Section 1.5. Each is stated as an open problem the source literature itself identifies, rather than as a claim that nothing has been done.

G1 Engagement is measured largely as quantity of use, and the field has not established what quantity means. Amount, fre-

quency, duration, and depth of use form the behavioural half of the definition of Perski et al. [8] and the basis of platform and marketing metrics [14]. The limits of this are stated within the literature itself: “more engagement does not necessarily equate to more effective engagement”, and effectiveness “may or may not require sustained engagement” [7], while the critical threshold “has yet to be established” [9]. Independent evidence points the same way: in daily-life studies of social media, users described getting lost in the feed, and more use than a participant’s usual went with fewer flow reports [4, 3], sustained attention fluctuates from trial to trial [2], and on a public-health TikTok account the length of a video, a property of the video distinct from watch time, was negatively associated with both likes and comments [15].

- G2 Physiological measurement is named as a route to the construct and little used within this literature.** Perski et al. list recording of physiological or psychophysical correlates as one of three routes to measuring engagement, but located such measures only in the computer-science and HCI literatures rather than in behavioural science [8]; the examples of physiological measures given by Perski et al. and by Yardley et al. do not name EEG, and validating “complementary and non-intrusive measures of effective engagement” is listed among Yardley et al.’s priority topics for future research [7].
- G3 The standard EEG measure of engagement was developed for a different kind of task.** The Engagement Index $\beta/(\alpha + \theta)$ was selected to decide when a computer task modelled on flight-deck work switches between manual and automated operation [5], and its classification accuracy has been reported as not optimal in a controlled learning task, where the index is noted to be used mainly in non-predictive applications [16]. Which spectral markers track engagement remains an active question: in one study, the Engagement Index ratio separated a sustained-attention test from a cognitive-engagement test, while markers chosen for engagement did not [17], and inter-subject variability is identified as an open problem across this literature [18]. The behaviour of the Engagement Index around self-paced decisions on a seconds timescale has not, to our knowledge, been reported.
- G4 There is no established method for placing an EEG label around a self-paced action.** Work on window length and window placement uses an externally cued event with a known onset [19, 20], and where two such parameters have been varied together rather than

one at a time, the group’s best time window was the best time window for only 31 of 109 participants [21]. A swipe has no cue, so the placement of the label relative to the action has to be argued rather than looked up.

G1–G3 produce the first two research questions. If quantity of use is not yet a validated measure of the construct, and the physiological route to it is largely unexercised in this literature, then the moment of disengagement is worth characterising directly at the level of the signal, and the field’s standard index is worth benchmarking rather than assuming. G4 produces the third: because the labelling scheme required to do this must be argued into existence, its parameters become an object of study rather than a settled configuration.

1.5 Research Questions

This thesis addresses three interrelated research questions:

1. **Separability:** Can EEG recorded during naturalistic, self-paced TikTok browsing be used to distinguish the period preceding a voluntary skip decision (SKIP) from continuous, non-skip viewing (STAY), using only a lightweight, consumer-grade, four-channel headband?
2. **Benchmark comparison:** How does the class-separation and predictive performance of individual, granular EEG features compare against the conventional Engagement Index, both as a descriptive effect-size measure (Cohen’s d) and as a predictive model (logistic regression), at both the intra-subject and inter-subject level?
3. **Parameter sensitivity:** How sensitive is any observed separability to the specific temporal parameters, namely window length, pre-swipe safety gap and imminent-skip period, used to define the STAY/SKIP labeling scheme itself?

Cohen’s d measures the difference between two class averages in one feature, using that feature’s own standard deviation as the scale. Because both inputs carry the same unit, the unit cancels out.

Answering these questions requires solving a chain of non-trivial methodological problems that do not have an established, off-the-shelf solution in the literature: a naturalistic recording protocol that preserves ecological validity while achieving millisecond-level behavioral synchronization; a principled,

literature-motivated labeling scheme for a self-paced (rather than externally cued) behavioral event; a feature space broad enough to allow genuinely comparative evaluation against a composite clinical baseline; and a cross-validation and significance-testing framework robust to the strong temporal autocorrelation and severe class imbalance inherent to naturalistic swipe data. The bulk of this thesis is accordingly devoted not only to answering the three questions above, but to building and documenting a reusable methodological pipeline capable of doing so reproducibly.

1.6 Approach

To address these questions, twenty-eight participants were recruited, of whom twenty-five ($n = 25$) were retained for analysis after exclusion of compromised or ineligible recordings (Section 3.1). Each participant was recorded with a consumer-grade Muse S EEG headband while freely browsing their personal TikTok feed for 25 minutes in a private, isolated setting designed to elicit naturalistic viewing behavior, preceded by a two-minute resting baseline used for per-participant signal normalization (Sections 3.2–3.3). Manual swipe events were synchronized to the continuous EEG stream via a custom keypress-logging architecture built on the Lab Streaming Layer protocol (Section 3.5).

A three-parameter labeling scheme was then developed to derive **STAY** and **SKIP** classes from this continuous, self-paced behavioral stream, with default parameter values grounded in the EEG window-length literature and in the readiness-potential and pre-movement EMG literature governing motor-preparatory contamination near the swipe action itself (Sections 3.4–3.4.2). Because, to our knowledge, no prior work had established these parameters for a self-paced, naturalistic swiping paradigm specifically, their influence on downstream class separability was additionally characterized through a one-at-a-time sensitivity analysis (Section 3.4.5) rather than assumed to be optimal by default.

From the resulting labeled dataset, a broad feature space of 236 univariate EEG descriptors was engineered per 1.0-second analysis window, spanning conventional band-power statistics, Hjorth parameters, and two features defined in the present thesis that count waves per second (Peak Frequency, Macro Frequency), alongside features drawn from the undecomposed raw broadband signal (Section 3.7). Feature-level and category-level class separability was quantified via Cohen’s d and benchmarked directly against the Engagement Index (Section 3.4.4), and the strongest candidate features were subsequently evaluated as standalone logistic regression classifiers against

both a chance baseline and the Engagement Index baseline, under both intra-subject (blocked temporal cross-validation) and inter-subject (leave-one-participant-out) evaluation, with Wilcoxon signed-rank testing used to assess statistical significance across the participant cohort (Sections 3.8.3–3.8.4).

1.7 Contributions

This thesis makes six principal contributions, each developed in detail in Chapter 4:

1. A fully naturalistic, ecologically valid EEG experiment synchronizing consumer-grade EEG with free TikTok browsing behavior in $n = 25$ participants, together with a reproducible experimental protocol (Section 4.1).
2. An open-source, reusable recording and analysis pipeline covering automated EEG acquisition, behavioral synchronization, artifact-aware pre-processing, feature engineering, and statistically rigorous model evaluation, transferable to related naturalistic EEG paradigms beyond the present study (Section 4.2).
3. A characterization of swipe-burst behavior, showing that skip events cluster into rapid, individually heterogeneous sequences rather than occurring as uniformly spaced discrete decisions (Section 4.3).
4. A direct, statistically grounded comparison between the conventional Engagement Index and granular, feature-level EEG descriptors in their ability to separate and predict **STAY** versus **SKIP** states, identifying specific features (raw TP10 variability, AF7 high-gamma activity, AF8 delta peak frequency) that outperform the Engagement Index despite the latter’s established status as a clinical and educational benchmark (Sections 4.4–4.4.2).
5. A neuroscientific, psychological, and behavioral interpretation of these results, together with an explicit, appropriately scoped discussion of their implications for public health communication, digital media policy, and health message design, and of the artifact- and hardware-related confounds that limit the strength of any causal claim (Section 4.5).

6. An updated, instrument-independent definition of engagement as a psychological construct, derived from the definitional gaps identified in Section 2.4 and constrained by the empirical results above, together with the reporting requirements it imposes on any engagement measurement and an explicitly hypothetical extension to lower-cost, field-deployable proxy measurement (Section 4.6).

1.8 Thesis Structure

The remainder of this thesis is organized as follows. Chapter 2 (Literature and State of the Art) situates the present work within the literature on attention and engagement in short-form video, parameter sensitivity and the handling of windows at label borders in EEG classification, class-separation metrics, the estimation of frequency from local maxima, EMG and powerline-noise contamination in consumer EEG, the role of delta in the awake brain, and the timing of movement preparation before a self-paced action. Chapter 3 (Materials and Methods) details the participant sample, data acquisition hardware and protocol, labeling strategy and its literature grounding, pre-processing pipeline, feature engineering, and the statistical framework used for model evaluation. Chapter 4 (Results and Discussion) reports the five contributions summarized above and closes with a discussion of their neuroscientific, psychological, behavioral, and applied implications, together with explicitly flagged limitations. Its closing section steps outside the paradigm and states, as a contribution of basic research, an updated definition of the engagement construct that the preceding results constrain, the measurement requirements following from it, and the limits of that definition. The thesis concludes with a set of concrete directions for future research, spanning validation and replication, mechanistic artifact clarification, theoretical development of the engagement construct, methodological expansion, and application to digital public health.

Chapter 2

Literature and State of the Art

The following sections outline the core theoretical claims, methodologies, and prior research required to contextualize the present thesis.

2.1 Adjacent Study

One published study sits closest to the present thesis: Rahman et al. record EEG while people watch short-form video, and train a classifier to tell short-form video viewing apart from other activities [22]. Rahman et al. present the classifier as a detector of Short-form Video Addiction (SVA), although the paper reports no measure of addiction, such as an addiction questionnaire. The present thesis calls the study of Rahman et al. the *adjacent study*. The adjacent study records EEG with a consumer-grade Muse S headband (the paper names both Gen 1 and Gen 2) at four scalp electrodes, TP9, AF7, AF8 and TP10, sampled at 256 Hz, plus two auxiliary channels of the headband, which the paper does not describe further; the band-wise analysis runs on all six channels. In a laboratory, a small sample of student volunteers ($n = 10$, ages 20–35) watches short-form video on their own phones, in their own feeds, TikTok included, and may swipe on at any moment. The present thesis calls self-paced viewing of the own feed on the own phone *naturalistic use*, and the feed itself the *infinite feed*.

The adjacent study labels whole blocks of time. A 15-minute block of short-form video viewing is the positive class; a block of other daily activity without media consumption on a smartphone is the negative class, balanced 1:1. The classifier of the adjacent study therefore answers one question: is this person inside the infinite feed, or not? Swipes play no role in the viewing-versus-not-viewing label. The two blocks may also differ in posture, gaze and hand movement: a person watching short-form video looks down at a phone

and moves a thumb, while a person doing something else may not. Part of the reported accuracy may therefore come from body movement as well as from brain activity.

The adjacent study pairs EEG with electrodermal activity, a measure of skin conductance, from an Empatica E4 wristband. Neural networks learn one shared representation of the EEG and the skin-conductance signal. The purpose of the shared representation is a sensor swap: learn with the head-band, then use the less obtrusive wristband alone. Evaluation repeatedly trains on 80% of the data and tests on the remaining 20% (k -fold cross-validation). The paper does not state whether windows from one 15-minute block can land in both the training data and the test data. If windows from one block land in both, the classifier can partly recognise the block itself, much like a student who has seen the exam questions during practice, and the reported accuracy may overstate performance on a new block. The adjacent study also reports no test on participants left out of training. Averaged over ten runs, the classifier reaches 76.3% accuracy from EEG alone and 84.2% from EEG and wristband together. For the sensor swap, the wristband alone reaches 71.5%, and 80.2% after training with help from the EEG.

On frequency, the adjacent study applies a band-stop filter at 50 Hz and the multiples of 50 Hz, and the paper names a “110V 50Hz” power supply. The ethics approval of the adjacent study comes from the University of Massachusetts Lowell, in the United States, where power lines carry alternating current at 60 Hz. The adjacent study then splits the signal into five bands: Delta 1–4, Theta 4–8, Alpha 7.5–13, Beta 13–30, and Gamma 30–44 Hz. The analysed spectrum stops at 44 Hz, so the mismatch between the 50 Hz filter and the 60 Hz power lines likely leaves the reported band results largely unaffected. Beta and Gamma are reported as the most informative bands, each above 80% accuracy.

Gap the present thesis fills: The adjacent study labels whether a person is inside the infinite feed; the present thesis on the other hand labels two states inside the infinite feed, anchored to each self-paced swipe: **STAY**, sustained viewing without a swipe, and **SKIP**, the seconds just before the swipe. Because the labels differ, the accuracies of the present thesis and of the adjacent study measure different things and are not directly comparable. More specifically, the present thesis asks whether EEG features with a known meaning, such as the strength of one frequency band, separate **STAY** from **SKIP** better than the Engagement Index does (Chapter 4). The adjacent study analyses the spectrum up to 44 Hz, whereas the present thesis analyses the spectrum up to 100 Hz, including the high-gamma (40–60 Hz)

and very-high-gamma (60–100 Hz) bands.

2.2 Secondary Adjacent Studies

Six further studies record EEG in connection with short-form video, all with laboratory EEG systems of 32 to 64 electrodes.

Short-video recommendation (Zhang et al.; He et al.) [23, 24]. *Research goal:* use EEG to improve short-video recommendation (Zhang et al.) and to detect how immersed a viewer feels (He et al.). *Design:* two papers from one research group report the same lab study with 30 participants. Each participant gets a fresh account on a real platform, trained by ten hours of browsing; the researchers then compose 15-minute sessions of 20–30 personalised, non-personalised and random videos. Participants swipe freely while a 64-channel cap records at 1,000 Hz, filtered to 0.5–50 Hz. After each session, participants re-watch the first seconds of each video and rate the video from 1 to 5 on several scales; Zhang et al. call six of the scales, among them valence, interest and immersion, *affective engagement*, and He et al. predict the immersion rating. *Result:* adding EEG raises the AUC of the recommendation models, the chance that a model scores a randomly picked positive case above a randomly picked negative case ($0.5 =$ guessing, $1 =$ perfect), by at most about 0.04, and in a few models lowers the AUC slightly. He et al. predict high versus low immersion with 69.1% accuracy without EEG and 70.1% with EEG. *Limits:* the ratings come from memory after the session, and Zhang et al. and He et al. do not report keeping each participant on one side of the training/test split, so viewings of one participant may land on both sides, the leak described for the adjacent study (Section 2.1).

Opinion polarisation (Du et al.) [25, 26]. *Research goal:* show whether one-sided short videos shift a viewer’s opinion of a historical or fictional figure, and whether EEG reveals the shift. *Design:* two papers report one dataset of 23 participants. The design has three steps. In a first lab session with EEG, a web page built by the researchers plays, for each of ten such figures, one one-minute clip praising the figure and one clip criticising the figure, mixed with film-clip distractors; a “Next One” button skips a clip after 30 seconds. Over the following six days, participants watch clips on the same web page daily without EEG, and each figure appears to a participant only in praising clips or only in criticising clips, decided at random per figure. A second lab session repeats the first. A classifier then guesses, for each participant–figure pair, whether the six days praised or criticised the figure. Input: alpha-band power at one forehead electrode, FPz, from a 64-channel

cap filtered to 0.5–50 Hz, averaged over the praising and the criticising clip of the figure in both lab sessions. *Result:* opinion ratings of a figure rise after six praising days and fall after six criticising days; the classifier reaches 64.1% accuracy and an AUC of 0.672. *Limits:* the five cross-validation folds are drawn at random over participant–figure pairs, so pairs of one participant can land in training and test data; FPz was chosen because FPz scored best in the same cross-validation, which likely makes the 64.1% optimistic; and the six browsing days carry no EEG.

Short-video advertising (Zhang and Zhang) [27]. *Research goal:* find which selling approach (transactional or relational) and which displaying style (high or low enthusiasm and humour) of an online celebrity evokes more “neural engagement”. *Design:* 25 participants watch the same 64 TikTok advertisements of 10–35 s each while a 64-channel cap records at 250 Hz. Neural engagement here means four EEG band-power ratios: emotion, relaxation, attention and cognitive load. The relaxation index is the Engagement Index, $\beta/(\alpha + \theta)$, turned into the negative logarithm, so a higher relaxation score means a lower Engagement Index. *Result:* transactional selling and the high displaying style evoke higher values on most of the four indices (analysis of variance). *Limits:* each advertisement plays to the end, so viewing is not self-paced, and the study compares group averages without predicting the state of a single viewer.

Remaining studies. Wade and Dumitru are unpublished student theses. *Wade asks:* does TikTok addiction impair response inhibition? EEG runs during a fixed 3-minute TikTok compilation and a following Go/NoGo task, in which a participant presses a button for one signal and holds back for another [28]. *Dumitru asks:* does TikTok-style content impair sustained attention? Participants scroll for 5 minutes through the own TikTok or Instagram feed; the EEG of the 5 minutes enters one left-versus-right alpha comparison, and a β/α engagement index is computed during attention tasks at baseline and after viewing [29]. *Yan et al. ask:* does a tendency toward short-video addiction go with weaker attention control? Yan et al. correlate an addiction questionnaire score with EEG recorded during an attention task, and record no viewing [30].

Gap the present thesis fills: In the six studies, no label or comparison is anchored to the moment of a swipe: Zhang et al. and He et al. log every swipe but leave the moment of the swipe unused, keeping only the share of each video watched, Du et al. allow a skip only after 30 seconds, Dumitru does not record the swipes, and the remaining studies involve no swipe. The present thesis anchors every SKIP label to one time-stamped, self-paced swipe

in the own feed of the participant, and analyses the spectrum above 50 Hz, which the 0.5–50 Hz filters of Zhang et al., He et al. and Du et al. remove. The present thesis also records with a consumer headband, which, unlike the laboratory systems of the six studies, is affordable for a single consumer and can be worn during naturalistic use outside a laboratory.

2.3 Recording Software

One route from a Muse headband to a laptop is `muselsl`, an open-source tool created by Barachant et al. [31]. `muselsl` connects to the Muse EEG headband over Bluetooth, and the command `muselsl stream` passes the EEG on to the laptop in near real time; further `muselsl` commands display and record the EEG. `muselsl` runs as a command-line tool and as an importable Python library, and the current documentation lists the Muse S used in the present thesis as supported.

`muselsl stream` sends the EEG into the Lab Streaming Layer (LSL), a system built to unify the collection of time-series data in research experiments. LSL puts each data source on the network as a separate stream, stamps every sample with a time, keeps the streams synchronized in time, and gives near-real-time access to every stream; the `muselsl` documentation calls LSL standard in EEG-based brain-computer interface research. As far as the `muselsl` documentation shows, `muselsl` offers no way to mark a keypress in the EEG recording and no automatic reconnection after a Bluetooth dropout.

A second route is Mind Monitor, an app that records the Muse on a phone; the adjacent study used Mind Monitor on Android [22]. The adjacent study labels whole 15-minute blocks (Section 2.1), so the adjacent study needs no marker for a single event such as a swipe.

Gap the present thesis fills: Every SKIP label of the present thesis is anchored to one swipe, marked by a keypress on the laptop, and every STAY label covers the remaining viewing time minus short gaps around each keypress that reduce muscle artifacts from the keypress (Sections 3.2 and 3.4.2), so the EEG recording needs a time mark for every keypress and a record of every Bluetooth dropout. A custom recording script by the author of the present thesis therefore starts `muselsl stream`, stamps each keypress with the laptop clock, converts the keypress time into LSL time, flags the EEG sample that falls within a defined tolerance window of the keypress, and restarts the stream in a new file after a Bluetooth dropout (Section 3.5). Pre-processing then measures each Bluetooth dropout from the timestamps be-

tween consecutive files, and no analysis window crosses a Bluetooth dropout (Section 3.6).

2.4 Defining Engagement: Existing Definitions and Their Gaps

The fields that measure engagement do not define engagement consistently. O’Brien and Toms found no agreed definition when O’Brien and Toms set out to deconstruct the term for human–computer interaction [32]. More than a decade later, a systematic review of social media engagement describes engagement as a term with many meanings and many dimensions [14]. In digital behaviour-change research, Yardley et al. call for “defining engagement more consistently and appropriately” [7]. Five fields bear on the present thesis.

Human–computer interaction. O’Brien and Toms interviewed 17 users of web search, online shopping, webcasting, and gaming applications. O’Brien and Toms define engagement as “a quality of user experience characterized by attributes of challenge, positive affect, endurability, aesthetic and sensory appeal, attention, feedback, variety/novelty, interactivity, and perceived user control”. In the model of O’Brien and Toms, engagement unfolds as a four-stage process: point of engagement, sustained engagement, disengagement, and re-engagement [32]. A recent review of EEG-based engagement research adopts the attribute list of O’Brien and Toms [18]. The four stages rest on the interviews, in which participants recall a past experience, and O’Brien and Toms note that a recalled experience is a “selected interpretation” of the event [32].

Marketing and platform analytics. Trunfio and Rossi, following Hollebeek, describe customer engagement as the positive, brand-related cognitive, emotional, and behavioural activity of a consumer during interactions with a brand. Trunfio and Rossi review social media engagement and find the behavioural dimension to be the most common way to measure engagement: counts of likes, comments, shares, and views, and indices derived from the counts [14]. Salminen et al. measure engagement with short-video advertising as viewer churn, the share of viewers who leave during playback, in 12 TikTok advertisements of one company. TikTok reports churn per quarter of a video, counted over all viewers of an advertisement. Most viewers left within the first quarter of an advertisement [33]. Churn shows in which quarter of a video viewers leave, while the state of each viewer before leaving stays unmeasured.

Human-factors neuroscience and later uses of the Engagement Index. Pope et al. tested candidate EEG indices with six participants in a computer task modelled on flight-deck work. A feedback loop switched a tracking task between the participant and the computer whenever an index rose or fell over a time interval. Among the candidate indices, the ratio of beta power to the sum of alpha and theta power, $\beta/(\alpha + \theta)$, the Engagement Index, reflected task engagement best [5]. Pope et al. judged each index by the switching pattern of the loop, and report no separate measure of engagement. Nuamah et al., citing earlier work, define task engagement as “effortful striving towards task goals”. Nuamah et al. used the Engagement Index as the input for recognising which of five mental tasks a participant performed, trained one classifier per participant, and report an average accuracy of 88.67% across six participants, where guessing among five tasks gives 20%. The recordings come from an existing dataset of 10-second trials, with 25 to 75 trials per participant [34]. In a controlled learning task, Apicella et al. classified two levels of cognitive engagement from the Engagement Index alone with 51 to 55% accuracy, close to the 50% that guessing between two levels gives, while features from twelve narrow frequency bands, combined across electrodes, reached about 77%. Apicella et al. note that EEG signals change over time and differ between people, which makes classification across participants difficult [16], and a recent review names the differences in EEG between people as a pressing issue in EEG engagement research [18]. In a conference abstract, de Carvalho et al. treat the Engagement Index ratio as a vigilance index, that is, a marker of sustained attention. In the study of de Carvalho et al. with 15 students, three of four markers chosen for sustained attention, the Engagement Index ratio among them, separated a sustained-attention test from a cognitive-engagement test, while the markers chosen for engagement did not. De Carvalho et al. attribute the failure of one engagement marker, theta power at the front midline of the head, mainly to a resting baseline recorded after the effortful tests [17].

Digital health and behaviour-change intervention science. Perski et al. define engagement with two components: the extent of use, such as the amount, frequency, duration, and depth of use, and a subjective experience characterised by attention, interest, and affect. Perski et al. describe engagement as “a dynamic process” that is expected to vary within and across individuals over time [8]. Yardley et al. describe engagement as a process that starts with a trigger and first use, followed by sustained engagement or disengagement, and separate moment-to-moment engagement with the intervention from engagement with the wider goals of the intervention. Yardley et al. note that an observed end of use can mean disengagement, or can mean that the user has learned enough to need the intervention no longer.

Yardley et al. shift the target to *effective engagement*, and note that “more engagement does not necessarily equate to more effective engagement” and that effective engagement “may or may not require sustained engagement” [7].

Learning analytics. Karimah and Hasegawa review automatic engagement estimation and follow Fredricks et al., who divide engagement into behavioural, emotional, and cognitive engagement. Emotional engagement, measured from affective cues, is the type studied most often. The largest share of the reviewed studies (43%) rates engagement by external observation, followed by self-report (20%). Karimah and Hasegawa note that self-report carries bias, and that external observation limits the judgement of the actual effort of learners [35].

Gap the present thesis fills: O’Brien and Toms reconstruct disengagement from recalled accounts, and an observed end of use, such as viewer churn counted per quarter of an advertisement or the end of use of a health intervention, shows roughly when people stopped but leaves the state of each person before stopping unmeasured. In small studies, the Engagement Index recognised five mental tasks well (Nuamah et al.), classified two engagement levels close to guessing (Apicella et al.), and behaved as a marker of sustained attention (de Carvalho et al.); classification across participants remains an open problem, and, to the author’s knowledge, the behaviour of the Engagement Index in the seconds before a self-paced decision has not been reported. The present thesis states a working definition anchored to the swipe, an observed end of viewing (Section 1.2), tests whether the Engagement Index and other EEG features separate the seconds before each self-paced swipe from sustained viewing, within and across participants, and later proposes an updated definition of engagement that names no platform and no measuring instrument (Section 4.6).

2.5 Measuring Engagement in the Health Communication Literature

The present section covers two bodies of work within health communication. The first body studies digital behaviour change interventions (DBCIs), such as health apps and health websites. The second body studies health messages on TikTok. Section 2.4 gave the definitions of engagement from the DBCI work. The present section asks how the two bodies of work *measure* engagement.

Three routes. In the DBCI work, engagement with a health intervention is a precondition for the effectiveness of the health intervention [7]. Perski et al. reviewed 117 articles on engagement with DBCIs and name three routes to measure engagement [8]. The present thesis calls the three routes the *ask route*, the *log route*, and the *body route*.

- The ask route: self-report questionnaires, interviews, or think-aloud studies.
- The log route: automated recording of use, such as logins and page views.
- The body route: recording of physiological or psychophysical correlates.

In the behavioural-science literature, the log route was the most common. Typical log-route measures were logins, time spent online, and the amount and type of content used. Body-route measures came from the computer-science and HCI literatures: cardiac activity, respiratory depth, electrodermal activity, and eye tracking. Perski et al. note two limits. Ask-route questionnaires typically measure engagement after a person has used the intervention, instead of during use. In the one study Perski et al. cite on the question, body-route measures and ask-route measures were only weakly associated [8].

The log route in practice. Yardley et al. judge that usage alone cannot be taken as a valid indicator of engagement. The evidence linking usage to health outcomes is mixed and largely correlational [7]. Evans et al. review digital media for behaviour change and name inconsistent, non-standardised measures, such as engagement and reach, as the research gap cited most often. Evans et al. count objective dosage and exposure data from platform analytics among the strengths of social-media interventions. Evans et al. recommend dose-response designs, which test whether more exposure goes with more behaviour change, and state that “what qualifies as a critical threshold for engagement in digital interventions has yet to be established” [9].

The body route in practice. Yardley et al. list body-route measures of task engagement: skin conductance, heart rate, facial expression, and fMRI. According to Yardley et al., body-route measures can show which parts of an intervention attract attention or evoke emotional arousal in the short term, and add an objective record of user reactions to ask-route measures. Yardley et al. also rate body-route measures as often intrusive, expensive, and hard

to scale. Body-route measures taken in a laboratory may lack ecological validity, meaning that laboratory results may not carry over to everyday use. Yardley et al. further warn that differences in attention “may not always translate into differences in intervention use” [7].

Health messages on TikTok. McCashin and Murphy systematically reviewed 24 studies of TikTok for public or mental health. In 20 of the 24 studies, the primary method was content analysis, which codes what the videos show [1]. Three studies describe health messages on TikTok through content analysis and the counts TikTok displays, such as views, likes, and comments.

Basch et al. coded 100 videos under #Coronavirus and all 17 posts of the World Health Organization (WHO) in April 2020. The 117 videos had 1,194,081,700 views in total. Fewer than 10% of the videos mentioned transmission, symptoms, or prevention. WHO videos covered transmission and symptoms more often than consumer videos, yet still in fewer than 10% of videos [10]. Very high reach therefore came with little health content. Basch et al. coded content and counts only, so the study does not show what viewers took in.

McCashin and Murphy also analysed the content of 29 Irish TikTok accounts: 2 public-health accounts, 5 charity accounts, and 22 personal creators. Both public-health videos came from one account, the Irish Health Service Executive. Public-health videos used humour in 0 of 2 videos and a popular sound in 0 of 2 videos. Personal creators used humour in 16 of 22 videos and a popular sound in at least 17 of 22 videos. Videos with a popular sound averaged 132,419 likes. Videos without a popular sound averaged 51,161 likes. McCashin and Murphy conclude that institutional accounts, meaning the public-health and charity accounts, used fewer TikTok features and received fewer likes, comments, and shares. The comparison uses no statistical test and rests on 2 public-health videos. McCashin and Murphy also ask whether TikTok analytics reflect in-depth user engagement that may lead to positive outcomes, and call for controlled studies that relate TikTok engagement to behaviour change [1].

Chen et al. analysed 355 COVID-19 videos from “Healthy China”, the TikTok account of the National Health Commission of China, from January to April 2020. Chen et al. measure engagement as the counts of likes, shares, and comments. Regression models for count data linked longer video length to fewer likes (incidence rate ratio, $IRR = 0.19$, $p < 0.001$) and fewer comments ($IRR = 0.39$, $p < 0.001$). An IRR below 1 means fewer counts, and an IRR above 1 means more counts. The emotion of the title changed the

link: longer videos with a positive title received more likes (IRR = 21.72, $p = 0.04$) and more comments (IRR = 10.14, $p = 0.047$) [15]. Chen et al. record the length of each video and do not record how long viewers watched.

The research agenda. Zenone et al. wrote when TikTok had passed one billion monthly users, and videos tagged #MentalHealth had 17 billion views. Zenone et al. propose six public-health research priorities for TikTok. Most of the six priorities concern the content on TikTok or TikTok as a company. One priority, on medical advice, also asks how users take up the health information received. Zenone et al. ask that research runs independently of TikTok, with conflicts of interest declared when TikTok funds or joins the research [11]. For DBCI research, Yardley et al. list a related priority: to “investigate and validate complementary and non-intrusive measures of effective engagement” [7].

Gap the present thesis fills: The health communication literature reviewed here measures engagement mostly through the log route, as counts of use and the counts TikTok displays, and the body-route examples of Perski et al. and Yardley et al. do not include EEG. To the author’s knowledge, no study in the health communication literature reviewed here records a body-route signal in the seconds before a viewer swipes a health message away, a moment that counts of use do not capture. The present thesis tests consumer EEG, recorded independently of TikTok and without data from TikTok itself (Section 4.2), as one candidate for the complementary measures Yardley et al. call for, keeping in mind that a headband remains a body-route measure that Yardley et al. rate as often intrusive and hard to scale.

2.6 Viewing Time as an Indicator of Engagement

Social media designers are tasked with maximising time on site [4], and Knierim et al. advise platforms to measure the quality of the experience rather than time on the platform alone [3]. To the author’s knowledge, no study tests directly whether long viewing of short-form video means engagement. Four studies bear on the question from two sides: two *daily-life studies* of social media use, and two *lab EEG studies* of attention within a task. None of the four studies records EEG during short-form video viewing.

Knierim et al. [3]. *Research goal:* test whether social media produces flow, a state of deep and effortless absorption. 40 university students took

part in a five-day field study. Tracking software logged the use of a preset list of phone apps. Three times a day, participants reconstructed the preceding hours and listed activities with flow. The survey did not ask about social media directly. Participants attributed 14 of 673 flow reports (2%) to social media. Knierim et al. sort apps, not behaviours, into two categories. Passive apps are built around feeds and videos, for example Instagram, TikTok, and YouTube. Active apps serve direct communication. When a participant used more social media than usual in the hours before a report, the participant reported fewer flow episodes. The drop in flow was significant for passive apps and not significant for active apps. More use than usual also went with lower mental readiness, a mix of mood, sleepiness, and motivation, for both app categories. Knierim et al. suggest physiological signals such as EEG to tell genuine flow apart from absorption that only resembles flow. *Limits:* the tracking software counts minutes per app, and participants recall flow hours later. No physiological signal is recorded.

Baughan et al. [4]. *Research goal:* learn how feed design shapes normative dissociation. Normative dissociation is full absorption with reduced self-awareness and a reduced sense of control. Baughan et al. count both mindless scrolling and deep absorption in the content itself as normative dissociation. 43 U.S. participants used a custom Twitter app for four weeks, and the Twitter app changed its design features every week. One in-app question measured dissociation, “I am currently using Chirp without really paying attention to what I am doing”, averaged per day. 11 participants were interviewed. Interviewees described getting “lost” in the feed and losing track of time. Custom lists and reading-history labels lowered dissociation. On days with more views of usage statistics, participants reported less dissociation. On days when participants closed the Twitter app through a time-limit prompt, participants reported more dissociation, and Baughan et al. read the time-limit prompt as a cue to notice the dissociation. Time spent reading, entered as a control variable, was not significantly associated with dissociation ($p = 0.52$). *Limits:* the feed shows text posts, dissociation is one self-reported item, and daily averages cannot show whether dissociation came before or after an action. No physiological signal is recorded.

deBettencourt et al. [2]. *Research goal:* separate two kinds of attention in memory. *Sustained attention* is attention held steady over time. *Spatial attention* is attention aimed at one place on the screen. In three lab experiments, a cue first points to one place on the screen. Images of objects then appear. Later, participants mark where each object appeared, or which colour each object had. In the EEG experiment ($n = 30$), a classifier reads the EEG after the cue and before the objects appear, and predicts whether an object will be remembered better or worse than the participant’s median,

at 57.2% accuracy against a 50% chance level. The classifier is trained and tested on averages over groups of trials, not on single trials. deBettencourt et al. read the 57.2% prediction as sustained attention rising and falling from trial to trial. During trials with low sustained attention, the EEG still showed spatial attention at the cued place. deBettencourt et al. therefore conclude that low sustained attention is not the same as dropping out of the task. *Limits:* each trial starts at a cue set by the experimenter, and the task offers no way to leave. The study does not test how sustained attention changes with time on task.

De Carvalho et al. [17]. *Research goal:* find EEG frequency markers that separate sustained attention from cognitive engagement. After the two tests described in Section 2.4, the 15 students watched a 27-minute science lecture video while a 32-channel research cap recorded the EEG. During the lecture video, several markers changed along curves over time. The Engagement Index ratio, measured over central and parietal sites, rose and then fell. Theta power at the back of the head fell and then rose. Alpha power over parietal and occipital sites, a marker chosen for engagement, rose steadily. De Carvalho et al. conclude that a marker averaged over a whole task hides changes within the task. *Limits:* the curves explain only about 1–2% of the variation in the markers, and no marker correlated with test scores. The lecture video offers no way to skip.

What the four studies show together. In the two daily-life studies, time spent in a feed was not significantly associated with dissociation, and more passive use than usual went with fewer flow reports. In the two lab EEG studies, attention changed within a task on two different timescales: from trial to trial in deBettencourt et al., and along slow curves over 27 minutes in de Carvalho et al. In deBettencourt et al., a drop in sustained attention did not mean leaving the task.

Gap the present thesis fills: To the author’s knowledge, no study has yet checked whether long viewing without a skip reflects engagement, and none of the four studies records a signal during short-form video viewing that could separate attention held on the content from continued viewing alone, a separation for which Knierim et al. suggest EEG. The present thesis records EEG during naturalistic use and puts all continuous viewing without a skip into one class, **STAY**, treated as a partial operationalisation of engagement rather than as engagement itself. The EEG analysis (Sections 3.7–3.8) tests what separates **STAY** from **SKIP**, keeping in mind that, as in deBettencourt et al., **STAY** may contain stretches of low attention that do not end in a swipe.

2.7 Parameter Choice in EEG Classification

An EEG classifier does not read a recording in one piece. The recording is first cut into windows. A window is a short slice of the recording. Each window becomes one example for the classifier. Two timing choices shape every window. The window length sets how long each window is. The window placement sets where each window sits relative to an event, such as a cue on the screen. Five studies show how window length and window placement change classification results. A separate line of work describes how window length and window placement are usually tested.

Moontaha et al. [19]. *Research goal:* classify emotions from wearable EEG in real time. In the main experiment, participants watched 16 short emotional videos, 8 with each of two consumer EEG devices. One of the two devices was the Muse S. Moontaha et al. tested window lengths of 1, 2, 3, 4, and 5 s, on the recordings of the study and on a public dataset (AMIGOS). The 1-second window scored best for every emotion dimension, classifier, and device. In most cases, the score fell as the window grew longer. Moontaha et al. read the fall as a sign that the classifier needs more examples. A shorter window gives more examples per video. *Limits:* the windows were shuffled before cross-validation, so neighbouring windows from the same video likely appear in both the training data and the test data, which may favour short windows.

Tzamourta et al. [36]. *Research goal:* detect Alzheimer’s disease from clinical EEG. 24 people took part: 14 patients and 10 healthy people of similar age. Participants sat relaxed with eyes closed. Window lengths ran from 5 to 12 s, in steps of 1 s. Accuracy rose at every step, in all six classification problems. The 12-second window scored best. *Limits:* the target is a disease, and a disease stays the same across the whole recording.

Window length: two directions. Moontaha et al. found shorter windows better. Tzamourta et al. found longer windows better. The finding of Moontaha et al. covers 1–5 s. The finding of Tzamourta et al. covers 5–12 s. The targets differ as well. An emotion can change within one video. A disease stays the same across a recording, so a longer window may simply average more of the same state.

Mi et al. [37]. *Research goal:* improve motor-imagery classification by combining several window lengths. In motor imagery, a participant imagines moving a hand, a foot, or the tongue after a cue. On a public dataset of nine participants, Mi et al. tested single window lengths from 0.5 to 3 s. The best window length differed from participant to participant. For one participant, accuracy differed by about 13 percentage points between window lengths. A combination of several window lengths beat every single window length.

Limits: the window-length test used one dataset of nine participants.

Rimbert et al. [20]. *Research goal:* test how the placement of a rest window changes motor-imagery results. 71 participants imagined left-hand or right-hand movements. A classifier separated imagined movement from rest. The rest window lasted 3 s and sat at one of four positions around the next cue, from 5 s before the cue to 1 s after the cue. The rest window furthest from the next cue gave the highest accuracy. The furthest rest window also sat closest to the end of the previous trial. The furthest rest window scored 4 percentage points above the rest window ending at the cue ($p < 0.05$), and 8 points above the rest window overlapping the cue ($p < 0.001$). Rimbert et al. warn that part of the gain may come from the previous trial. After a movement ends, real or imagined, beta power over the motor cortex rises above the resting level for a while. The rise peaks about 1.5 s after the movement. A rest window close to the previous trial can fall inside the rise. Measured against the raised level, the power drop during imagined movement looks larger than the true drop. Rimbert et al. therefore recommend at least 3 s between the end of one movement and the start of a rest window. The best position also differed between people. The furthest position was best for 52–55% of participants, and the second-furthest position for 22–30%. *Limits:* every rest window lasted 3 s, and the study covers one motor-imagery task.

How timing choices are tested. A sensitivity analysis asks how much a result changes when a setting changes. The simplest design changes one setting and holds every other setting fixed, like turning one knob on a mixing desk while the other knobs stay put. The design is called one-at-a-time (OAT) [38]. OAT never changes two settings together. OAT therefore cannot show an interaction, meaning a case where the effect of one setting depends on the value of another setting [38]. OAT is described as the most popular sensitivity-analysis practice in published modelling work [39]. Saltelli et al. argue that OAT explores the possible settings inefficiently, and recommend alternatives from statistical theory, such as experimental design and regression analysis [39]. Moontaha et al., Tzimourta et al., and Mi et al. each vary window length one setting at a time. Rimbert et al. vary window placement one setting at a time. McCartney et al. vary two settings together.

McCartney et al. [21]. *Research goal:* find the best combination of time window and frequency band for motor-imagery classification. On a public dataset of 109 people, McCartney et al. built one model per person for every combination of 5 time windows and 23 frequency bands. The time windows differed in length and in start after the cue. A repeated-measures ANOVA, a standard test for comparing the same people across conditions, found an effect of time window ($p < 0.001$) and an effect of frequency band ($p < 0.001$). The interaction of time window and frequency band gave $p = 0.044$, close

to the usual 0.05 threshold, and McCartney et al. report the interaction as significant. The best combination for the whole group was 0–4 s after the cue at 4–12 Hz. The best time window differed between people: 0–2.25 s was best for 34 of the 109 participants, and 0–4 s for 31. *Limits:* each per-person model learns from about 90 trials, and the best time window averages 57% accuracy, close to the 50% of guessing between the left and the right hand. The study is a preprint.

Gap the present thesis fills: Every study above either places its windows around an event set by the experimenter, such as a cue or a video, or cuts a resting recording with no event at all. EEG studies of self-paced movement do classify windows before the movement [40], but, to the author’s knowledge, no study varies window length and window placement before a self-paced action during media viewing, such as a swipe. The present thesis labels windows before each swipe as **SKIP** and varies three timing values one at a time: the window length, the gap length between the **SKIP** windows and the swipe, and the imminent-skip period, meaning the stretch that holds the **SKIP** windows (Section 3.4.5).

2.8 Windows at Label Borders

A labelled EEG recording consists of stretches. Each stretch carries one label. In an emotion study, one stretch can be one video, labelled with the rating of the participant. In the present thesis, one stretch is either **STAY** or **SKIP**. A window that crosses the border of a stretch holds EEG from both sides of the border. Yet the window can carry only one label. One EEG study and one review handle windows at a border.

Moontaha et al. [19]. Section 2.7 introduced the emotion study of Moontaha et al. Moontaha et al. cut the EEG into windows without overlap. Every window inside a video gets the rating the participant gave that video. Where possible, Moontaha et al. align the windows with the end of the video. Otherwise, Moontaha et al. use only windows that lie completely inside the time range of the video, so windows that cross the start or the end of a video stay unused. *Limits:* Moontaha et al. give no reason for leaving out windows that cross a video border, and report no test of windows at a border. The version read for the present thesis is the preprint.

Frénay and Verleysen [41]. *Research goal:* review what wrong labels do to a classifier, and how methods deal with wrong labels. Frénay and Verleysen name disagreement between experts as one source of wrong labels. For example, in electrocardiogram analysis, experts seldom agree on the exact

borders of the waves in a heartbeat recording. One reviewed electrocardiogram study therefore removes the labels of examples close to a border between two classes, because experts label less reliably near such a border. The chance of a wrong label can differ between examples. An example that resembles the other class may get a wrong label more often. Wrong labels can lower the accuracy of a classifier and raise the number of training examples a classifier needs. Several reviewed works found that removing examples with wrong labels worked better than correcting the labels. Removing too few or too many examples can both lower performance. *Limits:* the review treats wrong labels in general and includes no EEG study.

Gap the present thesis fills: In Moontaha et al., the experimenter sets every border, so the time of every border is known, and a window that crosses a border can be found and left out. In the present thesis, the viewer ends each video with a self-paced swipe, and the moment the decision to swipe begins is unknown, so even a window that lies completely inside one **STAY** or **SKIP** stretch may carry a wrong label. To the author’s knowledge, no EEG study tests where to place a label border whose time is unknown; the present thesis moves the border between **STAY** and **SKIP** by varying the length of the imminent-skip period, and reports how the separation between **STAY** and **SKIP** changes (Section 3.4.5).

2.9 Class Separation Metric

A study that compares features measured in different units, such as microvolts and hertz, cannot compare the raw differences between two class averages directly. Cohen’s d divides the difference between two class averages by the spread of the feature within the two classes, so the unit cancels out (Section 1.5). One book defines Cohen’s d , and two EEG studies use Cohen’s d to compare two groups of people.

Cohen [42]. *Research goal:* give behavioural researchers tables that show how likely a study with a given number of participants is to detect an effect of a given size. Cohen defines d for two groups as the difference between the two group averages, divided by the standard deviation within the groups. Cohen calls d a “pure” number, free of the original unit, and argues that d helps especially where the raw unit is arbitrary, such as points on an attitude scale. For two groups that spread differently, Cohen divides by the square root of the average of the two variances. For two measurements taken from the same people, such as a sensory threshold measured on both sides of the body of each patient, Cohen gives a second version of d , which first takes one

difference per person and then divides the average of the differences by the standard deviation of the differences.

Henao Isaza et al. [43]. *Research goal:* develop a way to enlarge small EEG samples, tested on telling carriers of a gene mutation that causes Alzheimer’s disease (PSEN1-E280A), who show no cognitive symptoms yet, apart from healthy non-carriers, using EEG recorded at rest. Henao Isaza et al. pool four EEG datasets and, aiming to balance age and sex, form samples with 2, 5, and 10 non-carriers per carrier. A decision tree selects the 100 most important features, and Cohen’s d then describes how far carriers and non-carriers differ in each selected feature. *Limits:* to reach the higher ratios, Henao Isaza et al. remove carriers, from 79 carriers at a ratio of 2 to 15 carriers at a ratio of 10, and the remaining carriers are older: the average age of the carriers rises from about 35 to 42 years, while the non-carriers stay near 30 years. Over the same steps, the largest d of an EEG feature rises from 1.22 to 2.28 in absolute value, and d for age rises from 1.10 to 2.78. The rise of d for age suggests that d changes with the make-up of the sample as well as with the EEG.

Kiarashi et al. [44]. *Research goal:* tell older adults with mild cognitive impairment (24 people) apart from cognitively normal older adults (19 people), using sleep EEG recorded at home over several nights with a device worn in the ear. A classifier gives each person a probability of mild cognitive impairment, and Cohen’s d measures how far the probabilities of the two groups lie apart. Averaging over all recorded nights, instead of using the first night alone, raises d for all three neural networks tested, for example from 0.65 to 0.96. Kiarashi et al. attribute the rise to the averaging, which reduces the night-to-night variation within each person while keeping the differences between people. Kiarashi et al. also compute d for single EEG features between the two groups. The study is a preprint.

Gap the present thesis fills: In Henao Isaza et al. and Kiarashi et al., Cohen’s d compares two groups of different people, with one value per person, and d changes with the people in the sample and with the variation within each person. In the present thesis, STAY windows and SKIP windows come from the same participants, yet Cohen’s d is computed with the two-group version over the windows of all participants together, instead of with the version for two measurements from the same people; the baseline z-scoring of each participant reduces the differences between participants (Section 3.6), but part of the differences may remain in the standard deviation and keep d small. The present thesis uses Cohen’s d to compare the seconds before a self-paced swipe (SKIP) with sustained viewing (STAY), both to rank single

features (Section 3.8.2) and to compare timing values (Section 3.4.5).

2.10 Estimating Frequency from Local Maxima

Peak Frequency, one feature of the present thesis, counts per second how often the strength of one frequency band peaks (Section 3.7.4). Each peak is a local maximum, a point higher than both neighbouring points. The count need not equal the frequency of the oscillation within the band. In EEG research, “peak frequency” often names the frequency with the largest value in the spectrum, meaning the strength of each frequency in the signal; in the present thesis, Peak Frequency names the count of local maxima per second. Three works estimate frequency, or information about frequency, from the signal over time.

Hjorth [45]. *Research goal:* describe the general character of an EEG trace with a few numbers, computed from the signal over time. One of the numbers is Mobility. Mobility is the square root of the ratio between the variance of the first derivative, meaning the slope of the signal, and the variance of the signal. Hjorth shows that Mobility equals the standard deviation of the spectrum along the frequency axis, and reads Mobility as a mean frequency. Computing Mobility from the signal over time needs no spectrum. Hjorth also notes that measuring the time between zero crossings, the moments the signal crosses zero, usually ends up as a frequency measure. *Limits:* Hjorth links Mobility to frequency by a mathematical derivation, and tests the link on no EEG recording. The only recording shown covers one night of sleep, and plots a mean frequency that Hjorth calls very similar to Mobility, and not Mobility itself.

Khambampati et al. [46]. *Research goal:* predict the recovery of patients in a coma after cardiac arrest from EEG, as an entry to a public competition, the PhysioNet Challenge 2023. Khambampati et al. compute three features from the spectrum, among them the dominant frequency, the frequency with the largest value in the spectrum. Khambampati et al. compute three features from the signal over time, among them the zero-crossing rate, the number of times the signal crosses zero. In a pure sine wave, each cycle crosses zero twice. Khambampati et al. describe the zero-crossing rate as information about the frequency of rapid changes in the EEG. A classifier, XGBoost, combines all features with patient data such as age, and predicts the outcome with 81% accuracy on the hidden test data. *Limits:* the zero-crossing rate is one of many features. The paper reports no separate contribution of the zero-crossing rate, and does not compare the zero-crossing rate with the frequency of the signal.

Niknazar et al. [47]. *Research goal:* measure how similar two nonlinear signals are, such as signals of chaotic systems, from the local extrema of the signals. A local extremum is a local maximum or a local minimum. Niknazar et al. use the height of each local extremum and the time between consecutive local extrema. Niknazar et al. state that local extrema carry information about the amplitude and the “global frequency” of a signal, and treat local extrema as a down-sampled version of the signal. Niknazar et al. also state that finding local extrema is highly sensitive to noise, especially high-frequency noise. Added noise raises the number of local extrema. On a public EEG dataset, the method of Niknazar et al. classifies seizure EEG, seizure-free EEG of patients, and EEG of healthy people in five tasks, with 90% to 97% accuracy depending on the task, and 99% to 100% when a second classifier is added. *Limits:* Niknazar et al. test the whole similarity method, and do not test the claim about frequency on its own.

Gap the present thesis fills: The three works count maxima, extrema or zero crossings of the signal itself. To the author’s knowledge, no study counts the local maxima of the strength of an EEG band, so Peak Frequency is a feature the present thesis defines without prior validation. Because added noise raises the number of local extrema [47], the present thesis computes Peak Frequency within each frequency band, where filtering removes noise outside the band. The present thesis tests only how well Peak Frequency separates STAY from SKIP, with Cohen’s d (Section 4.4.1) and with logistic regression (Section 4.4.3).

2.11 EMG Artifacts in EEG

Muscles produce electrical activity of their own. A recording of muscle activity is called an electromyogram (EMG). A scalp electrode records the EMG of nearby muscles together with the EEG of the brain. EMG and EEG share many frequencies, much like two singers who sing the same notes: a filter that keeps some frequencies and removes others cannot cleanly pull the two voices apart. Two head muscles are named among the main sources of EMG in the EEG [48, 49]: the frontalis, the forehead muscle that raises the eyebrows, and the temporalis, the muscle at the temple that closes the jaw. Four studies describe how EMG from head muscles enters the EEG, and how much EMG a cleaning method removes.

Goncharova et al. [48]. *Research goal:* describe which frequencies and which scalp sites the EMG of the frontalis and the temporalis reaches. 25 healthy adults wore a 64-electrode cap. Participants either relaxed or tensed

one muscle at a set strength, for example at 15% of the strongest possible tension. *Result:* EMG covered all frequencies from 0 to more than 200 Hz. Near the tensed muscle, EMG reached every EEG band, delta included. EMG was strongest at the electrodes closest to the tensed muscle, and spread over the whole scalp at higher frequencies. For the temporalis, EMG at temporal and nearby electrodes was highest at 40–80 Hz. On electrodes placed over the muscles themselves, EMG often showed a sharp peak that resembles an EEG beta rhythm, at 16 to 38 Hz for the frontalis and at 13 to 34 Hz for the temporalis, depending on the person. For the frontalis, the strongest EMG lay at the front edge of the scalp in two thirds of participants, usually at Fp1 or AF8. *Limits:* participants tensed each muscle on purpose, guided by feedback on a screen.

Whitham et al. [50]. *Research goal:* measure how much of the scalp EEG comes from muscles. Two men were recorded with 115 EEG channels twice: once normally, and once under a drug that paralysed all skeletal muscles except one arm, kept free for answering. Paralysis removes EMG, so the difference between the normal and the paralysed recording shows the share of EMG. *Result:* under paralysis, power above 20 Hz fell, by about 100-fold at 40–100 Hz. The largest drop, above 200-fold, lay at the lower edge of the scalp, near the frontalis, the temporalis and the neck muscles. Below 20 Hz, the normal and the paralysed recording did not differ. Whitham et al. conclude that most scalp EEG above 20 Hz comes from muscles, even in a relaxed person. *Limits:* the study has two participants. Whitham et al. note that the participants were highly alert during the procedure, which may have raised muscle activity, so the reported share of EMG may be on the high side.

Yilmaz et al. [49]. *Research goal:* test whether head muscles send EMG into the EEG during rest, and whether independent component analysis (ICA) removes the EMG. ICA is a common cleaning method: ICA splits the EEG into parts, and the parts judged to come from muscles or eyes are removed. In 13 healthy adults, fine wires inside the frontalis recorded single motor units, the smallest units of a muscle that fire together, while a 32-electrode cap recorded the EEG. For comparison, Yilmaz et al. reused recordings of the temporalis from an earlier study. *Result:* motor units of the frontalis fired continuously while participants rested. Averaged over many firings, many motor units left a small trace on nearby frontal electrodes, mostly Fp1, Fp2, F3 and F4. Traces from the temporalis spread wider, and were strongest at F7 or Fp1. At most electrodes, ICA made the traces of both muscles significantly smaller, yet did not remove the traces completely. *Limits:* the study covers two muscles, and ICA was tested only on the 19 of 58 frontalis motor units with a clear trace on more than one electrode.

Li et al. [51]. *Research goal:* remove more EMG by adding EMG recorded from head and neck muscles as extra input to ICA. *Result:* in 8 healthy adults clenching one hand, the method removed about 75% of the power at 40–100 Hz, a range dominated by EMG, where ICA alone removed about 49%. *Limits:* the method still cannot remove all EMG, partly because some head and neck muscles are hard to record from the skin.

What the four studies show together. Whitham et al. and Yilmaz et al. find EMG even at rest. Below 20 Hz, Goncharova et al. show EMG near a tensed muscle, while Whitham et al. found no difference below 20 Hz between the normal and the paralysed recording. ICA, with or without extra EMG input, reduces EMG without removing EMG completely.

The Muse S of the present thesis records at AF7 and AF8 on the forehead, near the frontalis, where Goncharova et al. and Yilmaz et al. found the strongest EMG, and at TP9 and TP10 behind the ears, at the lower edge of the scalp next to the small muscles around the ear [52], where Whitham et al. found the largest share of EMG. The present thesis applies no EMG correction, records no extra EMG, and removes no windows flagged as artifacts (Section 3.6.4). Facial muscle activity changes with mental effort and emotion [52], so if jaw or forehead tension changes before a swipe, the three features the present thesis tests further, TP10 raw standard deviation, AF7 high-gamma mean and AF8 delta peak frequency (Section 4.4.1), may reflect muscle activity, brain activity, or both. The present thesis cannot tell muscle activity and brain activity apart, and returns to the origin of the three features in Section 4.5.

2.12 Delta-Band Function in Awake Cognition

Delta is the slowest EEG band. The present thesis defines delta as 1–4 Hz (Section 3.6). Delta is best known from deep sleep. Two reviews describe what delta does in the awake brain, and one study relates delta to addiction to short-video apps.

Güntekin and Başar [53]. *Research goal:* review delta responses in healthy people and in several disorders. A delta response is a change in delta right after a stimulus, such as a tone or a picture. *Result:* delta responses take part in cognitive processes, mainly in decision making and attention, and in emotional processes. Delta responses are smaller in Alzheimer’s disease, mild cognitive impairment, bipolar disorder, schizophrenia and alcoholism. Güntekin and Başar therefore propose a smaller delta response as a possible general marker of impaired cognition. *Limits:* the review mainly covers delta right after a stimulus.

Knyazev [54]. *Research goal:* explain what delta does, from the view of evolution. *Result:* delta dominates the waking EEG of reptiles. In humans, delta is strong mainly early in development and in deep sleep. Delta rises with hunger, sexual arousal and pain, and in drug users, which links delta to motivation. In cognition, delta is linked to attention, to salience detection, meaning the noticing of what matters, and to perception below awareness. Knyazev concludes that delta reflects old, basic processes. In awake adults, newer processes tied to faster bands overshadow the old processes. The old processes rise when the newer processes lose priority, for example in sleep, early in development, or in disease. *Limits:* the conclusion is a hypothesis built on a review, and Knyazev notes that the evidence is scarce.

Xu et al. [55]. *Research goal:* test whether addiction to short-video apps goes with weaker working memory and weaker cognitive control in adolescents. Cognitive control was tested as the switching between two tasks. *Design:* 84 Chinese adolescents (mean age 12.3) filled in an addiction questionnaire and did two cognitive tests. One week later, an 8-electrode cap recorded 1 minute of rest and then 3 minutes of free browsing on Douyin, the Chinese version of TikTok. Xu et al. compared the power of each band during browsing with the power at rest. *Result:* delta rose over the whole head during browsing, which Xu et al. read as attention to the videos. Of all bands, only the change in delta predicted the addiction score ($F = 2.184$, $p = 0.040$, adjusted $R^2 = 0.115$). The more addicted an adolescent, the less delta rose at left temporal and left occipital electrodes. The delta pattern tied to addiction predicted lower working memory ($r = -0.226$, $p = 0.043$) and lower cognitive control ($r = -0.220$, $p = 0.047$). *Limits:* the effects are small, browsing lasted 3 minutes, and the paper reports no times of single swipes.

What the three works show together. Güntekin and Başar link delta right after a stimulus to attention and decision making. Knyazev links delta to motivation, attention and salience detection, drawing also on delta at rest and in sleep. Xu et al. link the change in delta during free browsing to a questionnaire score of addiction.

Gap the present thesis fills: None of the three works relates delta to the seconds before a self-paced swipe: Güntekin and Başar review delta mainly right after a stimulus, and Xu et al. report no times of single swipes. The three works also study mainly the strength of delta, whereas AF8 delta peak frequency, one of the three features the present thesis tests further (Section 4.4.1), counts how often per second the strength of the delta band peaks (Section 3.7.4). The present thesis tests how well the delta features,

AF8 delta peak frequency among them, separate the seconds before each swipe (SKIP) from sustained viewing (STAY), and compares the delta features with the features of the other bands and with the Engagement Index (Sections 4.4.1 and 4.4.3).

2.13 Notch Filtering and the High-Gamma Band

The recordings of the present thesis took place in Germany (Section 3.2). Power lines in Germany carry alternating current at 50 Hz. EEG electrodes pick up the electrical field of the power lines as powerline-noise, a sharp peak at 50 Hz. A notch filter removes a narrow band of frequencies around 50 Hz. The present thesis defines high gamma as 40–60 Hz (Section 3.6). The 50 Hz peak therefore lies in the middle of high gamma. A notch filter removes powerline-noise and any brain activity near 50 Hz together. One study compares a low-cost EEG amplifier with a research amplifier in the gamma range.

Pattisapu and Ray [56]. *Research goal:* test whether a low-cost, open-source EEG amplifier (OpenBCI) records gamma as well as a research amplifier (Brain Products). *Design:* 11 adults looked at full-screen striped patterns. Full-screen striped patterns are known to raise gamma. Each participant was recorded in two sessions, once with each amplifier. Both amplifiers recorded through the same cap of gel electrodes. *Result:* OpenBCI recorded a gamma response, a rise in gamma power from before the patterns to during the patterns, in almost all participants who showed a gamma response with the research amplifier. The alpha responses and gamma responses of the two amplifiers were highly correlated. In a test with the electrodes in salt water and no person attached, OpenBCI showed more powerline-noise than the research amplifier. OpenBCI also showed three extra peaks, at 14, 36 and 64 Hz. 36 and 64 Hz lie 14 Hz below and above 50 Hz. Pattisapu and Ray trace the extra peaks to a distortion of the powerline-noise inside the OpenBCI amplifier. The research amplifier showed no extra peaks. In the recordings of people, the extra peaks remained in some participants. The gamma response was smaller with OpenBCI, especially in fast gamma (35–66 Hz). Pattisapu and Ray name the extra peaks as one possible reason. *Limits:* the study used gel electrodes, unlike the dry electrodes of the Muse S.

In the bands of the present thesis, the three extra peaks would fall into beta (14 Hz), low gamma (36 Hz) and very high gamma (64 Hz). The present thesis did not test whether the Muse S shows the same extra peaks. To the author’s knowledge, no study measures for a consumer headband how much

of high gamma is powerline-noise and how much is brain activity. The present thesis applies no notch filter in any reported analysis. A notch-filtered copy serves only the artifact check (Section 3.6.4). Two of the three features the present thesis tests further may therefore partly reflect powerline-noise: AF7 high-gamma mean, and TP10 raw standard deviation, computed on the raw signal, the signal before the split into bands, which keeps the 50 Hz peak. If the Muse S shows the same extra peaks, beta, low-gamma and very-high-gamma results may also partly reflect the extra peaks (Section 4.5). Very high gamma (60–100 Hz) also reaches 100 Hz, twice 50 Hz, where powerline-noise can show a second peak.

2.14 Pre-Motor Window

The brain prepares a voluntary movement before the movement starts, and the muscles activate shortly before the movement. EEG recorded shortly before a movement may therefore partly reflect the preparation of the movement. Three works give the timing of brain and muscle activity before a self-paced movement.

Libet et al. [57]. *Research goal:* compare the start of brain activity before a voluntary movement with the moment a person becomes aware of wanting to move. *Design:* 5 participants flexed the fingers or the wrist of the right hand whenever they felt like it. Libet et al. measure all times from the start of muscle activity in the forearm, which comes just before the movement. Participants watched a spot circling like a clock hand. After each movement, participants reported where the spot had been when the wish to move first appeared. *Result:* the brain activity before the movement is called the *readiness potential*, or *Bereitschaftspotential*. The readiness potential is a slow negative shift in the EEG. In series without planning ahead, the readiness potential began on average about 550 ms before the muscle activity. In series where participants sometimes planned ahead, the readiness potential began on average about 1,025 ms before the muscle activity. The reported wish to move came about 200 ms before the muscle activity, so the readiness potential began about 350 ms before the reported wish without planning ahead, and about 800 ms before the reported wish with some planning ahead. Libet et al. conclude that the brain can start a voluntary movement before the person is aware of wanting to move. *Limits:* the study has 5 participants, the moment of awareness rests on what each participant recalled, and each EEG recording covered only 1.4 s before the muscle activity, so an earlier start of the readiness potential could not be seen.

Shakeel et al. [58]. *Research goal:* review methods that detect from

EEG that a person is about to move. *Result:* the readiness potential begins about 1 to 1.5 s before the movement. The readiness potential has an early part, which rises slowly, and a late part, which begins about 400–500 ms before the movement and rises more steeply. The start of the readiness potential varies considerably between conditions and between people. The summary of the review gives a start of about 2 s before the movement. *Limits:* the timing values come from earlier studies, and the review measures none of them.

Lew et al. [40]. *Research goal:* detect from EEG, in single trials, that a person is about to start a self-paced reaching movement. *Design:* in the first experiment, 8 healthy adults rested one hand on a home button. A sound named one of four targets. Participants then reached for the target whenever they wished, but not earlier than 2 s after the sound. Release of the home button marked the start of the movement. Electrodes on the skin recorded the EMG of the biceps, the muscle at the front of the upper arm. *Result:* the biceps activated on average 263 ± 40 ms before the start of the movement. A classifier detected the coming movement from the EEG above chance, on average about 460 ms before the start of the movement. *Limits:* the movement is an arm reach, and a sound set the target and the earliest start.

What the three works show together. The reported start of the readiness potential ranges from about 0.55 s before the movement in Libet et al. to about 2 s in the summary of Shakeel et al., and varies between people. All three works measure time from the movement, or from the muscle activity just before the movement.

The present thesis leaves the *pre-swipe gap*, the last 2.0 s before each keypress that marks a swipe, out of both **STAY** and **SKIP** (Section 3.4.1). The participant presses the key with the other hand at the moment of each swipe (Section 3.3), so the brain likely prepares two movements, the swipe and the keypress. If the keypress comes slightly before or after the swipe, the true gap before the swipe is slightly shorter or longer than 2.0 s. The pre-swipe gap exceeds the start of about 1 to 1.5 s in Shakeel et al. by about 0.5 to 1 s, and leaves no margin to the start of about 2 s in the summary of Shakeel et al. The present thesis does not test whether the **SKIP** windows are free of movement preparation.

Chapter 3

Materials and Methods

3.1 Participants

Twenty-eight ($N = 28$) participants were recruited for the study (14 female, 14 male). To participate, individuals were required to have normal or corrected-to-normal vision and no history of neurological disorders. Three participants were excluded prior to analysis: two due to a high number of self-reported keypress errors (greater than 10) which compromised label integrity, and one due to an undisclosed neurological condition. The experimenter's own data were recorded separately for pipeline validation and were never part of the recruited sample; they are excluded from all reported results. The final sample consisted of $n = 25$ participants (age range 20–30 years, $M = 24.6$, $SD = 2.9$, 13 female, 12 male). Figure 3.1 traces this selection from the recorded sessions through to the analysed sample. It also accounts for the participant-ID range used throughout this thesis: the first three recorded sessions (P1–P3) were pilot recordings made for pipeline validation and were never part of the recruited sample, so the recruited participants carry the identifiers P4–P31, and the retained identifiers still run up to P31 even though only $n = 25$ participants remain.

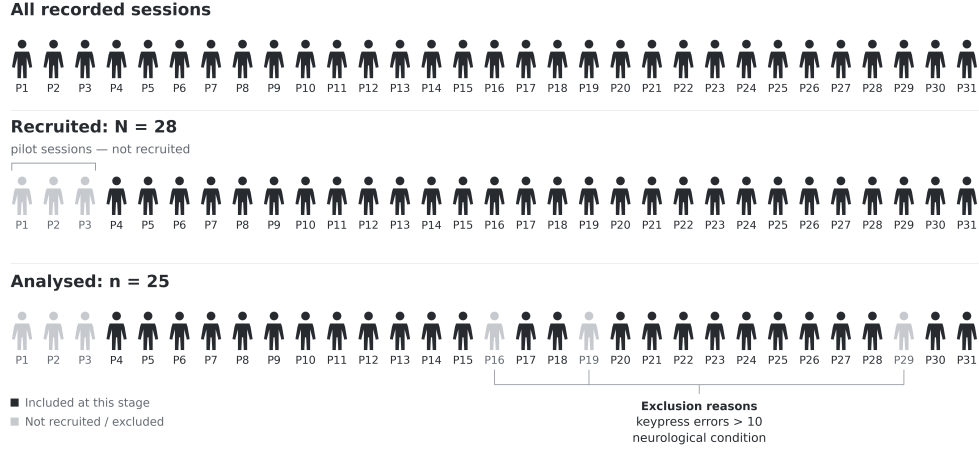


Figure 3.1: Participant selection from the recorded sessions to the analysed sample. Of the 31 recorded sessions, the first three (P1–P3) were pilot recordings for pipeline validation and were not part of the recruited sample, leaving $N = 28$ recruited participants carrying the identifiers P4–P31. Three of these were excluded prior to analysis (P16, P19, P29): two for self-reported keypress errors greater than ten, which compromised label integrity, and one for an undisclosed neurological condition. This yields the final sample of $n = 25$. Participant identifiers are kept unchanged throughout the thesis, which is why IDs reach P31 while the analysed sample size is 25.

Participants were collected over a total of five days and were conceptually divided into two groups based on recruitment timing: an initial smaller group was recorded in a separate room without financial compensation, while the subsequent main group (recorded a few days later) received a 20-euro voucher as an incentive - this may be noted as a potential limitation of this study, and subject of further investigation beyond the scope of this work. All participants provided informed consent and signed data protection agreements approved by the Ethics Committee of the University of Potsdam prior to the study (No. 5/147, 08 Sep 2025; “Optimizing Large Scale Public Health Interventions on Social Media by Creating Neurological Markers of Engagement and Addiction in TikTok to Drive Disease Prevention”). Because recruitment relied substantially on university mailing lists, the sample skews toward an academically-oriented population; the implications of this for generalizability are addressed in Section 4.5 and in Future Research.

ID	Sex	Age	Daily SFV Use	Sleep (h)	Alertness	Consumed	Status
P4	F	26	1-30m	6-7	Neutral	Caffeine	Valid
P5	F	30	1-2h	7-8	Alert	None	Valid
P6	F	24	1-30m	7-8	Neutral	None	Valid
P7	F	23	30-60m	6-7	Neutral	None	Valid
P8	F	23	30-60m	6-7	Neutral	None	Valid
P9	M	27	30-60m	6-7	Neutral	Caffeine	Valid
P10	F	24	1-30m	>8	Neutral	Caffeine	Valid
P11	F	26	30-60m	7-8	Neutral	Caffeine	Valid
P12	F	20	30-60m	5-6	Neutral	Energy Dr.	Valid
P13	M	28	1-30m	>8	Neutral	Caffeine	Valid
P14	F	22	1-2h	>8	Tired	None	Valid
P15	M	28	1-30m	7-8, >8	Neutral	Caffeine	Valid
P16	M	32	1-30m	7-8	Alert	None	Excluded
P17	F	30	30-60m	6-7	Neutral	Caffeine	Valid
P18	M	27	1-2h	7-8	Alert	Caffeine	Valid
P19	F	30	1-30m	7-8	Neutral	Caffeine	Excluded
P20	M	20	1-30m	>8	Alert	Caffeine	Valid
P21	M	21	30-60m	6-7	Neutral	None	Valid
P22	F	21	1-30m	>8	Alert, Neutral	None	Valid
P23	M	23	1-2h	>8	Neutral	None	Valid
P24	F	24	1-2h	>8	Neutral	Caff+Nic	Valid
P25	F	21	>3h	>8	Neutral	None	Valid
P26	M	24	30-60m	6-7	Neutral, Tired	None	Valid
P27	M	23	2-3h	7-8	Alert	None	Valid
P28	M	28	30-60m	7-8	Very alert	None	Valid
P29	M	22	1-2h	>8	Very alert	None	Excluded
P30	M	26	>3h	5-6, 6-7	Very alert	None	Valid
P31	M	27	1-2h	6-7	Alert	None	Valid

Table 3.1: Participant demographics and pre-session survey responses. SFV = Short-Form Video. P16, P19, and P29 were excluded from the final analysis.

3.2 Data Acquisition

Electroencephalography (EEG) data were recorded using a consumer-grade Muse S headband (Generation 2, see Figure 3.2), which has been used successfully in similar EEG-based tasks [59, 19] and has seen broader adoption in peer-reviewed EEG-based user engagement research [18]. The device features four dry electrodes placed at positions AF7, AF8, TP9, and TP10 according to the international 10-20 system, with the reference electrode located at Fpz. Validation work indicates that this system yields statistically quantifiable ERP components (N200, P300, reward positivity) at exactly these electrode sites, indistinguishable from those recorded with a research-grade 64-channel

system, though this validation is bounded to low-frequency (<15 Hz) ERP components [59]. Two physical units of the same model were used across the cohort; device-related effects therefore cannot be fully separated from participant-related effects (see Section 4.4.1). Data were continuously sampled at a rate of 256 Hz, consistent with the hardware specifications of the Muse S headband [22]. All recordings took place in Germany.



Figure 3.2: Schematic of the consumer-grade Muse S (Gen 2) EEG headband that was used for data acquisition, shown in the worn position with the frontal sensor block across the forehead and the band passing over the ears.

The EEG signals were transmitted via Bluetooth and recorded through the Lab Streaming Layer (LSL) protocol using the `muselsl` library [31]. A custom Python recording script was developed to handle data collection (https://github.com/coffeeandcoffee/eeg_tiktok_pipeline). This script featured real-time frequency monitoring and an automatic reconnection mechanism to seamlessly manage any Bluetooth drops without disrupting the session.

To synchronize the behavioral data (video transitions) with the continuous EEG stream without relying on video monitoring of the participant, keystroke markers were mapped directly into the recording script. Participants used their non-dominant hand to press a specific key on a laptop keyboard provided in the room: the “B” key to denote the start of the baseline period, and the “A” key to mark every manual swipe (skip event) during the TikTok browsing session. The recording cabin and the physical arrangement of phone, laptop, and headband are shown in Figures 3.3 and 3.4.



Figure 3.3: The experimental setup cabin ensuring privacy and a distraction-free environment for the participant.



Figure 3.4: Close-up schematic sketch of the recording setup showing the researcher’s phone and the laptop used for keypress synchronization by the participant wearing the EEG headband while naturalistic TikTok browsing.

3.3 Experimental Protocol

The experimental sessions were scheduled for 45-minute timeslots, with the actual experimental procedures taking approximately 35 to 40 minutes. Upon arrival, participants were greeted, provided with an information sheet, and asked to sign the ethics consent forms. Afterward, they completed a pre-

experiment demographic and basic screening survey, which took approximately five to ten minutes.

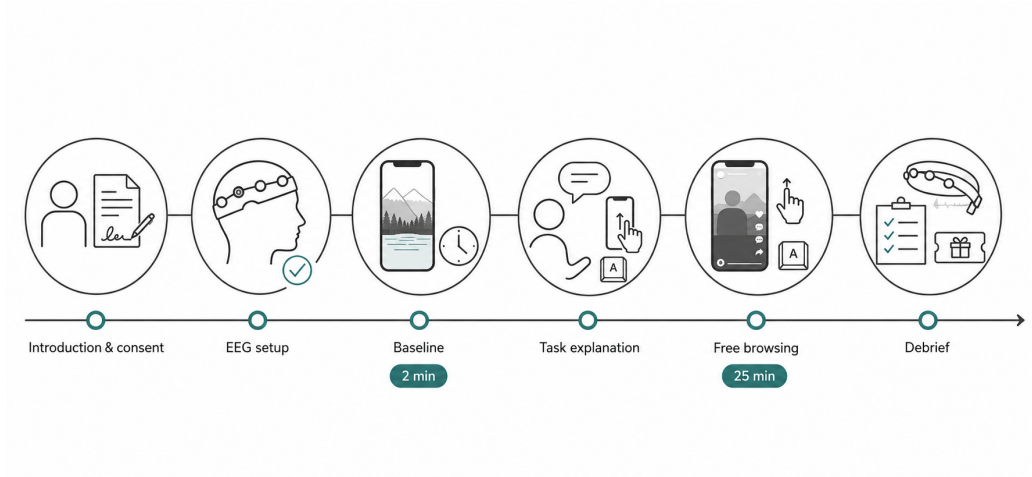


Figure 3.5: Experimental protocol flow: Introduction and consent, EEG setup, 2-minute baseline video, task explanation, and 25-minute TikTok free browsing session, debrief.

Figure 3.5 summarizes the resulting session structure. Participants were then instructed on the tasks. The Muse S headband was mounted on their head and adjusted to ensure optimal skin contact, particularly on the forehead and behind the ears for participants with long hair. The experimenter verified the signal quality and ensured the device was adequately charged.

The experimental paradigm consisted of two main phases: a baseline recording and a free-browsing session. For the baseline phase, participants were provided with the researcher’s iPhone, and a two-minute nature video on YouTube was started. Participants were instructed to relax and simply watch the video. Concurrently, the recording script was initiated, and the “B” marker was pressed. To ensure ecological validity while keeping the participant focused, the iPhone’s “Guided Access” feature was enabled to prevent exiting the application. Participants were left alone in the room to afford them privacy. After two minutes, the Guided Access lock visually timed out, signaling the conclusion of the baseline period.

The experimenter then returned to explain the second phase: a naturalistic short-form video browsing task using the TikTok application. To ensure a standardized yet personalized starting state, the app’s “For You” feed was refreshed using the “Reset feed” option in the content preferences settings.

Participants were instructed to consume TikTok content as they normally would, watching videos they found interesting and swiping away those they found boring. To restrict the action space strictly to scrolling behavior, they were asked to refrain from using the “like” function, writing comments, or visiting user profiles.

Crucially, participants were tasked with pressing the “A” button simultaneously with every swipe motion on the phone to accurately synchronize their micro-decisions with the EEG data. For instance, a participant holding the phone in their right hand used their left hand to press the designated key on the adjacent laptop keyboard; Figure 3.6 illustrates this synchronization step.

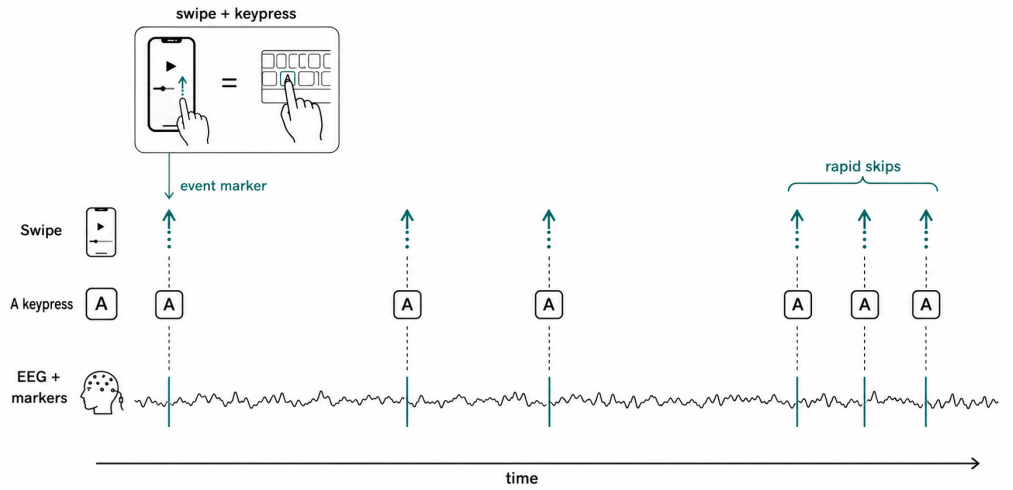


Figure 3.6: Schematic representation of the manual synchronization process: the participant swipes to skip the video and simultaneously presses the ‘A’ button to mark the event in the EEG stream.

The Guided Access feature was reactivated and set to a 25-minute limit for the TikTok app. Participants were reminded that the headband script would automatically reconnect if a Bluetooth disconnection occurred. They were then left alone in the room with the door closed, creating an isolated viewing environment designed to elicit natural responses to the video feed. This unconstrained free-viewing design follows standard practice in naturalistic neuroimaging, where subjects freely viewing continuous audiovisual material, with no fixation constraint, trial structure or controlled stimulus onset, nonetheless produced extensive, highly significant intersubject correlation [60]. They were instructed to knock on the glass door if any errors

or issues arose; however, no significant interruptions occurred during the sessions.

Once the 25-minute viewing task concluded, the experimenter entered the room, and the participant removed the headband. They then answered a few post-session survey questions, specifically self-reporting if they made any keypress synchronization errors during the session. Participants in the compensated group then signed the required forms to receive their 20-euro voucher. The post-measurement debrief and surveying process took roughly two to four minutes. Participants were thanked for their time, briefly debriefed about the broader context of the study if interested, and dismissed.

3.4 Labeling Strategy

To quantify class separability under different labeling choices, Cohen’s d was computed as the standardized mean difference between the STAY and SKIP distributions. A one-at-a-time parameter sensitivity analysis was then performed in which gap length, window length, and imminent-skip period were each varied while the remaining two were held fixed (Section 3.4.5).

3.4.1 Operational Definition of Labeling Parameters

The labeling strategy was derived directly from the research questions and the aim of isolating neural states relevant to sustained engagement and imminent disengagement during naturalistic browsing. Accordingly, the computational target was constrained to two discrete classes: sustained engagement (STAY) and impending disengagement (SKIP), reflecting the distinction between sustained attentional maintenance, understood as a fluctuating internal state rather than a fixed behavioral duration [2], and passive or dissociative browsing states in which continuous consumption need not indicate engagement [4, 3].

The default labeling scheme was defined by three temporal parameters. The gap length was set to 2.0 seconds, the window length was set to 1.0 second, and the imminent-skip period was set to 3.0 seconds; a fixed 0.5-second post-skip orientation grace period was additionally excluded from both classes. These defaults were not chosen arbitrarily: the 1.0-second window matches the value found to maximize classification performance in a direct 1–5 s window-length sweep on wearable EEG, where accuracy generally declined as window length increased [19]. The 2.0-second gap exceeds the onset of the readiness potential, reported at about 1 to 1.5 s before self-paced movement with a late, steeper component beginning 400–500 ms before onset

[58], so the gap was set beyond this window with a margin intended to reduce contamination of the **SKIP** epochs by motor-preparatory activity. The gap also exceeds the directly measured onset of arm muscle (EMG) activity in a self-paced reaching task, on average 263 ± 40 ms before movement onset [40]. Because the start of the readiness potential varies between people, and the summary of the same review gives a start of about 2 s before the movement [58], the margin is small (Section 2.14). The 3.0-second imminent-skip period was not derived from a specific literature value and is treated as an heuristic starting point, addressed directly by the parameter sensitivity analysis in Section 3.4.5. In the default configuration, the **SKIP** class therefore comprised the 1.0-second windows tiled across the 3.0-second imminent-skip period preceding the 2.0-second gap before each registered A-keypress (swipe event), yielding a label placement that aligned with the pre-skip attentional state while avoiding the most immediate motor-preparatory interval. The **STAY** class comprised all remaining continuous viewing segments that satisfied the exclusion criteria described below.

3.4.2 Operational Definition of **STAY** and **SKIP**

The **SKIP** class was drawn from the imminent-skip period, defined as the 3.0-second interval terminating 2.0 seconds before the timestamp of each registered A-keypress (swipe event); the labeled **SKIP** epochs are the 1.0-second windows tiled within this interval. This interval captures neural activity preceding the physical act of skipping a video while placing the **SKIP** window beyond the readiness potential, the slow EEG shift of motor preparation, which has been reported to begin about 1 to 1.5 s before movement onset [58]; the 2.0-second gap preceding the swipe marker therefore exceeds this interval, with limited margin at the earliest reported onsets of the readiness potential. This gap also exceeds the directly measured onset of arm muscular (EMG) activation in a self-paced reaching task, reported at 263 ± 40 ms before movement onset [40], indicating that the gap is unlikely to be contaminated by the physical motor execution itself, though residual overlap with the readiness potential cannot be fully excluded given between-subject variability in the start of the readiness potential.

The **STAY** class represents continuous, non-skip viewing periods, which are treated as a behavioral proxy for sustained engagement. It is noted that continuous viewing time is an imperfect proxy: passive dwell time can also reflect disengaged or dissociative states rather than effortful attention [4, 3], and sustained attention itself is better understood as a fluctuating internal state than a fixed behavioral duration [2]. No EEG marker has been shown to positively validate duration-based labeling as a proxy for cognitive engagement;

recent work directly comparing sustained-attention and engagement spectral markers found engagement-sensitive markers failed to discriminate task conditions and that duration-aggregated metrics obscure real-time fluctuations in cognitive state [17], though the authors attribute the frontal-midline-theta null result mainly to a baseline confound, and further to median averaging and to a similar load in both tasks, rather than to an unqualified failure of the marker. The **STAY** class was defined by extracting all remaining continuous viewing periods during the naturalistic TikTok consumption phases that were i) outside the 3.0-second imminent-skip period, ii) outside the 2-second pre-skip motor-preparation buffer, and iii) outside the 0.5-second post-skip orientation grace period.

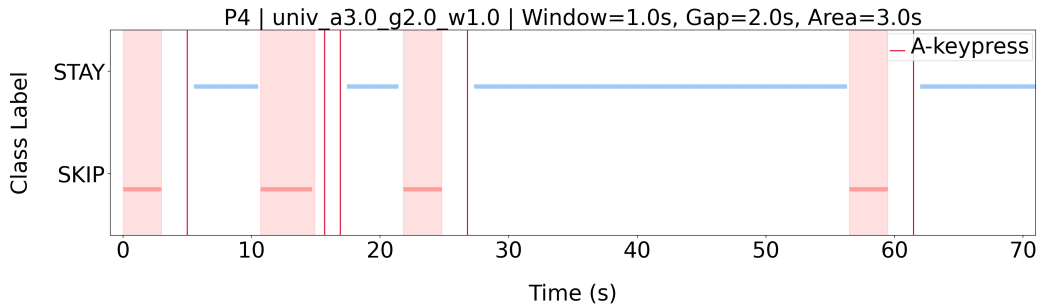


Figure 3.7: Schematic overview of the labeling strategy under the default parameterization. The figure illustrates the placement of the **STAY** and **SKIP** labels relative to the registered press markers, including the default gap length, window length, and imminent-skip period.

3.4.3 Baseline Exclusion

The manually categorized pre-task resting baseline phases were excluded from the **STAY**/**SKIP** classification dataset and used only for per-participant signal normalization. Consequently, the algorithms were trained to discriminate between continuous non-skip viewing (**STAY**) and the period preceding a swipe-away event (**SKIP**) during browsing, rather than between resting and viewing. Figure 3.7 illustrates the default labeling logic, including the press markers and the temporal placement of the labeling windows.

In addition to baseline exclusion, the labeling procedure omitted segments affected by boundary overlap between **STAY** and **SKIP** definitions to preserve clear class assignment and avoid ambiguous temporal regions near each swipe event, similar to Moontaha et al., who, where windows cannot be aligned with the end of a video, use only windows that lie entirely within the video

[19]. In the label-noise literature, removing mislabelled examples has often worked better than relabelling them, although removing too few or too many examples can both lower performance [41]; the exclusion was kept despite the resulting reduction in sample count. The final label construction therefore prioritized temporal purity of the target classes over maximal sample count.

3.4.4 Class Separation Metric

Cohen’s d measures how far the **STAY** windows and the **SKIP** windows lie apart in one feature, in units of the standard deviation of the feature (Sections 1.5 and 2.9). A larger $|d|$ means a stronger separation between **STAY** and **SKIP**. The denominator is the square root of the average of the two class variances, the version Cohen gives for two groups that spread differently [42]:

$$d = \frac{\bar{x}_{\text{STAY}} - \bar{x}_{\text{SKIP}}}{\sqrt{\frac{s_{\text{STAY}}^2 + s_{\text{SKIP}}^2}{2}}} \quad (3.1)$$

where $\bar{x}_{\text{STAY}}$ and $\bar{x}_{\text{SKIP}}$ denote the means of the feature over all **STAY** windows and over all **SKIP** windows, and s_{STAY}^2 and s_{SKIP}^2 the corresponding variances.

For each parameter configuration, Cohen’s d was computed over the windows of all participants together, once for every feature (e.g. `TP10_raw_std`), yielding one $|d|$ value per feature. The two levels at which $|d|$ is reported in the feature-ranking analyses, exact per-feature values and category means, are described in Section 3.8.2.

3.4.5 Parameter Sensitivity Analysis

To assess how label definition affects class separability, a controlled one-at-a-time (OAT) sensitivity analysis [38, §2.4.2] over three temporal parameters was conducted: in each experiment two parameters were held constant while the third was varied, so that any change in separability is attributable to the varied parameter alone. This follows established practice in EEG decoding, where window length [19, 36, 37] and window placement [20] have been varied in isolation to quantify their effect on classification performance. Two limitations follow from the design. First, attribution is local to the operating point at which the remaining parameters were fixed [38]. Second, interaction effects are not identifiable under an OAT scheme by construction; resolving them would require a factorial sweep of the kind used for time window and frequency band in motor-imagery decoding [21]. The direction of window-length effects is mixed: Tzamourta et al. found EEG-classification accuracy increased step-wise from 5 s to 12 s windows across six classification problems

with no exceptions [36], whereas Moontaha et al. found performance generally declined from 1 s to 5 s windows [19]. In MI-BCI paradigms, rest windows positioned further from the upcoming task cue yielded higher classification accuracy than rest windows positioned closer to the cue, a gain Rimbert et al. attribute partly to the rebound in beta power after the previous movement [20]. Window-length effects also differ between people: a multi-scale CSP study found subject-specific optimal window sizes with accuracy differences of up to 13% depending on window choice [37], reinforcing that the present sweep should be read as suggestive rather than as evidence of a universal optimum.

The influence of window size (ws), gap size (gs), and imminent-skip period (isp) on data preprocessing and segmentation is illustrated in three example-configurations: (i) default parameters (ws = 1s, gs = 2s, isp = 3s; Figure 3.7), (ii) reduced window size (ws = 0.3s, gs = 2s, isp = 3s; Figure 3.8), and (iii) reduced gap with extended imminent-skip period (ws = 1s, gs = 0.5s, isp = 4s; Figure 3.9). The swipe event is indicated by vertical red lines. All figures show the same randomly selected 70-second segment from a single participant. Time (seconds) is plotted on the x-axis, and class labels, STAY (blue) and SKIP (red), on the y-axis.

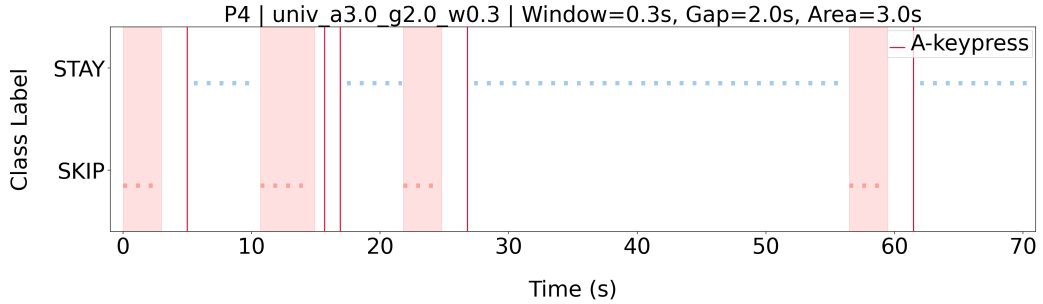


Figure 3.8: Window parameter exploration (default parameters, except a smaller window size).

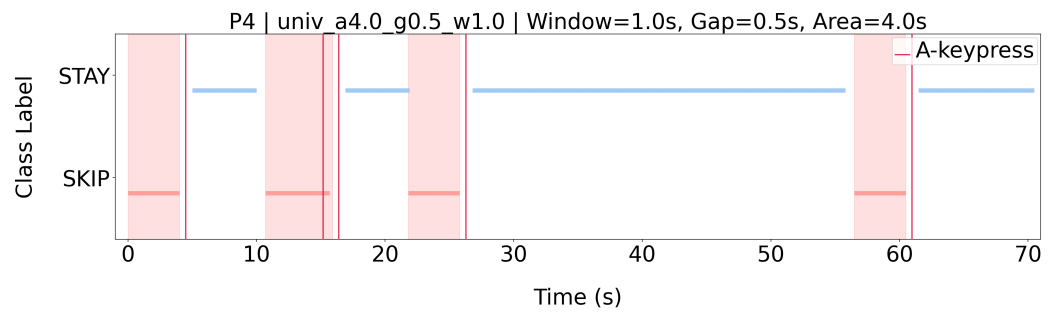


Figure 3.9: Window parameter exploration (default parameters, except a smaller gap size & a longer imminent-skip period (Skip-Area)).

Window Exploration: Maximizing Feature Separation

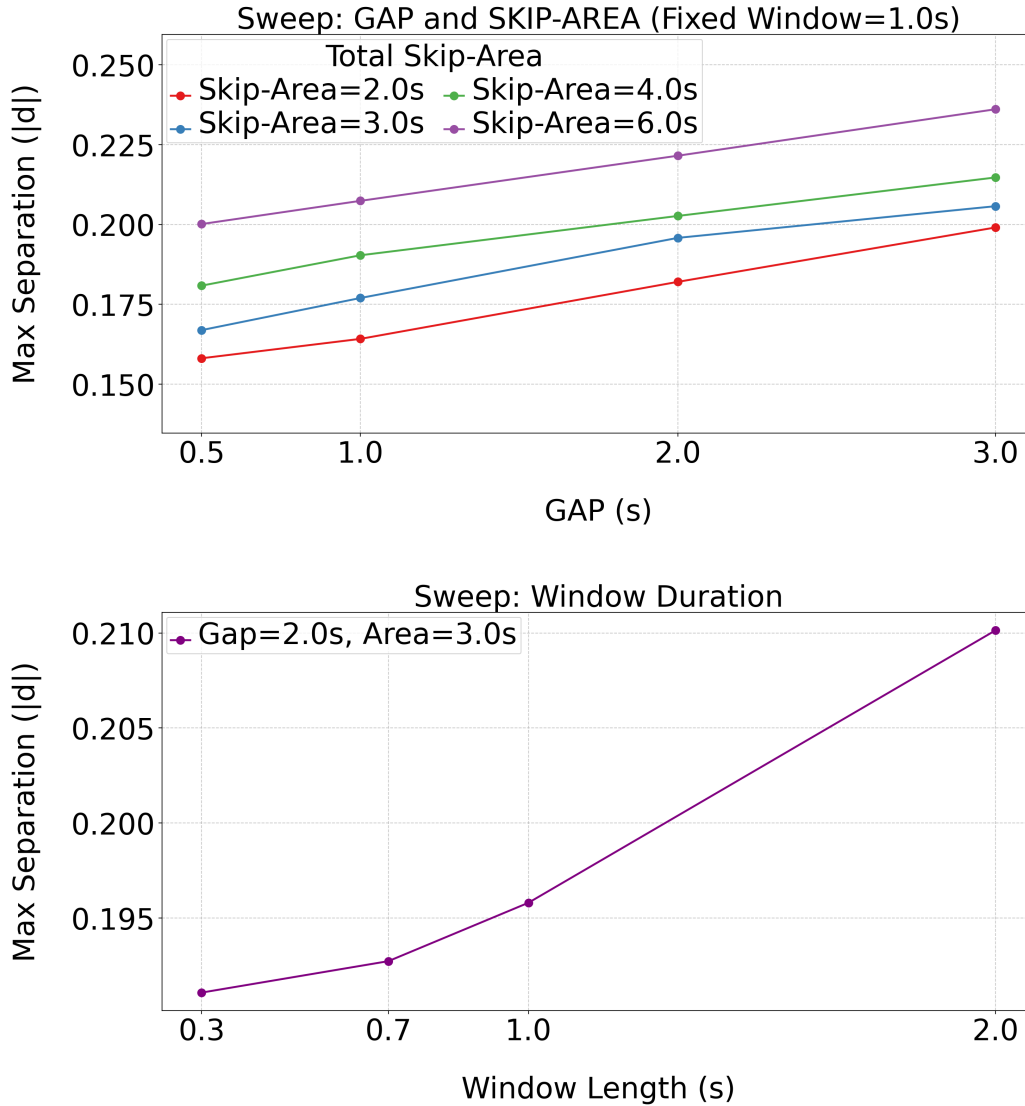


Figure 3.10: Parameter sweep of the window exploration analysis. The sweep evaluates only individual component features and does not use the Engagement Index.

The key result is that, with fixed imminent-skip period and window size, increasing the gap length consistently increases Cohen's d (Figure 3.10, top), indicating greater separability as the SKIP interval moves further from the swipe event. A similar effect is observed for longer imminent-skip periods,

where larger SKIP intervals also improve separability. For fixed gap and imminent-skip period, increasing window size is monotonically associated with larger Cohen’s d (Figure 3.10, bottom), again corresponding to increased class separation. The window-length trend matches Tzimourta et al., who found accuracy rising with window length over 5–12 s [36], and runs against Moontaha et al., who found performance generally falling over 1–5 s [19]. The gap-length trend resembles the finding of Rimbert et al., where rest windows further from the upcoming cue gave higher accuracy, a gain Rimbert et al. attribute partly to the rebound in beta power after the previous movement [20]. Because swipes often come in bursts (Section 4.3), a longer gap may likewise move SKIP windows toward the previous swipe. Optimal window length has also been shown to be subject-specific rather than fixed [37]. While all these results strongly suggest better separability with increased overall data-availability (bigger windows, and areas) and simultaneously better separability the further we move away from the swipe, interpreting these relationships and their implications for optimal parameter selection remains beyond the scope of this thesis as it requires separate analysis, and is therefore a promising direction for future work. To maintain focus and scope, this analysis proceeds with the locked, default parameters:

- Window size (ws) = 1 second
- Gap size (gs) = 2 seconds
- Imminent-skip period (isp) = 3 seconds

The sweep was run after these parameters were set and changed none of them; it measures what the fixed choice costs. Its standing relative to the main results is noted at the start of Chapter 4.

3.5 Universal Recording Pipeline

To facilitate rigorous and reproducible data collection, a robust Python-based recording architecture (https://github.com/coffeeandcoffee/eeg_tiktok_pipeline/data_collection/recording_script.py) was developed. This script orchestrates the simultaneous capture of raw EEG telemetry from the Muse S headband and has the capability of recording synchronized behavioral video from a webcam which was not used in this experiment, yet may be of value to future investigations. Its autonomous capabilities ensure that the experimental protocol can be executed reliably without extensive technical oversight.

For this study’s recordings the command `python recording_script.py --nocamera --duration` was used to call the script. The two flags simply mean (i) bypassing camera recording, as it is not in the scope of this study yet kept for future investigations and (ii) the recording duration is $1800\text{s} = 30$ minutes. The pipeline’s core features include:

- **Automated Hardware Discovery:** Upon launch, the script automatically scans the local Bluetooth environment for active Muse devices and enumerates all connected camera peripherals, allowing the researcher to select the optimal hardware interactively without requiring hardcoded MAC addresses or device indices.
- **Precision Synchronization and Event Logging:** A dedicated, threaded keypress listener (`pynput`) monitors the A and B keys used by participants to mark swipe events and baseline respectively. Crucially, the script computes a dynamic time offset between the system clock and the LSL stream clock. Keypresses are retroactively aligned to this LSL timeline within a defined tolerance window and embedded directly into the EEG data rows as binary flags, ensuring absolute temporal synchrony between cognitive events and physical interactions.
- **Simultaneous Multimodal Capture:** The script utilizes the `pylsl` library to interface with the Lab Streaming Layer (LSL), pulling raw EEG data from the Muse headband at 256 Hz. Unless deactivated with the flag, it concurrently employs OpenCV to capture and encode synchronized webcam video using the H.264 (avc1) codec, providing visual ground truth of participant behavior.
- **Resilient Auto-Reconnection:** To combat the inherently unstable nature of Bluetooth Low Energy (BLE) connections, the script features an active connection-loss detection loop. If the LSL stream halts (no samples received within a timeout period), the script automatically terminates the hanging subprocess, rescans the Bluetooth environment, restarts the stream, and initiates a sequential data fragment (e.g., `_2.csv`, `_3.csv`). This guarantees that transient connection drops do not corrupt the entire recording session.
- **Concurrent Multi-Threading:** To ensure that blocking operations (such as writing high-resolution video frames to disk) do not disrupt the high-frequency polling required for EEG telemetry, the script isolates the LSL pulling loop, the OpenCV capture loop, and the keypress listener into independent Python daemon threads.

This open-source compatible architecture ensures that any independent researcher equipped with a standard webcam and a consumer-grade Muse S headband can fully replicate the data collection methodology utilized in this thesis.

3.6 Universal Pre-Processing Pipeline

The following subsections describe step-by-step exactly what the Python code in the https://github.com/coffeeandcoffee/eeg_tiktok_pipeline directory executes to process the raw EEG signals into a uniform shape suitable for analysis.

3.6.1 Data Synchronization and Interpolation

The preprocessing pipeline began with the chronological alignment and concatenation of all recorded CSV files for each participant, ensuring a continuous temporal sequence (executed via https://github.com/coffeeandcoffee/eeg_tiktok_pipeline/data_analysis/step01_preprocess.py). The raw data streams collected from the LSL (Lab Streaming Layer) exhibited slight variations in intrinsic sampling rates. To establish a uniform temporal grid suitable for across-session and across-participant analysis, the true sampling rate was derived by calculating the median time difference (Δt) between consecutive timestamps. Subsequently, a linear interpolation algorithm was applied to upsample and regularize the data to a constant sampling frequency of exactly $f_s = 256$ Hz. This process guarantees that any subsequent spectral analysis relies on a strictly uniform time axis, avoiding mathematical artifacts stemming from temporal jitter. Figure 3.11 confirms that all participants' signals operate at exactly 256 Hz after interpolation.

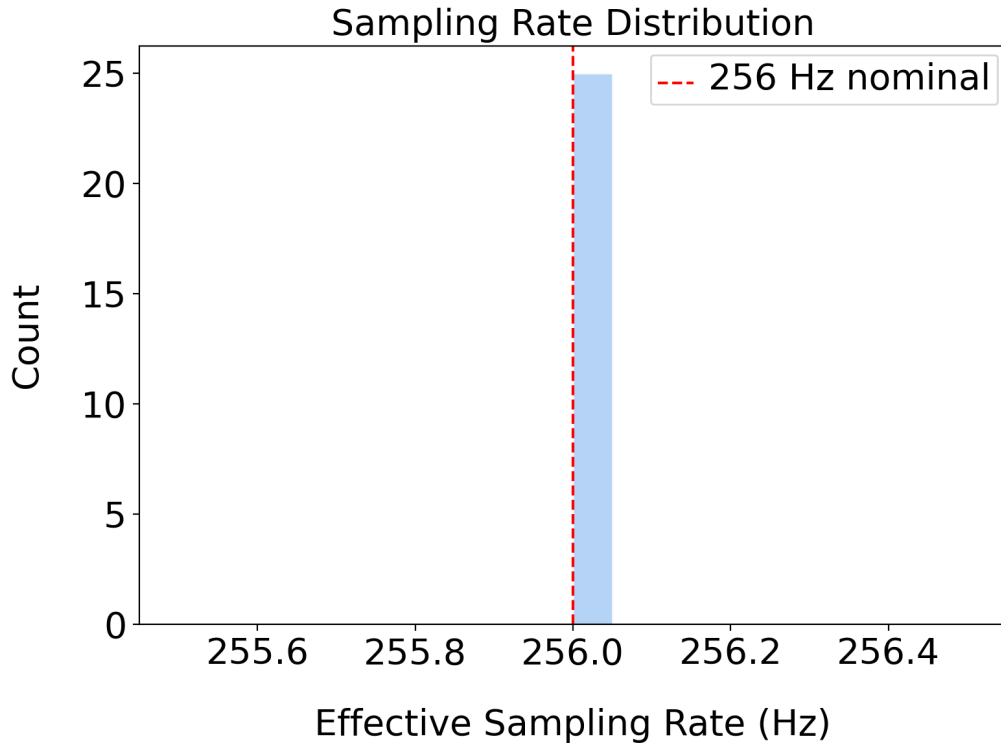


Figure 3.11: Sampling Rate Distribution across participants, demonstrating that after interpolation all signals operate precisely at 256 Hz, which is a mathematical prerequisite for the correct extraction of spectral band power.

During this temporal alignment process, Bluetooth transmission dropouts were identified by locating unnatural temporal gaps between consecutive CSV boundaries. To ensure data integrity, the duration of these dropouts was logged, and any potential data windows traversing such boundaries were strictly excluded in later extraction phases. Figure 3.12 shows the distribution of these events across the experimental timeline.

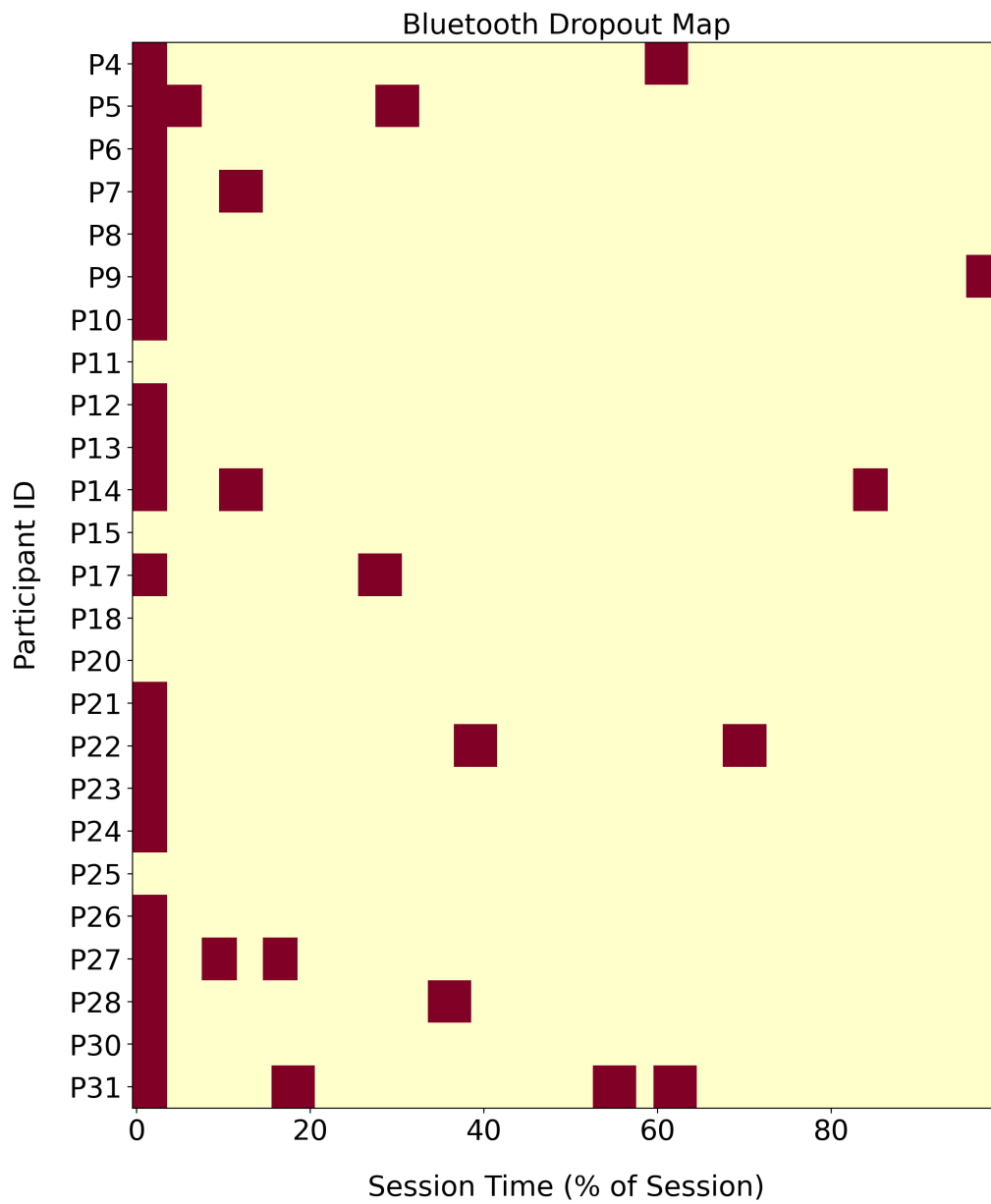


Figure 3.12: Bluetooth Dropout Heatmap indicating signal loss events across the experimental timeline. Data windows overlapping with these dropouts were removed from the analysis pipeline.

3.6.2 Baseline Extraction and Normalization

Following temporal synchronization, a baseline resting state was extracted for each participant to account for individual physiological differences, such as varying skull thickness or inherent resting cortical activity. The baseline period was rigidly defined as a 90-second continuous window commencing 10 seconds after the participant’s initial B keypress (executed via https://github.com/coffeeandcoffee/eeg_tiktok_pipeline/data_analysis/step01_preprocess.py).

The interpolated signal was then separated into seven standard EEG frequency bands using a 4th-order Butterworth bandpass filter: Delta (1–4 Hz), Theta (4–8 Hz), Alpha (8–13 Hz), Beta (13–30 Hz), Low Gamma (30–40 Hz), High Gamma (40–60 Hz), and Very High Gamma (60–100 Hz). The undecomposed broadband signal was retained alongside these bands and is referred to throughout as the *raw* signal. The instantaneous amplitude envelope A for each band was then obtained via the Hilbert transform. The linear envelope is used directly; it is not squared to power.

To normalize the extracted features across participants, a z-score normalization was applied against the participant-specific baseline. For every temporal sample, the normalized band amplitude z was computed as:

$$z = \frac{A - \mu_{base}}{\sigma_{base}} \quad (3.2)$$

where μ_{base} and σ_{base} denote the mean and standard deviation of the band amplitude envelope derived exclusively from the aforementioned 90-second baseline period. This normalization ensures that all subsequent statistical evaluations and machine learning models operate on normalized deviations from a participant’s own resting state, effectively mitigating absolute inter-subject voltage variance.

3.6.3 Windowing and Class Labeling

Following normalization, the continuous data streams were segmented and labeled into discrete **STAY** and **SKIP** classes based on the participant’s interaction events (A keypresses) as previously established in Sections 3.4.1 and 3.4.2. To reduce motor artifact contamination arising from the physical swipe action, a **2.0-second gap** was enforced immediately preceding the A keypress, and a **0.5-second gap** was enforced immediately following it. No data was extracted from these gap periods.

The imminent-skip period was rigorously defined as the **3.0-second interval** immediately preceding the aforementioned 2.0-second pre-swipe gap,

and all SKIP windows were drawn from within it (executed via https://github.com/coffeeandcoffee/eeg_tiktok_pipeline/data_analysis/step03_label_windows.py). This temporal framing is intended to isolate the cognitive decision-making phase leading up to the swipe from the physical execution of the swipe itself; the 2.0-second gap exceeds both the typical reported onset of the readiness potential (about 1 to 1.5 s before movement [58]) and the directly measured onset of arm muscular activation (263 ± 40 ms before movement) reported in a self-paced reaching task [40], though full elimination of motor-preparatory contamination cannot be guaranteed given individual variability in onset timing. The STAY class was defined as all remaining valid temporal periods outside of the SKIP windows and safety gaps.

Finally, discrete 1.0-second analysis windows were extracted using a sliding window with no overlap (stride = window length). No two windows overlap, and a window is kept only if it lies entirely within one STAY or SKIP region, so no window crosses a class boundary or enters a safety gap, thereby preventing data leakage and artificially inflated model performance, consistent with the practice of excluding windows that do not lie entirely within a single labeled interval [19].

3.6.4 Artifact Analysis

To ensure the integrity of the analyzed EEG signals, an automated outlier detection approach was implemented to flag potential artifacts (executed via https://github.com/coffeeandcoffee/eeg_tiktok_pipeline/data_analysis/step02_artifact_flag.py). The preprocessing pipeline was executed twice, once without and once with a 50 Hz notch filter. The artifact-detection procedure described in this section relies on peak-to-peak amplitude thresholding, a coarse, broadband measure targeting large-amplitude deflections from blinks and muscle activity; for this purpose only, the notch-filtered variant was used so that a constant, non-physiological powerline-noise component would not itself inflate the amplitude threshold. All analyses reported elsewhere in this thesis, including the frequency-band and gamma-range results in Chapter 4, instead use the un-notched data: because the 50 Hz powerline-noise sits inside the conventional high-gamma range (40–60 Hz), notch filtering is not a neutral operation, and this thesis prioritizes preserving the full, unaltered frequency content over pre-emptively discarding a band on the assumption that its content near 50 Hz is purely artifactual. Specifically, the 95th percentile of the peak-to-peak amplitude computed on the notch-filtered signal was utilized as a dynamic, participant-specific threshold. Frontal channels (AF7, AF8) were evaluated for ocular artifacts (e.g., blinks), while temporal channels (TP9, TP10) were evaluated for electromyography.

genic (EMG) artifacts, consistent with prior evidence that EMG is strongest at electrodes near head muscles [48, 50]. The frontal channels lie near the forehead muscle and are exposed to EMG as well (Section 2.11). Near a tensed muscle, EMG reaches every EEG band, delta included [48], so EMG is treated here as a risk for all bands, and not only for high-gamma or broadband features. Absolute voltage thresholds (e.g., $150\mu\text{V}$) were initially investigated but were ultimately discarded as they did not yield generalizable or useful results across the dataset.

No data points were excluded from the final machine learning pipeline based on these artifact flags. Incorporating such filtering would introduce additional experimental conditions and substantially expand the analysis beyond the scope of this proof-of-concept study. Instead, the approach is documented to highlight awareness of artifact-related confounds and to provide a foundation for more advanced rejection strategies in future work. The resulting artifact-rate distribution is shown in Figure 3.13.

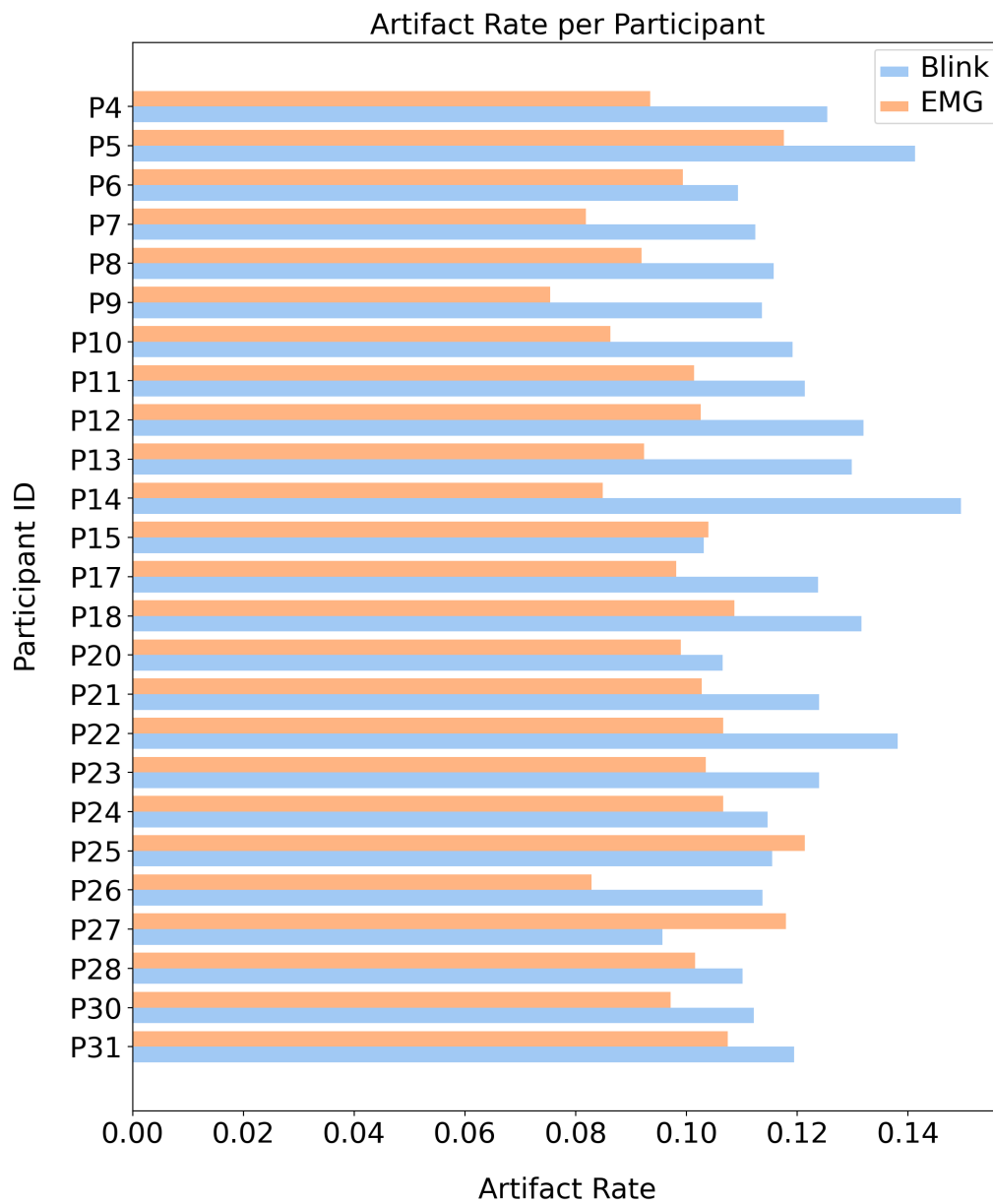


Figure 3.13: Artifact rate distribution and signal quality flagging based on the 95th percentile peak-to-peak outlier approach.

3.6.5 Behavioral Burst Analysis

In addition to physiological artifacts, behavioral anomalies termed “bursts” were identified. A burst is defined as a sequence of rapidly successive skip events (e.g., multiple A keypresses merging into a single contiguous temporal block), reflecting frantic swiping rather than deliberate evaluation. Flagging these events aims to ensure that the **SKIP** class captures deliberate pre-swipe evaluation periods rather than high-momentum, rapid-succession scrolling behavior.

Although bursts were successfully detected, they were not excluded from the final analysis. Figure 3.14 visualizes the detection mechanism via the distribution of inter-swipe intervals (i.e., dwell time) across all participants. The vertical threshold marks the 6.0-second merge boundary (twice the 3.0-second imminent-skip period). Consecutive swipes below this threshold cause their respective **SKIP** windows to overlap and merge into a single contiguous block, thereby defining a burst. The shaded region highlights the frequency of such rapid, consecutive events.

It is important to note that the imminent-skip period is defined independently of the dwell time distribution; the 3.0-second value is an intuitive choice. As such, its influence on the resulting burst structure warrants further investigation. Incorporating burst rejection would introduce additional analytical complexity and is therefore beyond the scope of this proof-of-concept study. Instead, its inclusion highlights a relevant behavioral confound and motivates future work.

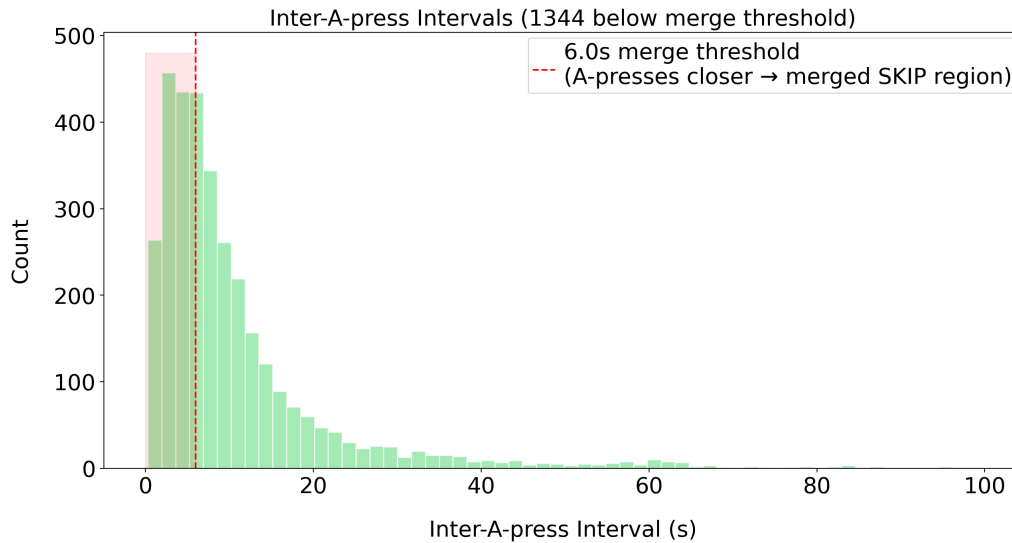


Figure 3.14: Distribution of inter-swipe intervals (view times of each video across all participants, or dwell time). The shaded region highlights intervals below the 6.0-second merge threshold, where consecutive **SKIP** windows overlap to form behavioral burst blocks.

3.7 Feature Engineering

3.7.1 Band Power and Statistical Moments

Following window extraction, a comprehensive feature set was computed for each valid temporal window (executed via https://github.com/coffeeandcoffee/eeg_tiktok_pipeline/data_analysis/step05_feature_engineering.py). Two distinct extraction schemes are applied. For each of the seven frequency bands at each of the four electrodes, the mean and standard deviation of the normalized envelope are computed, alongside the relative power, Hjorth, and peak/macro frequency features described in the following subsections. Minimum and maximum were computed during development but are excluded from the final feature set. The undecomposed raw broadband signal is treated separately and is described at the end of this section. An example of the extracted statistics for one 1-second window of one frequency band (alpha) at one electrode of one participant is depicted in Figure 3.15.

In addition to these four moments, a Relative Power feature was computed for each band and electrode as the mean power of that band divided by the summed mean power across all bands at the same electrode. This expresses

each band’s contribution as a proportion of the total spectral power measured at that electrode and is therefore invariant to overall amplitude scaling.

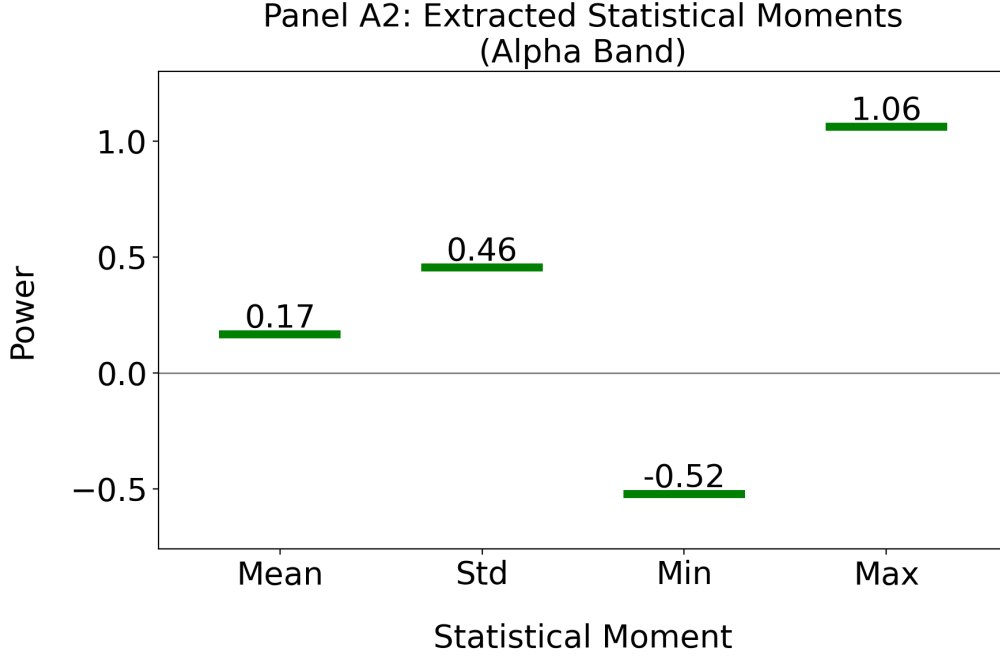


Figure 3.15: Extraction of standard statistical moments from the normalized EEG frequency bands.

3.7.2 Engagement Index Calculation

To provide an established reference measure from human-factors neuroscience, the Engagement Index (EI) was calculated for each analysis window. Pope et al. selected the EI from candidate EEG indices of attention and vigilance, using band power at central and parietal sites, to decide when a computer task modelled on flight-deck work switches between manual and automated operation [5]. The EI is mathematically defined as:

$$EI = \frac{\beta}{\alpha + \theta} \quad (3.3)$$

In this pipeline, exactly one scalar Engagement Index (EI) is computed per 1.0-second analysis window per participant. To calculate this single value, the within-window variance of the baseline-normalized band signal is first extracted for the Beta (β), Alpha (α), and Theta (θ) bands across each

of the four individual electrodes. Because the normalized signal is z-scored and therefore centred near zero, the ratio is formed from variances rather than means; the variance is strictly non-negative, so the denominator cannot approach zero and the numerical instability of a mean-based ratio is avoided. These four spatial variances are then averaged together to produce a single, whole-head β , α , and θ scalar for that specific window, which are finally fed into the EI ratio formula. This single composite scalar serves as the primary clinical benchmark against which the predictive power of individual statistical features is evaluated in the machine learning phase. An example calculation of the EI for a single sample window of one participant can be observed in Figure 3.16, visually showing the three spatially-averaged scalars that are utilized in the Index calculation in Equation 3.3.

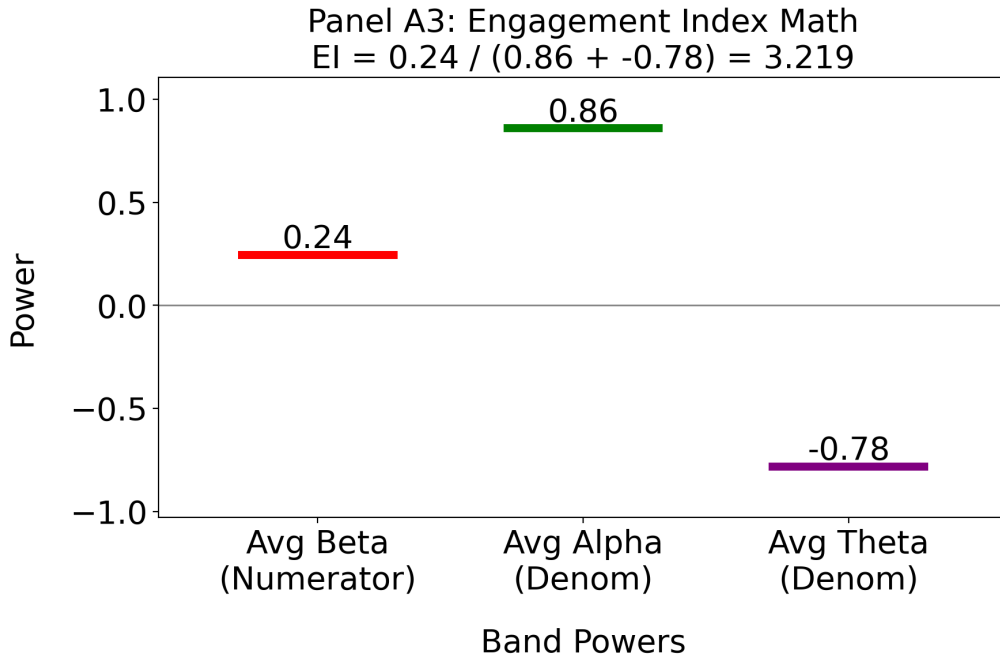


Figure 3.16: Engagement Index (EI) calculation deriving a single scalar engagement metric from the Beta, Alpha, and Theta band powers.

3.7.3 Hjorth Parameters

In addition to basic descriptive statistics, Hjorth Parameters (Activity, Mobility, and Complexity) were computed to capture the sequence-dependent temporal dynamics of the signal within the 1.0-second window. While stan-

standard statistical moments characterize the amplitude distribution independently of their temporal order (i.e., they are time-invariant), Hjorth parameters depend explicitly on the time-domain derivatives, thus capturing spectral properties, such as mean frequency and bandwidth, directly from the time-domain signal.

For a discrete EEG signal vector $y(t)$ of length $N = 256$ representing one frequency band at one channel within a window, the three parameters are mathematically defined as follows:

Activity

Activity measures the total signal power and is defined simply as the variance of the signal amplitude:

$$\text{Activity} = \sigma_y^2 = \frac{1}{N} \sum_{t=1}^N (y(t) - \mu_y)^2 \quad (3.4)$$

where μ_y is the mean of $y(t)$. The input dimension is the normalized amplitude and the output dimension represents the squared amplitude.

Mobility

Mobility serves as a time-domain proxy for the mean frequency of the power spectrum. It is defined as the square root of the ratio of the variance of the first derivative $y'(t)$ to the variance of the primary signal $y(t)$:

$$\text{Mobility} = \frac{\sigma_{y'}}{\sigma_y} = \sqrt{\frac{\text{var}(y'(t))}{\text{var}(y(t))}} \quad (3.5)$$

Since the differentiation of a signal with respect to time yields a rate of change, the resulting output dimension of Mobility is expressed in Hertz (Hz).

Complexity

Complexity measures the bandwidth of the signal and indicates how closely the signal's shape resembles a pure sine wave (where a pure sine wave converges to a complexity of 1). It is defined as the ratio of the Mobility of the first derivative $y'(t)$ to the Mobility of the primary signal $y(t)$:

$$\text{Complexity} = \frac{\text{Mobility}(y')}{\text{Mobility}(y)} = \frac{\sigma_{y''}/\sigma_{y'}}{\sigma_{y'}/\sigma_y} \quad (3.6)$$

where $\sigma_{y''}$ is the standard deviation of the second derivative of the signal. Because it is a ratio of two frequency-domain (Hz) equivalents, Complexity is a dimensionless scalar.

Together, these three parameters output exactly three scalar features per frequency band per channel, expanding the univariate feature space by explicitly encoding the temporal structural variations that basic amplitude statistics fundamentally miss.

3.7.4 Peak and Macro Frequency Extraction

Beyond traditional amplitude statistics, two specialized time-domain features were engineered to capture the dominant oscillatory and structural patterns within each frequency band of the 1.0-second analysis window. Both features are calculated on $y(t)$, the baseline-normalized amplitude envelope z of one band (Equation 3.2), of length $N = 256$ and total duration $T = 1.0$ s, as follows:

Peak Frequency

Rather than extracting the maximum bin from a Fourier transform, Peak Frequency is calculated directly in the time domain by identifying the total number of local maxima (peaks) within the signal. A point $y(t)$ is considered a peak if it is strictly greater than its immediate neighbors. Let P be the set of all peak indices. The Peak Frequency f_{peak} is defined as the absolute peak count normalized by the window duration:

$$f_{peak} = \frac{|P|}{T} \quad (3.7)$$

The input is the amplitude envelope of one band, and the output is the count of local maxima of the envelope per second, in Hertz (Hz). Because $y(t)$ is the envelope, and not the band-pass filtered oscillation itself, Peak Frequency counts how often per second the strength of the band peaks, which need not equal the frequency of the oscillation within the band. The z-score normalization shifts and scales the envelope, and leaves the count of local maxima unchanged. Section 2.10 places Peak Frequency among earlier works that estimate frequency from the signal over time; to the author's knowledge, no study counts the local maxima of the strength of an EEG band, so Peak Frequency is a feature the present thesis defines without prior validation.

Macro Frequency

While Peak Frequency captures every local maximum, it is highly sensitive to noise and high-frequency jitter [47]. To isolate the true, underlying macroscopic oscillatory bursts, a novel Macro Frequency algorithm was developed.

First, a dynamic amplitude threshold θ_{peak} is calculated as the mean amplitude of all previously identified peaks:

$$\theta_{peak} = \frac{1}{|P|} \sum_{p \in P} y(p) \quad (3.8)$$

The signal is then converted into a binary sequence $B(t)$ where $B(t) = 1$ if $y(t) \geq \theta_{peak}$ and 0 otherwise.

Because a single macroscopic wave might contain slight amplitude dips that prematurely break a contiguous block of ones, the algorithm calculates the average length of all zero-sequences (gaps), denoted as $\bar{G}$. Any gap in $B(t)$ with a length $g \leq \bar{G}$ is mathematically bridged (converted to ones), producing a smoothed binary sequence $B_{smooth}(t)$.

Finally, the Macro Frequency f_{macro} is defined as the total number of distinct, contiguous blocks of ones in $B_{smooth}(t)$, normalized by the window duration:

$$f_{macro} = \frac{\text{BlockCount}(B_{smooth})}{T} \quad (3.9)$$

The output is a stabilized, macroscopic frequency in Hertz (Hz) that robustly captures the count of dominant structural waves within the cognitive window, effectively ignoring micro-fluctuations.

3.7.5 Raw Broadband Features

The undecomposed raw broadband signal is not passed through the band-specific feature set described above; instead, a separate set of time-domain descriptors is extracted directly from the raw signal. Treating the raw signal as a distinct feature class is motivated by evidence that broadband/aperiodic activity can carry decodable information independent of, and sometimes exceeding, band-limited oscillatory features [61, 62], and that band-power ratio measures can conflate periodic and aperiodic components [62]. Three time-domain descriptors are extracted per electrode: the mean, the standard deviation, and the linear slope of the signal across the window, the latter obtained as the gradient of a least-squares linear fit and expressed per second. Together these capture the amplitude level, the amplitude variability, and the direction and steepness of any sustained drift within the window.

3.7.6 Resulting Feature Space

Taken together, the two extraction schemes yield the final feature space. For each of the seven frequency bands at each of the four electrodes, eight statistics are computed, namely mean, standard deviation, relative power, the three Hjorth parameters, and the peak and macro frequencies, giving $7 \times 4 \times 8 = 224$ features. For the raw broadband signal, three time-domain statistics are extracted at each electrode, giving a further $1 \times 4 \times 3 = 12$ features. The complete feature space therefore comprises 236 univariate features per 1.0-second analysis window.

3.8 Result Synthesis Pipeline

3.8.1 Temporal Cross-Validation and Data Balancing

Before proceeding to statistical evaluation and modeling, the inherent class imbalance depicted in Figure 3.17, where the **STAY** class (blue) heavily dominates the **SKIP** class (red), had to be addressed (executed via https://github.com/coffeeandcoffee/eeg_tiktok_pipeline/data_analysis/step04_balance_split.py). To prevent the models from trivially predicting the majority class, the **STAY** class was randomly undersampled for each participant to achieve a strict 50/50 balance, per participant, as depicted in Figure 3.18.

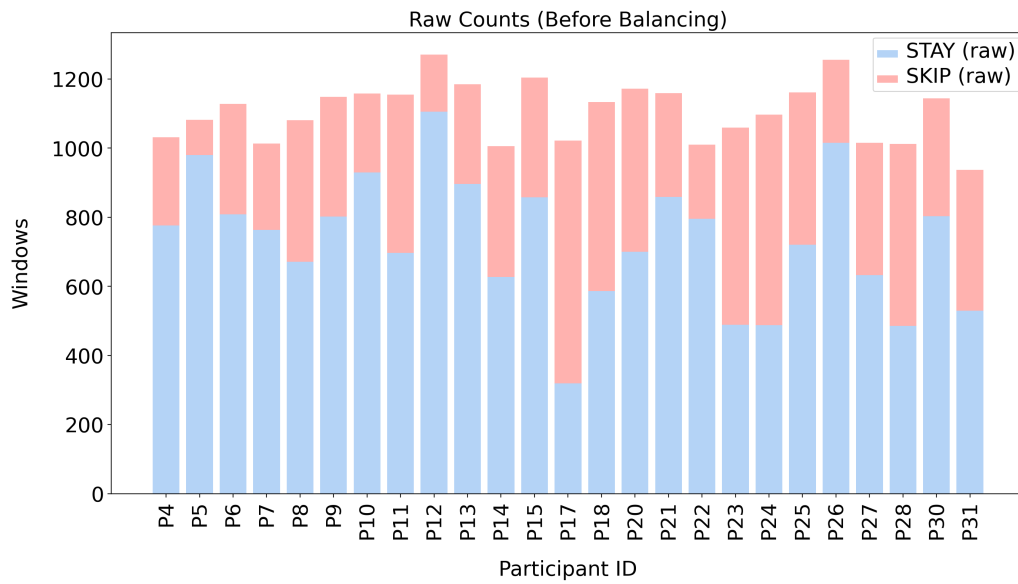


Figure 3.17: Raw window counts before balancing showing the extreme STAY dominance.

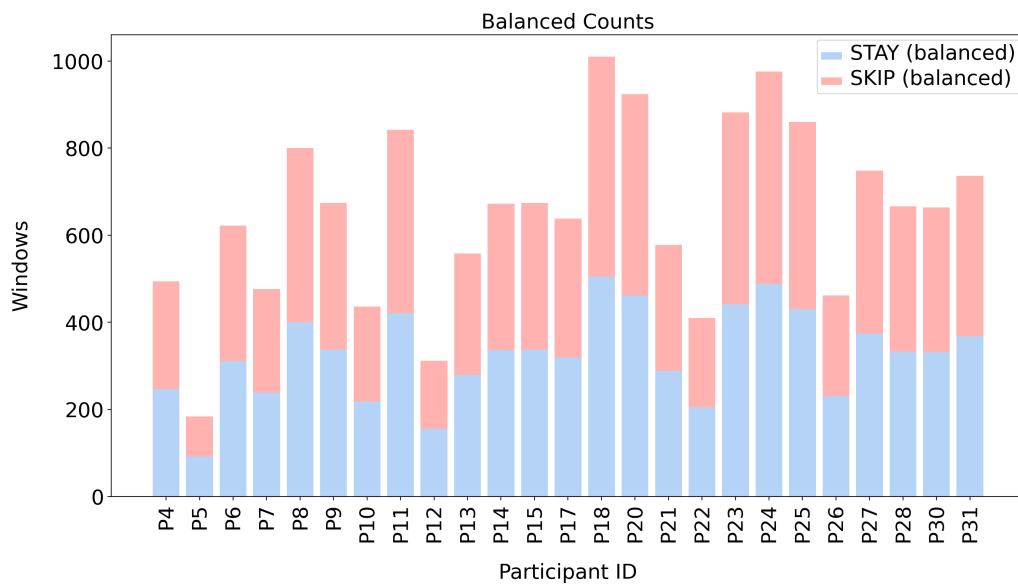


Figure 3.18: Window counts after undersampling, achieving a strict 50/50 ratio per participant.

Two cross-validation schemes are used throughout. The intra-subject setting applies blocked temporal 5-fold cross-validation within each participant; the inter-subject setting applies Leave-One-Group-Out cross-validation across participants, training on 24 participants and evaluating on the held-out 25th. To mitigate temporal leakage due to autocorrelation between adjacent EEG windows in the intra-subject setting, blocked temporal 5-fold cross-validation was applied. Each participant’s data were partitioned chronologically into five folds with equal counts of minority-class (**SKIP**) instances. Figure 3.19 shows minority-class occurrences (red) and fold boundaries (black), defined by equal splits of these events, per participant.

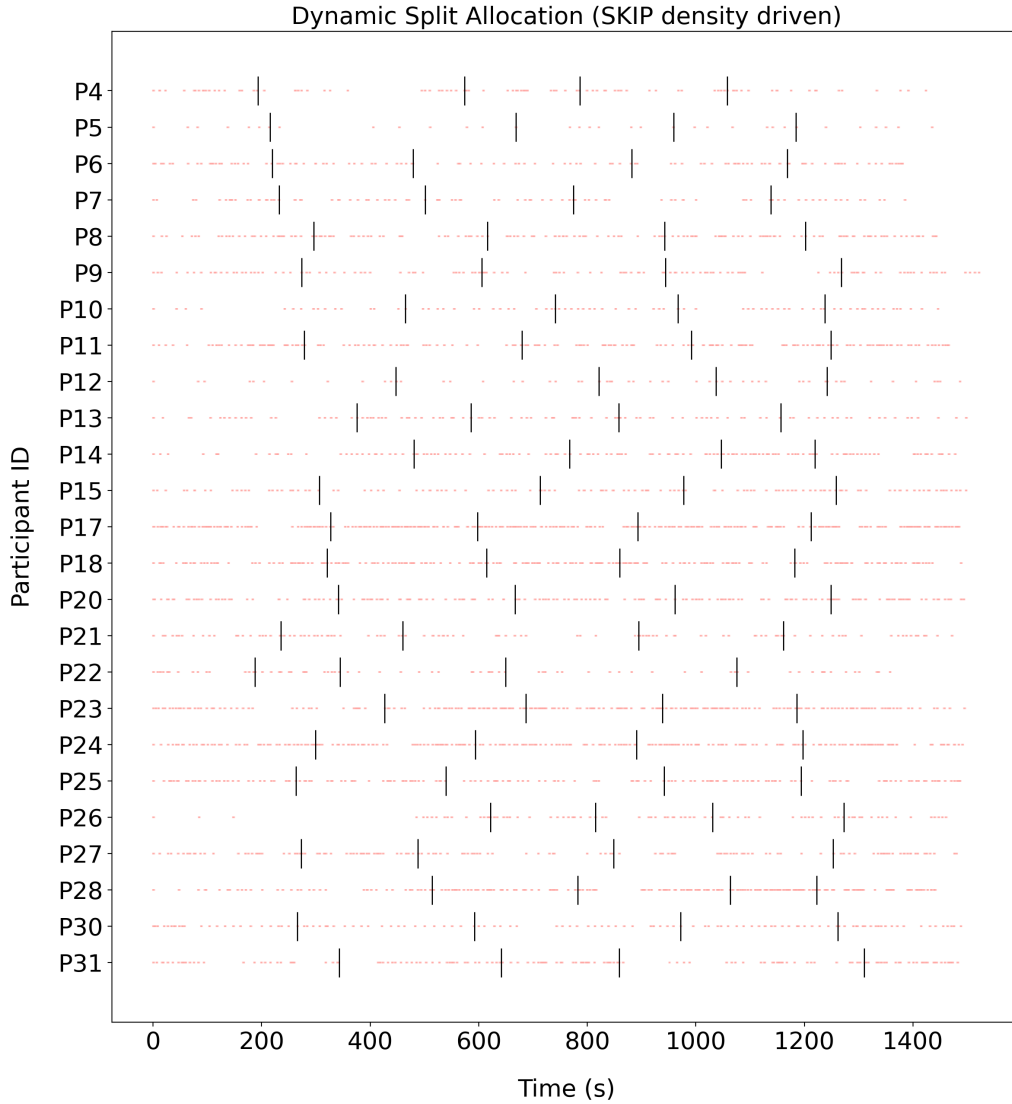


Figure 3.19: Each participant’s data is temporally split into 5 folds containing equal amounts of the minority SKIP class.

To rigorously prevent temporal leakage, a minimum separation of 3 seconds between all training and testing windows was strictly enforced at every fold boundary. This constraint was implemented as a hard criterion: any violation automatically triggered pipeline termination, ensuring that no autocorrelated samples could enter the training or test sets. As verified in Figure 3.20, all fold boundary gaps (red points along the x-axis) exceed the predefined 3-second threshold (red dashed line), confirming compliance

across the dataset. This procedure produces valid minority-balanced folds while preserving strict independence of the test data; an example for a single participant is shown in Figure 3.21, where the separation gaps between folds (colored blocks) are clearly visible.

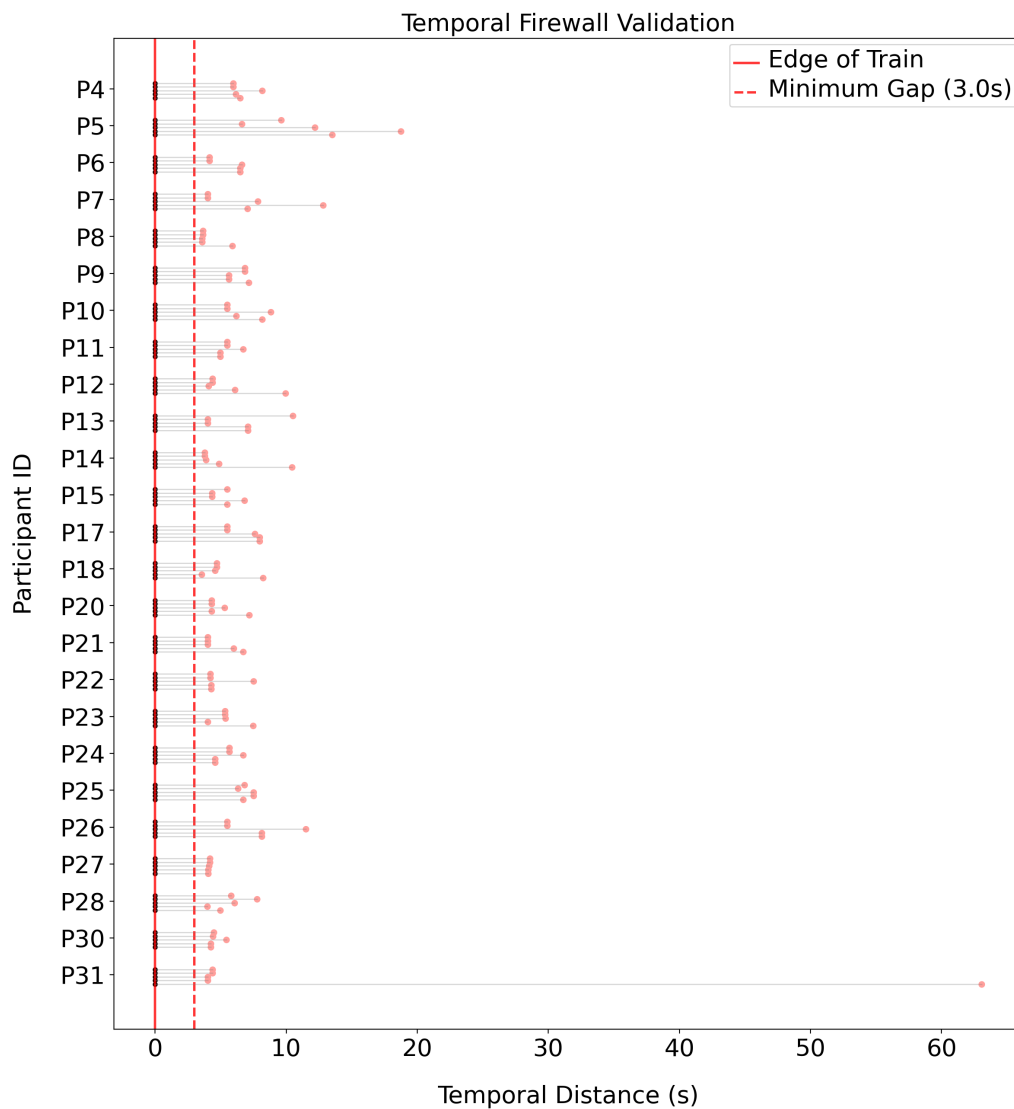


Figure 3.20: Temporal minimum required safety distance of 3 seconds between train and test windows is strictly fulfilled.

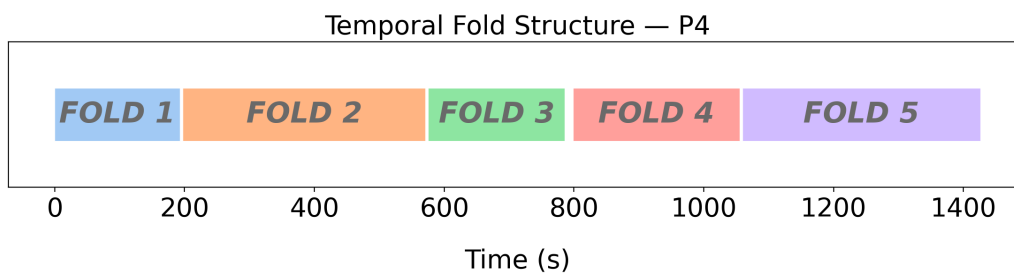


Figure 3.21: Temporal cross-validation split visualization, detailing the visible gaps between folds that successfully prevent data leakage.

3.8.2 Feature Separation Analysis

To mathematically evaluate the discriminative power of individual features prior to modeling, Cohen’s d was computed to measure the standardized effect size between the **STAY** and **SKIP** distributions for each engineered feature (executed via https://github.com/coffeeandcoffee/eeg_tiktok_pipeline/data_analysis/viz17_feature_ranking.py). Cohen’s d is computed as defined in Equation 3.1. As a standardized effect size, Cohen’s d provides a scale-invariant measure of class separation that is independent of the absolute units of the feature (e.g., microvolts versus power spectral density). It quantifies how many standard deviations separate the centers of the two classes, thereby capturing theoretical separability prior to any model training.

By calculating the absolute effect size $|d|$, the pipeline systematically ranked all extracted features (e.g., `TP10_raw_std`) according to their individual ability to separate the **STAY** and **SKIP** classes. Aggregated plots grouped by frequency band, electrode, or statistic report the arithmetic mean of $|d|$ across all constituent features in that category and are shown without confidence intervals; the band $\times$ electrode profile is shown as a point plot with bootstrapped 95% confidence intervals.

Confidence intervals are reported only for this band-electrode profile, where each point represents a single well-defined feature combination; category-aggregated means (by band, electrode, or statistic alone) intentionally omit confidence intervals, as they average over heterogeneous and unequally-sized feature sets whose spread reflects between-feature heterogeneity rather than sampling uncertainty of a single estimand. Ranked individual-feature ($|d|$) values (Table 4.1) are likewise reported as a descriptive screening criterion for candidate selection; formal inferential claims of superiority over baseline models are instead established via the Wilcoxon signed-rank tests in Section 3.8.4.

Feature-level rankings report the exact $|d|$ of a single named feature. Because the raw broadband signal contributes three statistics per electrode whereas each frequency band contributes eight, category means rest on unequally sized feature sets, and comparisons between them are indicative rather than exact. Features demonstrating the highest absolute $|d|$ values were designated as Top Features and isolated for independent predictive baseline modeling.

3.8.3 Logistic Regression Modeling and Baselines

To benchmark the true predictive capability of the engineered EEG features against chance and the Engagement Index baseline, three distinct model-

ing approaches were evaluated on the temporally isolated test sets (executed via https://github.com/coffeeandcoffee/eeg_tiktok_pipeline/data_analysis/step07_train_intra.py and synthesized in https://github.com/coffeeandcoffee/eeg_tiktok_pipeline/data_analysis/step16_parallel_universes.py).

1. Coinflip Baseline: A uniform dummy classifier (`DummyClassifier(strategy='uniform')`) was utilized to establish an empirical chance baseline. Given the rigorous 50/50 balancing of the datasets, this approach reliably generated predictions hovering around the 50% chance level, verifying that no underlying structural bias existed within the cross-validation folds.

2. Engagement Index (EI) Baseline: The pre-calculated EI scalar (derived from $\beta/(\alpha + \theta)$) was scaled utilizing a standard scaler (subtracting the mean and scaling to unit variance) and fed as a singular feature into a Logistic Regression model (`LogisticRegression(class_weight='balanced')`). This established a clinical baseline, indicating how well classical engagement theory alone could predict the behavioral swipe decision in this specific short-form video context.

3. Top Feature Logistic Regression: To test the predictive power of individual statistical features independently, the features ranked highest via Cohen’s d (e.g., specific band powers or statistical moments) were similarly scaled and evaluated using isolated Logistic Regression models. This isolated approach allows for direct comparison of individual feature efficacy without the confounding complexities of high-dimensional multivariate models.

To ensure a strict one-to-one comparison with the one-dimensional Engagement Index baseline, each macro “Top Parameter” (band, electrode, statistic) was represented by its strongest single constituent feature.

3.8.4 Metrics and Significance Testing

To comprehensively evaluate model performance, standard classification metrics were computed for each cross-validation fold: Accuracy, Precision, Recall, and the F_1 -score. In this context, correct prediction of the **STAY** class reflects successful detection of continuous non-skip viewing, whereas correct prediction of the **SKIP** class corresponds to identifying the period preceding a swipe-away decision within a 2–5 second pre-action window.

The F_1 -score was used as the primary performance metric, as it balances Precision and Recall and is robust to class imbalance. Metrics were computed separately for each class by treating the respective class as the positive label

(i.e., one-vs-rest evaluation for **SKIP** and **STAY**).

Metric definitions: Let TP, FP, TN, and FN denote true positives, false positives, true negatives, and false negatives, respectively. To achieve optimal classification performance, the correct predictions (TP and TN) must be maximized while the incorrect predictions (FP and FN) are minimized.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

$$\text{Precision} = \frac{TP}{TP + FP}$$

$$\text{Recall} = \frac{TP}{TP + FN}$$

$$F_1 = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$

Case 1: SKIP as the positive class

- **TP:** correctly predicted **SKIP** instances
- **TN:** correctly predicted **STAY** instances
- **FP:** **STAY** instances incorrectly predicted as **SKIP**
- **FN:** **SKIP** instances incorrectly predicted as **STAY**

Case 2: STAY as the positive class

- **TP:** correctly predicted **STAY** instances
- **TN:** correctly predicted **SKIP** instances
- **FP:** **SKIP** instances incorrectly predicted as **STAY**
- **FN:** **STAY** instances incorrectly predicted as **SKIP**

A class-specific F_1 -score substantially exceeding chance level indicates that the model captures discriminative structure beyond random guessing for that class. For reference, under balanced random predictions, the expected F_1 -score is approximately 0.5, whereas under strong class imbalance, this baseline decreases and should be interpreted relative to the class distribution.

To rigorously evaluate whether the Logistic Regression models utilizing individual EEG features outperformed the established baselines across the participant cohort, non-parametric statistical testing was employed. Specifically, the Wilcoxon signed-rank test was used to compute the statistical significance (p -value) of the performance differences between paired model distributions (e.g., Top Feature model vs. EI Baseline model, or EI Baseline model vs. 50% chance level) at the participant level. For each participant, the performance metric (e.g., F_1 -score) of the evaluated model was contrasted with that of the baseline, yielding 25 paired differences, one per participant.

The algorithm employs a two-sided Wilcoxon test with a standard significance threshold ($\alpha = 0.05$). Because a two-sided test detects significant differences in either direction, a directional validation check is applied post hoc: a model is deemed statistically significantly superior only if two conditions are met simultaneously, namely (i) the test yields $p < 0.05$, and (ii) the arithmetic mean of the model’s scores across all participants is strictly greater than the arithmetic mean of the corresponding baseline. The rank-biserial correlation (r) is also computed to quantify the effect size of the Wilcoxon test, indicating the strength of the model’s superiority over the compared baseline. This structured chain of statistical proof robustly determines whether individual EEG features can provide a significant improvement over both pure chance and classical clinical engagement metrics when attempting to predict real-world short-form video interactions. Because these comparisons are repeated across a number of feature–baseline pairs, metrics, and evaluation settings, the reported p -values are not corrected for multiple comparisons; individual comparisons are therefore best read as exploratory screening criteria rather than confirmatory hypothesis tests, with family-wise error control identified as a priority for future replication (see Future Research).

Chapter 4

Results and Discussion

This chapter presents the results of a multi-stage experimental investigation.

This thesis delivers *six core scientific contributions*, all reported in this chapter:

1. **Contribution #1: Primary EEG Experiment under Naturalistic Conditions**

The foundation of this work is the primary experiment (Methods Section 3.1 - 3.3, Results Section 4.1), in which participants were recorded via EEG while engaging in naturalistic TikTok browsing. This data collection phase constituted a substantial portion of the overall effort, requiring careful experimental design, technical implementation, and data acquisition under ecologically valid conditions. Not only the resulting dataset, but also the easily reproducible nature of the developed experimental method forms the basis for all subsequent analyses and should be understood as a central contribution of this thesis.

2. **Contribution #2: Reusable EEG Processing and Analysis Pipeline**

In addition to the experimental dataset and process, a further key contribution lies in the development of a comprehensive script library and analysis pipeline (Methods Sections 3.4–3.8, Results Section 4.2). This framework enables fully reproducible processing, from automated EEG recording on standard laptop hardware to feature extraction, class separation, and machine learning-based prediction. As such, it not only supports the present analysis but also provides a transferable methodological toolset that can be readily adapted to related experimental paradigms.

3. Contribution #3: Behavioral Swipe-Burst Phenomenon

We continue by focusing on the behavioral burst analysis (Methods Section 3.6.5, Results Section 4.3). As no burst rejection was implemented, due to scope constraints outlined in Section 3.6.5, this component is treated as a secondary result rather than the central analytical pathway.

4. Contribution #4: Evaluating the Engagement Index’s Predictive Capacity

The primary results emerge from a multi-layer analysis (Methods Sections 3.7–3.8, Results Section 4.4): Using the Cohen’s d method described in Section 3.8.2, first class separation capability of the conventional Engagement Index was evaluated, then specific features were identified that most effectively differentiate between the SKIP and STAY classes (Section 4.4.1) and which features outperform the Engagement Index’s class-separation capability of SKIP and STAY. Building on this, a set of machine learning models as described in Section 3.8.3 is applied to predict scrolling behavior based on these class distinctions using the Engagement Index with a Logistic Regression and comparing it to a random-chance baseline, as well as challenging the conventional Engagement-Index by using previously identified superior features with the same Logistic Regression approach and comparing it to the Engagement Index. Best-performing models and features are reported (Section 4.4.2).

As described in Section 3.8.4, these results are carefully introduced along their statistical significance, a crucial measure considering the sample size of $n=25$ participants, in order to give result metrics such as the F1-score the appropriate context for meaningful interpretation for discussion and further use. While this prediction task serves as an initial proof of concept, it already indicates which features may provide sufficient discriminatory power.

5. Contribution #5: Neuroscientific and Behavioral Interpretation

The results (Section 4.4.1) offer insights into the brain regions and frequency bands most predictive of this behavior, thereby informing both theoretical interpretation and potential practical applications. Therefore, in the final section (Section 4.5), these findings are interpreted in the context of neuroscience, psychology, and behavioral research, highlighting their implications and real-world relevance.

6. Contribution #6: A General Definition of Engagement

Finally, the results are used for a purpose beyond the present paradigm. The closing section of this chapter (Section 4.6) treats them as constraints on the engagement construct itself and states an updated, instrument-independent definition of engagement, together with the measurement requirements that follow from it and the limits of that definition. This contribution is conceptual rather than empirical; it does not alter the analyses presented here.

Findings that are reported in the Methods chapter

Five empirical findings appear in Chapter 3 rather than here. Because, Four of them describe the dataset rather than answer a research question. Each is reported next to the part of the apparatus it checks: the effective sampling rate across participants (Figure 3.11), the Bluetooth dropout events across the sessions (Figure 3.12), the blink and EMG artifact rates per participant (Figure 3.13), and the distribution of intervals between consecutive swipes (Figure 3.14). The distribution of intervals between consecutive swipes also carries behavioral meaning and return as a result in Section 4.3.

The fifth result is the parameter sensitivity analysis in Section 3.4.5 likewise reports a measured outcome (Figure 3.10), but belongs there because the labeling parameters were fixed beforehand from the literature (Section 3.4.1). The parameters were not informed by these results in Section 3.4.5.

4.1 Key Findings from the Experimental Paradigm

The experimental paradigm proved feasible in a semi-controlled but still naturalistic setting, and several practical lessons emerged from its execution. Recruitment via email outreach and student mailing lists worked well, especially when paired with a clear and modest incentive. Participants were generally receptive to the study, and most were academically oriented, which appeared to increase interest in the research question and reduce concerns about signing the consent and data-protection forms.

A particularly useful element was the use of iOS Guided Access during both the baseline and TikTok phases. It provided a simple way to constrain the task environment, prevent app switching, and automatically track the duration spent in each app, which matched the temporal structure of the design well. This was especially helpful for the 2-minute baseline and the 25-minute scrolling phase, where reliable timing was essential.

At the same time, the study also highlighted several practical challenges. One recurring difficulty was maintaining a strict and consistent script during participant instruction, especially when explaining what participants should do when the task started, paused, or ended. Another challenge was ensuring that the instructions around synchronization, keypress behavior, and task boundaries were communicated clearly and uniformly across sessions. These issues did not invalidate the experiment, but they do suggest that the protocol would benefit from a more standardized briefing script and a more formalized session flow in future replications.

4.2 The Reusable Methodological Pipeline

A major practical contribution of this thesis is the reusable analysis and recording pipeline, which can be accessed through the associated GitHub repository, first mentioned in Section 3.2 and further described in Section 3.5. In the present study, the pipeline ran mostly smoothly after multiple months of optimization, and finally enabled the full workflow from recording to preprocessing, feature extraction, and analysis. The main technical interruptions were occasional Bluetooth dropouts, which were handled by the recording architecture without causing major session failures, but which still underline the need for continued robustness improvements.

The development process also revealed an important software-engineering lesson: a substantial portion of the code base became obsolete during the iterative refinement across multiple months. Precisely 69.6% (representing a reduction from 37,189 to 11,311 total lines of code) of the scripts were temporary, diagnostic, or superseded by improved versions during development. While this is normal in an exploratory research pipeline, it also shows that the project would benefit from a more systematic cleanup strategy in future iterations. A lightweight script-archiving or pruning mechanism should therefore be introduced next time to preserve clarity, reduce maintenance burden, and make the final codebase easier to reuse and extend.

More broadly, the pipeline demonstrates that a relatively compact and open methodological framework can support consumer-grade EEG recording, synchronized behavioral logging, preprocessing, and downstream feature analysis in a reproducible way. That makes it not only useful for this thesis, but also transferable to other studies with similar naturalistic EEG and behavior-synchronization requirements.

4.3 Behavioral Swipe-Burst Phenomenon

As established in the methodology (Section 3.6.5), a burst is defined as a sequence of rapidly successive skip events reflecting frantic swiping. To quantify this burst severity, overlapping SKIP windows were grouped into “burst blocks.” The block size n denotes the number of consecutive rapid swipes before a participant pauses to engage. A block of $n = 1$ represents standard behavior, whereas larger values (e.g., $n = 3$) indicate repeated rapid skipping, possibly without meaningful engagement due to high frequency and action-state inertia.

Figure 4.1 shows this behavior at the participant level, revealing a clear dichotomy: some participants exhibit frequent large burst blocks (“fast scrollers”, e.g. P17), while others predominantly show $n = 1$ behavior (“careful scrollers”, e.g. P15). Figure 4.2 aggregates these patterns, confirming that while $n = 1$ dominates, a long tail of burst behavior persists.

Finally, Figure 4.3 illustrates the sensitivity to the chosen window size by repeating the analysis with a ± 5.0 -second window (10.0-second merge threshold). The wider threshold captures a larger portion of the dwell time distribution and merges many isolated swipes into larger burst blocks, substantially reducing the proportion of $n = 1$ events, simply by changing the definition of a “burst block”. Specifically, under the default ± 3.0 s window, 67.4% of swipe events are isolated singletons (participant-level mean $69.1\% \pm 16.2\%$), while widening to ± 5.0 s reduces this to 48.9% (participant-level mean $46.9\% \pm 20.8\%$). The large between-participant standard deviation is itself informative: burst tendency appears to be a strongly individual trait rather than a fixed population parameter, foreshadowing the precision-public-health argument developed in Section 4.5. Specifically, under the ± 3.0 s window paradigm, 67.4% of all swipe events are isolated $n = 1$ sequences (participant-level average: $69.1\% \pm 16.2\%$). In contrast, widening the window to ± 5.0 s reduces the overall proportion of isolated swipes to 48.9% (participant-level average: $46.9\% \pm 20.8\%$). This underscores the strong dependence on parameter selection and the need for systematic optimization in future work.

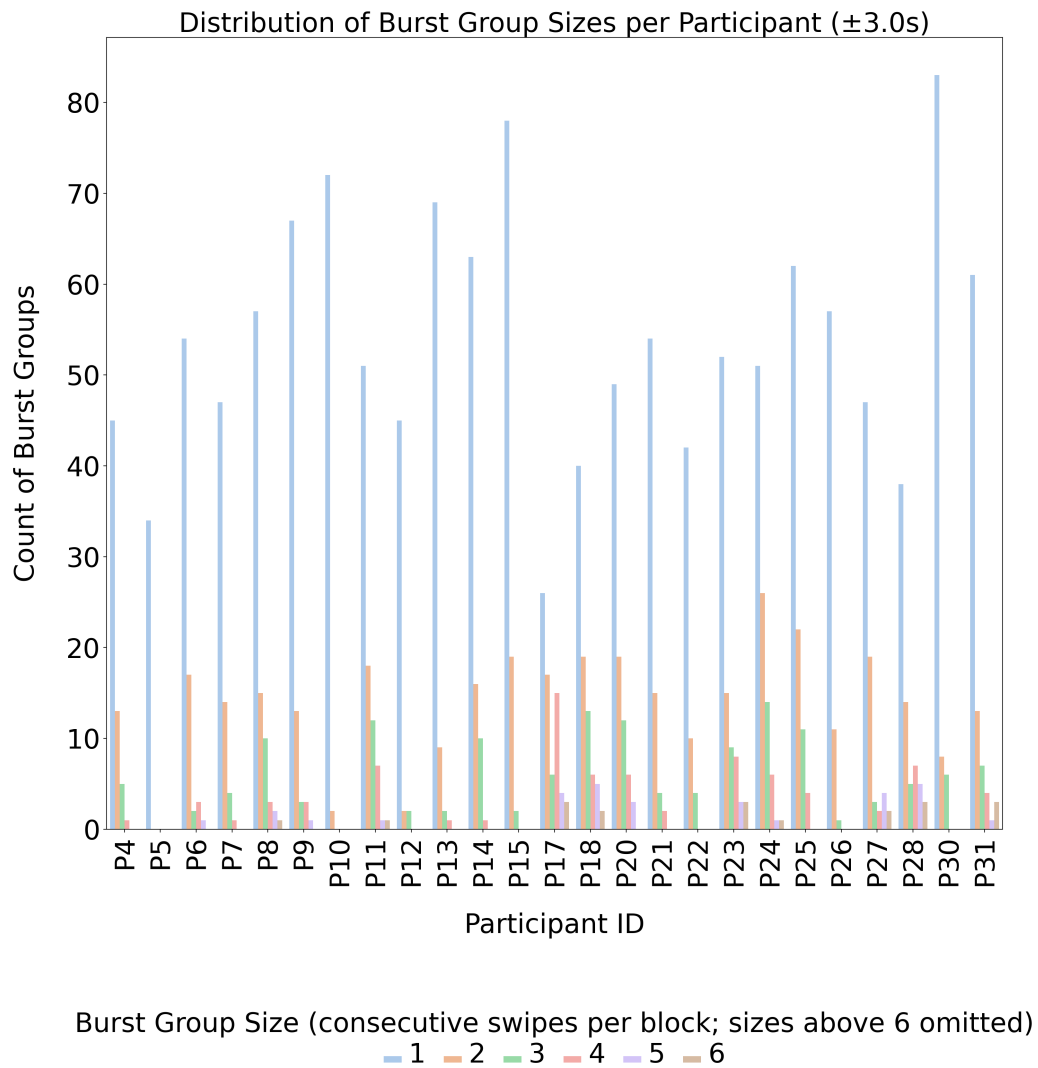


Figure 4.1: Distribution of burst group sizes per participant (using the $\pm 3.0s$ window). This highlights the distinct behavioral difference between “fast scrollers” (larger burst blocks) and “careful scrollers” (predominantly isolated $n = 1$ swipes).

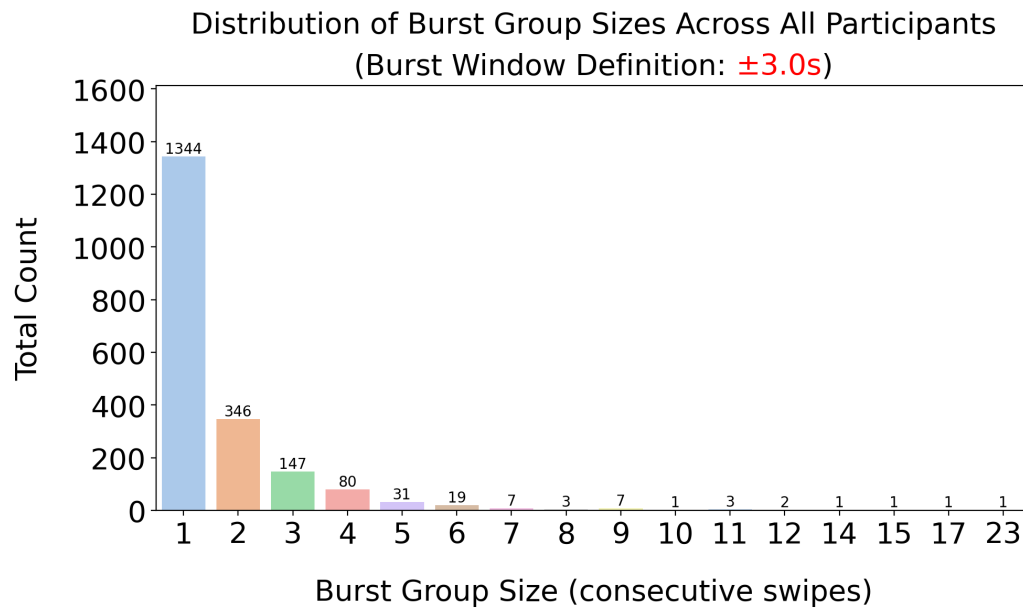


Figure 4.2: Total burst group sizes across all participants ($\pm 3.0s$ window).

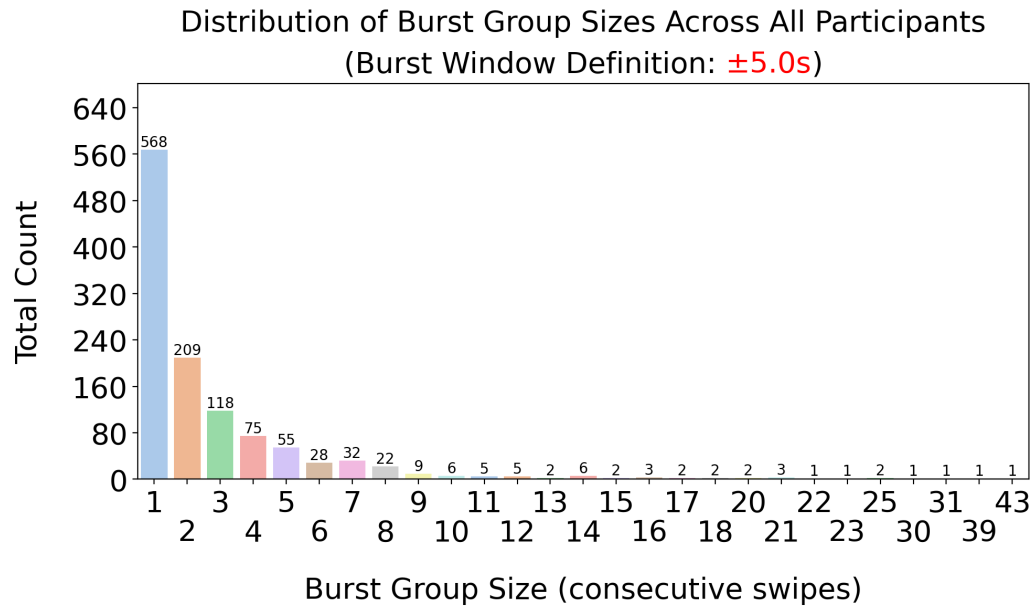


Figure 4.3: Total burst group sizes across all participants ($\pm 5.0s$ window). The wider threshold computationally merges many normal swipes into larger burst blocks.

4.4 The Engagement Index’s Swipe-Prediction Performance

4.4.1 Feature Importance and Neurophysiological Correlates

In Figure 4.4, we observe that class separation, quantified via Cohen’s d as described in Section 3.8.2, is consistently stronger for several individual frequency bands (blue bars) than for the conventional Engagement Index (red dotted line). In particular, the α -band (8–13 Hz) achieves an effect size approximately 0.005 standard deviations higher than the Engagement Index, and the high- γ band (40–60 Hz), as the strongest candidate, exceeds it by more than 0.03 standard deviations. The Engagement Index, which combines three bands into a single scalar, itself reaches just above 0.02 and is therefore weaker in class separability than any of the individual frequency bands and the raw signal. This suggests that aggregating multiple bands into a single composite metric may dilute discriminative information that is more clearly expressed within specific, well-chosen frequency ranges.

Moreover, class separation reaches values above 0.04 only for a small subset of bands: the raw, broadband signal (without frequency decomposition), the δ band (1–4 Hz), the low- γ band (30–40 Hz), and the high- γ band (40–60 Hz). These results indicate that both very low-frequency activity and specific γ -range oscillations carry comparatively strong discriminative structure for the SKIP versus STAY decision. The prominence of the γ range is consistent with gamma activity (30–100 Hz) being associated with high levels of cognitive processing, perception, and attention [18]. The prominence of the δ band is consistent with evidence that delta responses take part in decision making and attention [53], and that delta is linked to attention and salience detection [54], although the delta responses reviewed by Güntekin and Başar follow a stimulus, whereas the SKIP windows precede a self-paced swipe (Section 2.12).

Taken together, these findings highlight three key points. First, individual frequency bands exhibit stronger class-separation capability than the conventional Engagement Index, suggesting that analyses based on raw or band-specific signals may be more informative than combined engagement metrics. Second, the raw broadband signal reaches a higher mean $|d|$ than all separated bands except high γ , consistent with prior EEG decoding work showing broadband/undecomposed signal can match or exceed band-limited features depending on task [61]. This comparison should be read with care, since the raw signal is summarized by only three time-domain statistics whereas each

band is summarized by eight (Section 3.7); it nevertheless motivates closer examination of both broadband information and high- γ activity in particular.

Third, the pattern across bands follows a clear up-down-up trend: separability is high at low frequencies, drops to a minimum in the α range (8–13 Hz), then rises to a global maximum in the high- γ range before decreasing again for very high γ frequencies (60–100 Hz). Non-monotonic, band-specific separability profiles of this kind have been reported in EEG classification more broadly; band-wise performance in EEG-based person verification has been shown to peak in non-adjacent frequency ranges rather than scaling smoothly with frequency [63]. This systematic profile offers a frequency-resolved view of where discriminative information is concentrated in the EEG. The α minimum within this profile is notable in itself: alpha activity is classically interpreted as cortical idling, the synchronized rhythm of regions that are neither processing sensory input nor generating motor output, which desynchronizes once a region is recruited for active processing [64]. A rhythm indexing momentary passivity rather than content appraisal would be expected to separate the two classes poorly.

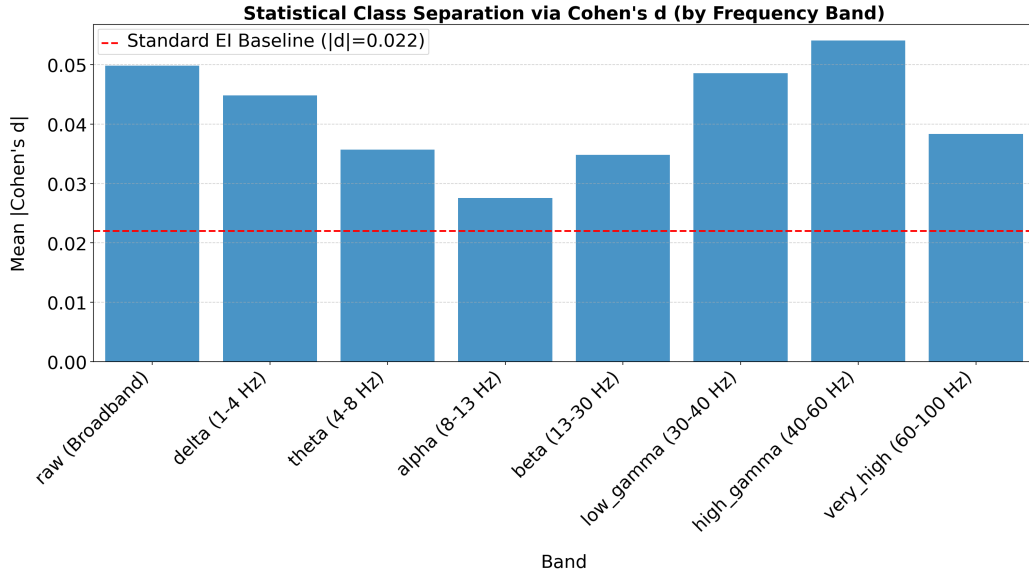


Figure 4.4: Mean $|d|$ by frequency band, aggregated across electrodes and statistics. The dotted line marks the Engagement Index reference value; error bars are omitted.

Similarly, Figure 4.5 shows that separating the signal by electrode of origin, namely TP9, AF7, AF8 and TP10, yields higher class separability than

the conventional Engagement Index in all cases. The improvement ranges from just under 0.01 standard deviations for TP9 and AF8 to more than 0.03 standard deviations for AF7 and TP10. This suggests that distinguishing between individual brain regions provides more informative class separation than collapsing the signal into a single composite metric, and that the different electrodes contribute unequally to this discriminative task.

Moreover, the figure reveals a clear dominance of AF7 (left anterior frontal) and TP10 (right temporal pole) over TP9 (left temporal pole) and AF8 (right anterior frontal) in terms of Cohen’s d . This pattern suggests that the frontal and right temporal regions contribute more strongly to the SKIP-versus-STAY separation than the left temporal pole and right anterior frontal regions, pointing to a potentially lateralized and region-specific neurophysiological signature of scrolling behavior.

Among the electrodes, AF7 (left anterior frontal) reaches the highest mean $|d|$, exceeding the value shown for the raw signal in Figure 4.4 by just over 0.005 standard deviations. This comparison is indicative rather than exact, because the two means aggregate over different and unequally sized feature sets: the electrode mean averages across all signal types and statistics at AF7, whereas the raw-signal mean averages across all electrodes but over only the three time-domain statistics extracted for the broadband signal (Section 3.7).

Potential systematic differences related to the recording hardware cannot be excluded at this stage, as two physical units of the headband were used across the $n = 25$ participants (Section 3.2).

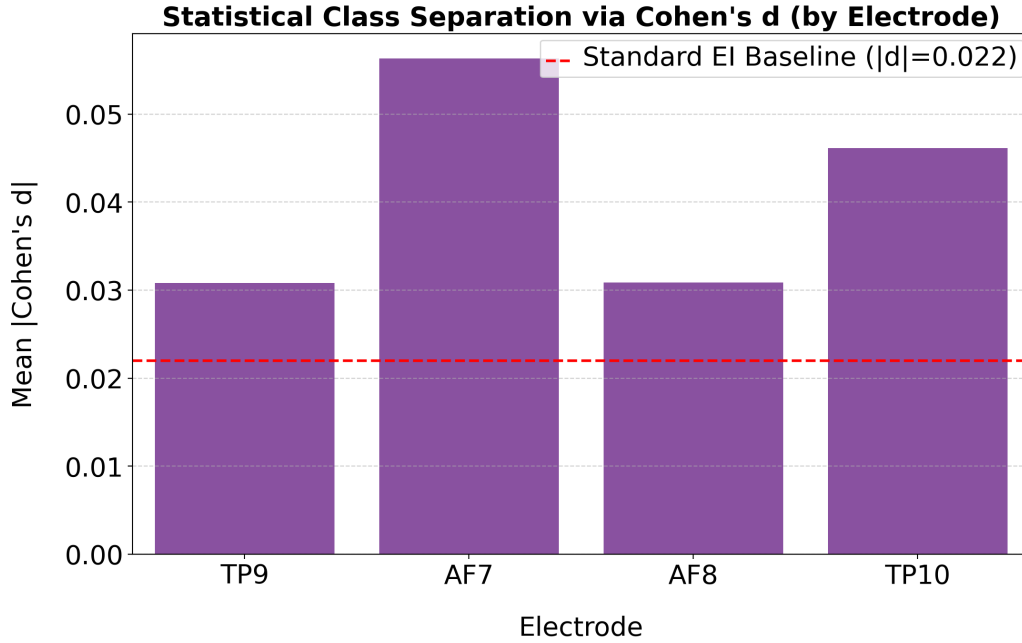


Figure 4.5: Mean $|d|$ by electrode, aggregated across signal types and statistics. The dotted line marks the Engagement Index reference value; error bars are omitted.

Figure 4.6 shows that, among the statistical moments and related features considered, only a small subset exceeds the conventional Engagement Index by at least 0.01 standard deviations in terms of Cohen’s d class separation. These features are Hjorth activity, Hjorth mobility, the mean, the standard deviation, Hjorth complexity, and peak frequency. Taken together, this suggests that simple time-domain descriptors and Hjorth-derived measures retain meaningful discriminative information, but only for a limited number of feature types.

Furthermore, only two statistics - Hjorth complexity and peak frequency - reach a higher mean $|d|$ than the raw signal, again noting that these means aggregate over different feature sets and are therefore only indicatively comparable. This indicates that, while the unprocessed signal already contains substantial discriminative structure, certain higher-order descriptors can still extract additional information beyond the broadband signal alone.

Across the three feature families examined here, namely frequency bands, electrodes and statistical moments/metrics, the statistical moments and metrics show the strongest overall class-separation performance. In particular, peak frequency stands out with a mean $|d|$ of nearly 0.07, placing it al-

most 0.05 standard deviations above the conventional Engagement Index and nearly 0.02 above the mean for the raw signal. This makes peak frequency the strongest single feature in the present comparison and suggests that oscillatory structure within the signal may be especially informative for distinguishing between the two classes. At the same time, these findings should be interpreted cautiously, as the observed advantages are modest in absolute terms and may be influenced by dataset-specific factors.

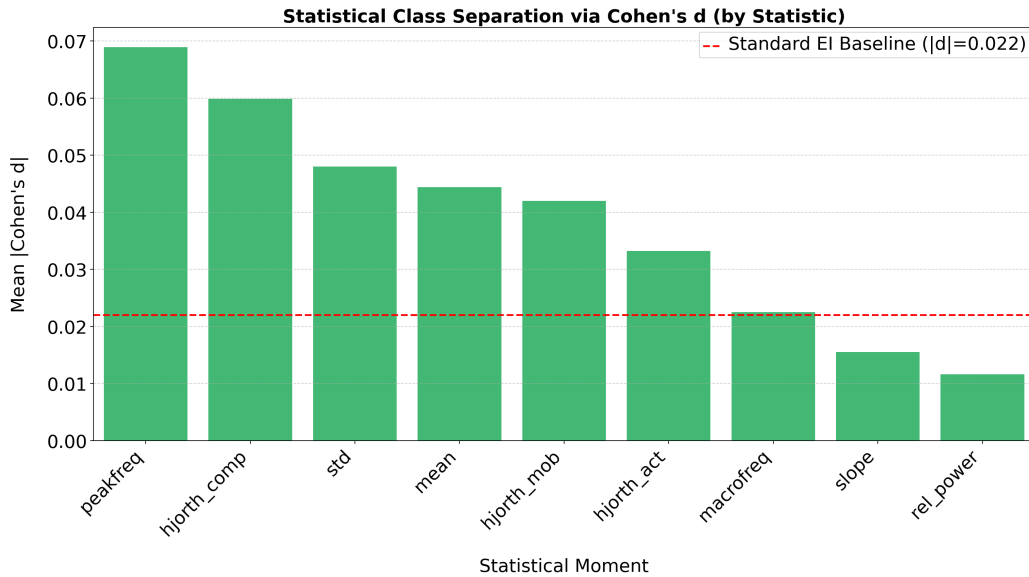


Figure 4.6: Mean $|d|$ by statistic, aggregated across electrodes and signal types. The dotted line marks the Engagement Index reference value; error bars are omitted.

While the comparison of features up to this point has already provided informative detail, the highest level of resolution is reached when each parameter is specified individually: (i) electrode, (ii) band or raw signal, and (iii) statistic or metric. Table 4.1 illustrates the value of this finer-grained perspective. In the preceding analyses, peak frequency emerged as the strongest class separator according to Cohen's d , when considering all bands and electrodes jointly. In those comparisons, the only variable under inspection was the parameter being held constant, while the remaining parameters varied across features. This resulted in aggregated Cohen's d values for each parameter group, as shown in Figures 4.4, 4.5, 4.7, and 4.6. These results raise a natural question: are there individual features, defined across all three parameters, that exceed the currently best observed class-separation perfor-

mance of peak frequency, at approximately 0.07?

Table 4.1 shows that such features do indeed exist. In fact, the top 30 features all outperform the peak-frequency result in terms of Cohen’s d , indicating that a more specific feature definition can yield substantially stronger class separability. The strongest feature is TP10 raw standard deviation, with a Cohen’s d of 0.196, corresponding to an improvement of nearly ten-fold over the conventional Engagement Index in terms of class separation.

This shows that class separability is not only determined by broad feature families such as frequency bands, electrodes, or statistical moments, but can increase substantially when the signal is described at the level of individual feature combinations. It further suggests that the conventional Engagement Index captures only part of the discriminative structure present in the data, whereas more specific features can reveal considerably stronger separation between classes. At the same time, these findings should be interpreted as evidence of dataset-specific predictive potential rather than as a universal hierarchy of feature importance.

Rank	Feature Name	Cohen's d
1	TP10 raw std	0.196
2	AF7 high gamma mean	0.168
3	AF7 high gamma hjorth mob	0.145
4	AF7 delta hjorth comp	0.141
5	AF8 delta peakfreq	0.140
6	AF7 high gamma peakfreq	0.139
7	TP9 delta hjorth comp	0.137
8	TP9 delta peakfreq	0.136
9	TP10 delta hjorth comp	0.131
10	TP10 delta peakfreq	0.130
11	AF7 theta hjorth comp	0.129
12	AF7 raw std	0.129
13	AF7 theta peakfreq	0.124
14	AF7 low gamma hjorth comp	0.124
15	AF7 delta peakfreq	0.123
16	TP10 low gamma hjorth act	0.119
17	AF8 high gamma mean	0.119
18	AF7 low gamma mean	0.118
19	AF7 beta mean	0.115
20	AF8 delta hjorth comp	0.115
21	TP10 low gamma std	0.114
22	TP10 high gamma hjorth mob	0.110
23	AF7 very high mean	0.109
24	TP10 beta hjorth act	0.108
25	TP9 theta peakfreq	0.103
26	TP9 high gamma std	0.103
27	TP10 beta std	0.100
28	AF8 theta peakfreq	0.100
29	AF7 beta hjorth comp	0.100
30	TP9 high gamma hjorth act	0.098

Table 4.1: Top 30 Features by Separation (|Cohen's d |).

For all features, Figure 4.7 shows Cohen's d values sorted jointly by frequency band and electrode. For the raw, broadband signal (no frequency filtering), the TP10 combination shows the highest mean $|d|$ in the visualization, consistent with the top individual feature "TP10 raw std" originating from this combination. Overall, similar trends to the previous figures can be observed, but the combined view makes the joint band-electrode profile more apparent. In particular, there is a pronounced spike in class-separation

performance for the broadband signal at TP10 and AF7.

Moreover, the AF7 electrode stands out by exhibiting a clear upward trend in Cohen’s d from the α band (8–13 Hz) through β (13–30 Hz) to low γ (30–40 Hz), followed by a plateau across the broader γ range (30–100 Hz; low, high, and very high γ) with a slight decline toward the highest frequencies. The peak for AF7 lies in the low- γ band (30–40 Hz), where its class-separation ability is highest. This enhanced separability for the SKIP-versus-STAY distinction at AF7 (left anterior frontal), particularly in the γ and high- γ range (30–100 Hz), may suggest that left anterior frontal oscillatory activity in this frequency range is especially sensitive to the neural processes that differentiate **STAY** from **SKIP** behavior in this task. Such sensitivity would be mechanistically plausible, as gamma-band synchronization has been proposed as the fundamental mechanism through which attentional selection is implemented in cortical circuits, with attended stimuli producing stronger and higher-frequency gamma responses [65]. These interpretations should, however, be viewed as preliminary and require replication in independent datasets.

Furthermore, the superior class-separation ability of the TP10 broadband signal, relative to TP10’s performance within individual frequency bands, suggests that discriminative information may extend beyond the conventional 1–100 Hz decomposition. This could indicate that additional signal components, including potential artifacts such as electromyographic activity from jaw movements, contribute to the SKIP-versus-STAY separation alongside, or in some cases instead of, purely neural activity. This possibility is not merely speculative: electrodes near head muscles pick up EMG, and TP10 lies at the lower edge of the scalp, where EMG is largest; above 20 Hz, EMG is present even at rest, and standard ICA-based correction reduces EMG without removing EMG completely [48, 50, 49] (Section 2.11). These possibilities highlight the need for careful artifact characterization and control in future work.

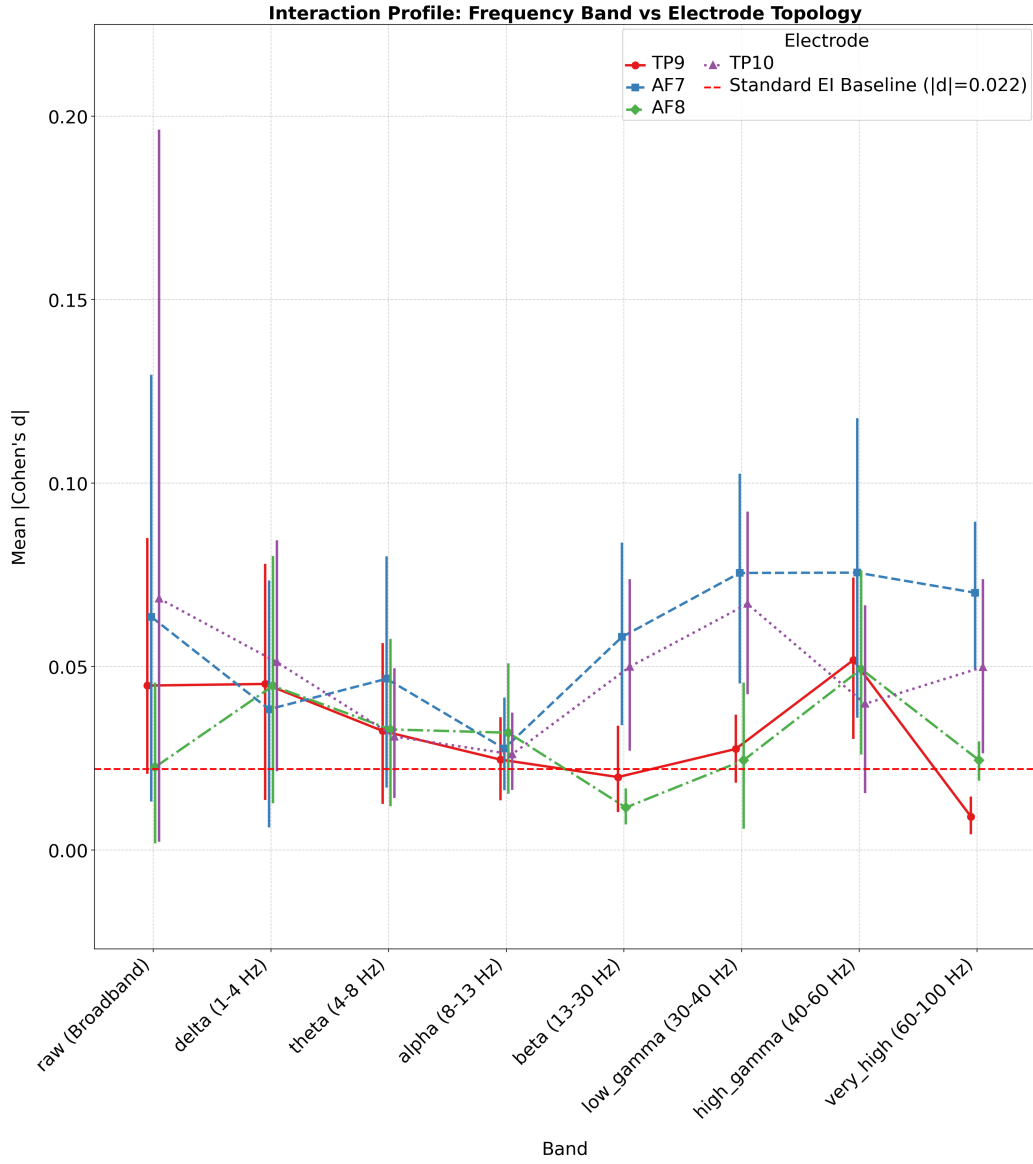


Figure 4.7: Mean $|d|$ for each frequency-band $\times$ electrode combination (band–electrode separation profile). Error bars represent bootstrapped 95% confidence intervals.

As shown in Figure 4.8 the top features were selected for each category: i) overall top feature #1, ii) overall top feature #2, iii) top #1 feature of top band, iv) top #1 feature of top electrode, v) top #1 feature of top statistic/metric.

The feature “TP10 raw std” ($|d| = 0.196$) is placed for i) feature #1,

whereas “AF7 high gamma mean” ($|d| = 0.17$) simultaneously represents all 3 categories ii), iii) and iv). The “AF8 delta peakfreq” ($|d| = 0.14$) feature takes the place of category v), which is the #1 feature amongst the top statistics, namely the Peak-frequency.

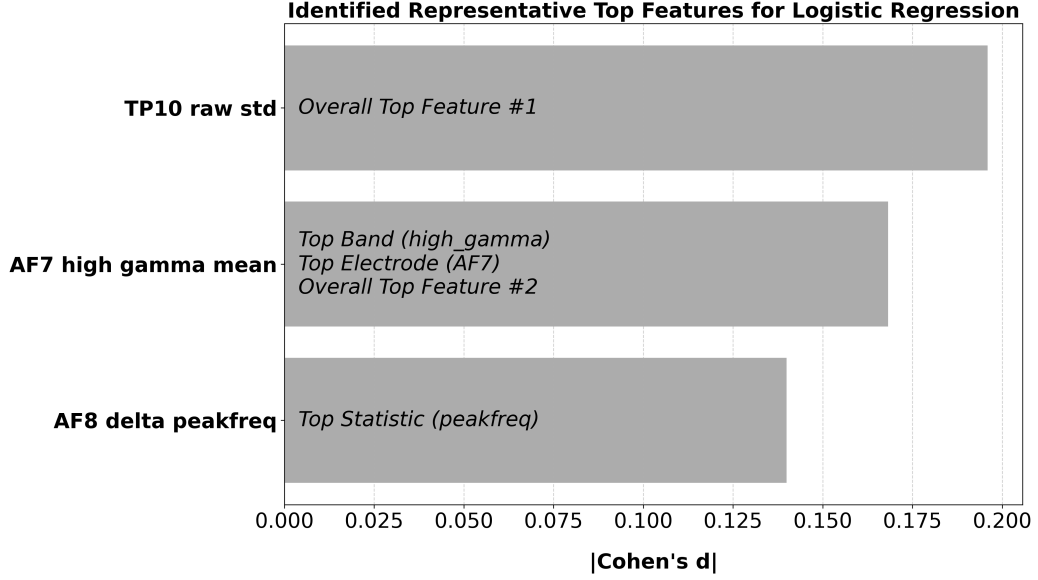


Figure 4.8: Selection of the three candidate features carried forward to the Alternative Index models: overall top feature, top feature of the strongest band, and top feature of the strongest statistic.

4.4.2 Engagement Index Model Performance and Benchmark Comparison

The modeling framework described in Section 3.8.3 was used to evaluate whether the engineered EEG features could predict **SKIP** versus **STAY** behavior beyond chance and beyond the conventional Engagement Index. First, a coin-flip baseline established the empirical chance level. Second, the Engagement Index was evaluated as a one-dimensional logistic regression baseline. Third, the strongest features identified in Section 4.4.1 were tested in isolated logistic regression models to assess whether they could improve predictive performance beyond the baseline models.

To evaluate predictive performance, standard classification metrics were computed as described in Section 3.8.4. In particular, Accuracy, Precision, Recall, and the F_1 -score were calculated for each cross-validation fold, sep-

arately for both classes via one-vs-rest evaluation. The F_1 -score served as the primary metric because it balances Precision and Recall and is therefore more informative than Accuracy under the present class structure.

Statistical significance was assessed using the Wilcoxon signed-rank test, as outlined in Section 3.8.4. The coin-flip baseline serves as the reference for chance-level performance, whereas the Engagement Index is used as the clinical baseline for subsequent comparisons. Performance is reported both for the intra-subject evaluation setting, in which models were trained and tested within the same participant, and for the inter-subject evaluation setting, in which models were generalized across participants. The corresponding results are shown in Table 4.2 and Table 4.3, respectively. In these tables, green values indicate metrics that are both significantly different from the baseline ($p < 0.05$) and higher than the baseline on average, thus reflecting genuine performance gains rather than mere statistical deviation.

At the inter-subject level shown in Table 4.3, the Engagement Index logistic regression yields a statistically significant improvement over the coin-flip baseline only for **STAY** as the positive class. The most prominent effect is Recall, which rises to 89.2% compared with 51.7% for the coin-flip baseline. This indicates that the model is highly sensitive to **STAY** windows and rarely misses them, but it does not necessarily imply balanced or reliable class discrimination on its own. In fact, high Recall in combination with only moderate F_1 -scores suggests a strong tendency to capture the majority of **STAY** instances while still leaving room for false positives, meaning that the model may be better at recognizing **STAY** periods than at sharply distinguishing them from **SKIP**.

The corresponding F_1 -scores provide a more nuanced interpretation. At 62.0% for training and 62.5% for testing, the model performs above chance and remains relatively stable across folds, which argues against severe overfitting in the inter-subject setting. However, the gain over the coin-flip baseline is modest in absolute terms, especially when viewed alongside the comparatively large improvement in Recall. This combination suggests that the Engagement Index carries some predictive structure, but that this structure is unevenly distributed across metrics: it may identify **STAY** behavior reasonably well, yet it does not provide a uniformly strong decision boundary between the two classes.

Importantly, these results also clarify what the Engagement Index can and cannot do in this dataset. Its limited predictive utility here is consistent with prior work: in a controlled learning-engagement task, EI-based classification yielded accuracy performances described as “not optimal,” with the authors noting the index is mainly used in non-predictive applications [16]. It appears capable of capturing a limited but real signal related to

the **STAY** class specifically, but it is not sufficient as a standalone descriptor of the full SKIP-versus-STAY decision. The fact that the model improves clearly in Recall while remaining only moderately strong in F_1 -score raises a central interpretive question: does the Engagement Index reflect a broad engagement-related tendency, or is it primarily sensitive to a subset of windows that are easier to classify as **STAY**? This ambiguity motivates the next analytical step in Section 4.4.3, namely to examine which specific features, electrodes, and frequency ranges contribute more directly to class separability and whether they offer a more interpretable explanation as an “Alternative Index” over the aggregate Engagement Index. In that sense, the present numbers do not merely indicate performance; they identify the precise limitation of the baseline and justify moving toward feature-level interpretation and model comparison.

Metric	Coin Flip		EI LR	
	Val	Val	Sig	
F1 Train (STAY)	50.6%	53.8%	TRUE	
F1 Test (STAY)	52.4%	50.9%	TRUE	
Precision (STAY)	50.1%	49.1%	TRUE	
Recall (STAY)	55.0%	52.9%	TRUE	
F1 Train (SKIP)	49.4%	49.6%	FALSE	
F1 Test (SKIP)	47.5%	46.9%	FALSE	
Precision (SKIP)	50.1%	48.9%	TRUE	
Recall (SKIP)	45.2%	45.1%	FALSE	

Table 4.2: Step 1 (Intra-Subject): EI vs Coin Flip. Significance tested using Wilcoxon signed-rank test against Coin Flip (two-sided, $\alpha = 0.05$, $N=25$).

Metric	Coin Flip	EI LR	
	Val	Val	Sig
F1 Train (STAY)	50.3%	62.0%	TRUE
F1 Test (STAY)	50.8%	62.5%	TRUE
Precision (STAY)	49.9%	49.4%	FALSE
Recall (STAY)	51.7%	89.2%	TRUE
F1 Train (SKIP)	49.6%	20.6%	TRUE
F1 Test (SKIP)	48.9%	10.7%	TRUE
Precision (SKIP)	49.9%	23.5%	TRUE
Recall (SKIP)	48.0%	9.6%	TRUE

Table 4.3: Step 1 (Inter-Subject): EI vs Coin Flip. Significance tested using Wilcoxon signed-rank test against Coin Flip (two-sided, $\alpha = 0.05$, $N=25$).

When assessing the individual participants’ performance in the cross-participant model setting, as shown in Figure 4.9, the pattern further supports the interpretation that **STAY** events are systematically and significantly easier for the Engagement Index to recognize than **SKIP** events are to detect reliably. This asymmetry suggests that the model captures a broad tendency toward the **STAY** class, but lacks equally strong sensitivity to the more transient **SKIP** class.

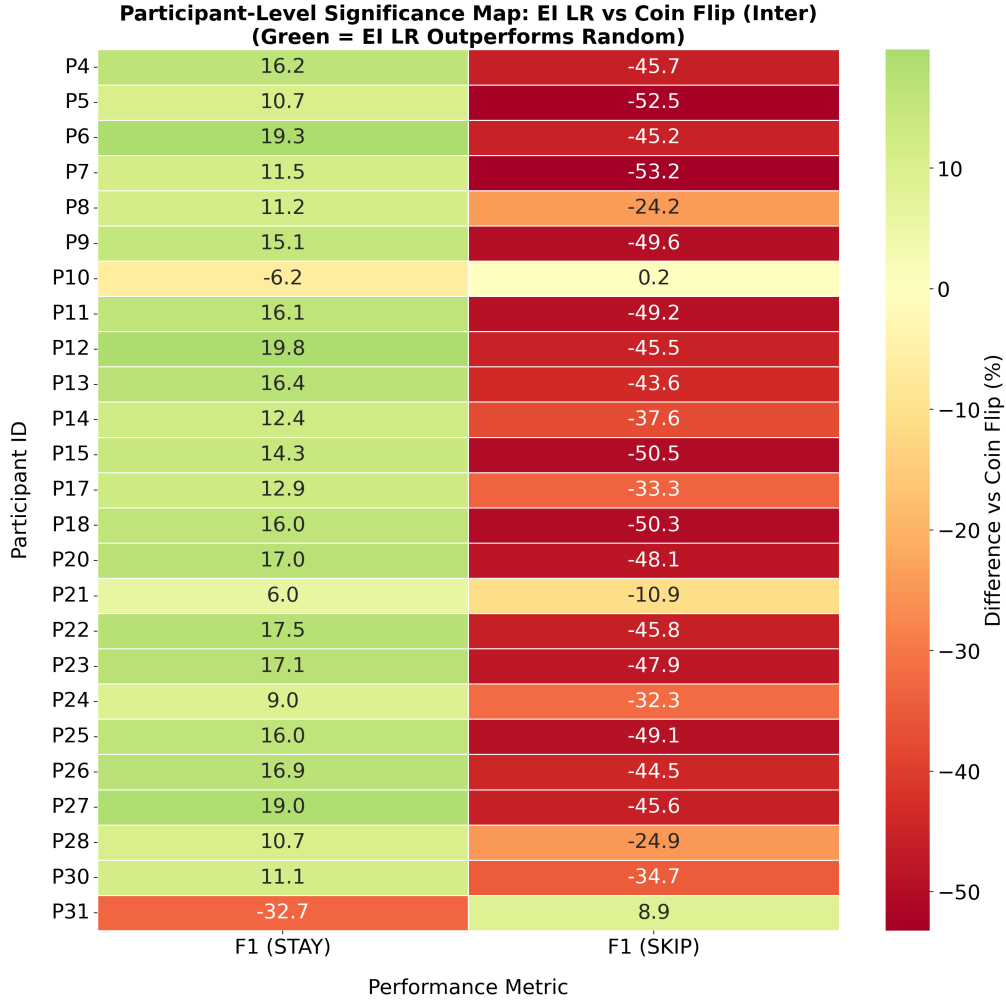


Figure 4.9: Inter-subject significance heatmap comparing F_1 -test performance of the Engagement Index and the coin-flip baseline for SKIP and STAY as positive classes.

4.4.3 Alternative Index Exploration

The Alternative Index exploration evaluates the three features with the highest $|d|$ identified in Section 4.4.1, namely “TP10 raw std”, “AF7 high gamma mean” and “AF8 delta peakfreq”, which are fed into a Logistic Regression model, and compared directly to the Engagement Index Logistic Regression model performance from the previous step in Section 4.4.2.

Intra-Subject (Per-Participant Level)

The performance of TP10 raw standard deviation relative to the Engagement Index is shown in Table 4.4. At the intra-subject level, the only significant improvement is observed for F_1 -train with **SKIP** as the positive class, where TP10 raw standard deviation reaches 53.6% compared with 49.6% for the Engagement Index. Because this advantage does not carry over to the test set, it should be interpreted cautiously and is more consistent with participant-specific overfitting than with robust predictive structure. Nevertheless, the result is still informative: it suggests that raw signal variability at TP10 may contain class-related information that the Engagement Index does not capture, even if that information is not stable enough to generalize within participants.

Metric	EI LR			TP10 raw std		
	Val			Val		Sig
F1 Train (STAY)	53.8%			52.4%		TRUE
F1 Test (STAY)	50.9%			49.3%		TRUE
Precision (STAY)	49.1%			48.2%		TRUE
Recall (STAY)	52.9%			50.4%		TRUE
F1 Train (SKIP)	49.6%			52.6%		TRUE
F1 Test (SKIP)	46.9%			46.8%		FALSE
Precision (SKIP)	48.9%			47.9%		TRUE
Recall (SKIP)	45.1%			45.7%		FALSE

Table 4.4: Step 5 (Intra-Subject): Proposed Feature (TP10 raw std) vs EI. Significance tested using Wilcoxon signed-rank test against EI (two-sided, $\alpha = 0.05$, $N=25$).

The performance of AF7 high- γ mean relative to the Engagement Index is shown in Table 4.5. Here, several small but significant improvements emerge: Precision for **STAY** rises to 49.9% from 49.1%, F_1 -train for **SKIP** increases to

52.9% from 49.6%, and Precision for **SKIP** reaches 49.8% compared with 48.9% for the Engagement Index. Although these gains are modest in absolute terms, their consistency across both classes suggests that AF7 high- γ activity may carry slightly more class-specific structure than the Engagement Index. In other words, this feature does not dramatically outperform the baseline, but it does appear to encode a more balanced and slightly more sensitive representation of the **SKIP**/**STAY** distinction.

Metric	EI LR AF7 high gamma mean		
	Val	Val	Sig
F1 Train (STAY)	53.8%	52.6%	TRUE
F1 Test (STAY)	50.9%	52.1%	FALSE
Precision (STAY)	49.1%	49.9%	TRUE
Recall (STAY)	52.9%	54.6%	FALSE
F1 Train (SKIP)	49.6%	52.9%	TRUE
F1 Test (SKIP)	46.9%	47.2%	FALSE
Precision (SKIP)	48.9%	49.8%	TRUE
Recall (SKIP)	45.1%	45.0%	FALSE

Table 4.5: Step 5 (Intra-Subject): Proposed Feature (AF7 high gamma mean) vs EI. Significance tested using Wilcoxon signed-rank test against EI (two-sided, $\alpha = 0.05$, $N=25$).

The performance of AF8 delta peak frequency relative to the Engagement Index is shown in Table 4.6. In this case, both F_1 -train scores are significantly higher, reaching 83.0% and 83.1% for **SKIP** and **STAY** as the positive class, respectively. However, the corresponding test performance drops sharply toward chance level, which strongly suggests overfitting. Interestingly, F_1 -test, Precision, and Recall for the **SKIP** class are still slightly and significantly better than the Engagement Index, indicating that some task-relevant structure is present, even though it is not stable enough to generalize cleanly within

participants. This pattern is important because it shows that a feature may carry real discriminative information while still being too context-specific to function as a dependable standalone predictor.

Metric	EI LR	AF8 delta peakfreq	
	Val	Val	Sig
F1 Train (STAY)	53.8%	83.0%	TRUE
F1 Test (STAY)	50.9%	49.6%	TRUE
Precision (STAY)	49.1%	50.3%	TRUE
Recall (STAY)	52.9%	48.9%	TRUE
F1 Train (SKIP)	49.6%	83.1%	TRUE
F1 Test (SKIP)	46.9%	51.0%	TRUE
Precision (SKIP)	48.9%	50.3%	TRUE
Recall (SKIP)	45.1%	51.7%	TRUE

Table 4.6: Step 5 (Intra-Subject): Proposed Feature (AF8 delta peakfreq) vs EI. Significance tested using Wilcoxon signed-rank test against EI (two-sided, $\alpha = 0.05$, $N=25$).

Inter-Subject (Cross-Participant Level)

The performance of TP10 raw standard deviation relative to the Engagement Index is shown in Table 4.7. For SKIP as the positive class, all performance metrics improve significantly over the Engagement Index. F_1 -train rises from 20.6% to 52.4%, F_1 -test increases from 10.7% to 44.3%, Precision improves from 23.5% to 43.0%, and Recall rises from 9.6% to 50.8%. It is worth stating explicitly that an F_1 -test of 44.3% remains below the approximate 0.5 chance-level reference for balanced binary classification (Section 3.8.4); the gain reported here is therefore an improvement over a weak Engagement Index baseline rather than evidence of exceeding chance-level discrimination in the cross-participant setting. The pattern is nevertheless meaningful: unlike the

Engagement Index, TP10 raw standard deviation appears substantially more sensitive to the transient **SKIP** state. This suggests that raw signal variability at TP10 may preserve information that is more directly aligned with the **SKIP** class than the aggregate Engagement Index score does.

Metric	TP10 raw std		
	EI LR	Val	Sig
F1 Train (STAY)	62.0%	49.6%	TRUE
F1 Test (STAY)	62.5%	46.0%	TRUE
Precision (STAY)	49.4%	47.5%	FALSE
Recall (STAY)	89.2%	51.0%	TRUE
F1 Train (SKIP)	20.6%	52.4%	TRUE
F1 Test (SKIP)	10.7%	44.3%	TRUE
Precision (SKIP)	23.5%	43.0%	TRUE
Recall (SKIP)	9.6%	50.8%	TRUE

Table 4.7: Step 5 (Inter-Subject): Proposed Feature (TP10 raw std) vs EI. Significance tested using Wilcoxon signed-rank test against EI (two-sided, $\alpha = 0.05$, N=25).

The performance of AF7 high- γ mean relative to the Engagement Index is shown in Table 4.8. Again, the **SKIP** class is predicted more effectively than by the Engagement Index. F_1 -train increases from 20.6% to 45.5%, F_1 -test from 10.7% to 35.5%, Precision from 23.5% to 52.4%, and Recall from 9.6% to 42.7%. While this does not yet establish a fully reliable cross-participant classifier, it does show that AF7 high- γ activity captures a more informative decision-related pattern than the Engagement Index baseline. The improvement in Precision is particularly notable, as it suggests that when the model predicts **SKIP**, it does so with substantially greater confidence and reduced false-alarm behavior than the Engagement Index.

Metric	EI LR	AF7 high gamma mean	
	Val	Val	Sig
F1 Train (STAY)	62.0%	54.3%	TRUE
F1 Test (STAY)	62.5%	45.5%	TRUE
Precision (STAY)	49.4%	44.3%	FALSE
Recall (STAY)	89.2%	58.1%	TRUE
F1 Train (SKIP)	20.6%	45.5%	TRUE
F1 Test (SKIP)	10.7%	35.5%	TRUE
Precision (SKIP)	23.5%	52.4%	TRUE
Recall (SKIP)	9.6%	42.7%	TRUE

Table 4.8: Step 5 (Inter-Subject): Proposed Feature (AF7 high gamma mean) vs EI. Significance tested using Wilcoxon signed-rank test against EI (two-sided, $\alpha = 0.05$, $N=25$).

The performance of AF8 delta peak frequency relative to the Engagement Index is shown in Table 4.9. Here, too, the **SKIP** class is predicted more effectively than by the Engagement Index. Precision for **STAY** increases from 49.4% to 52.8%, while for **SKIP**, F_1 -train rises from 20.6% to 54.1%, F_1 -test from 10.7% to 45.5%, Precision from 23.5% to 53.1%, and Recall from 9.6% to 48.7%. Among the three candidate features examined here, AF8 delta peak frequency therefore provides one of the clearest improvements over the Engagement Index in the cross-participant setting. This is particularly important because it suggests that a feature derived from low-frequency timing structure may be more informative for **SKIP** detection than the conventional engagement metric, even if the absolute performance remains moderate. Taken together, these results support the idea that the Engagement Index can capture a limited engagement-related tendency, but that it is not sufficient as a full account of the **SKIP**/**STAY** distinction. More specifically, it appears to be relatively better at reflecting sustained **STAY**-like behavior than at identifying the more transient and behaviorally decisive **SKIP** state.

This creates a clear next question for Section 4.5: what neurophysiological or behavioral mechanism is actually being captured by the Engagement Index, and which of the more specific features are better aligned with the underlying decision process?

Metric	EI LR	AF8 delta peakfreq	
	Val	Val	Sig
F1 Train (STAY)	62.0%	57.4%	TRUE
F1 Test (STAY)	62.5%	48.3%	TRUE
Precision (STAY)	49.4%	52.8%	TRUE
Recall (STAY)	89.2%	54.6%	TRUE
F1 Train (SKIP)	20.6%	54.1%	TRUE
F1 Test (SKIP)	10.7%	45.5%	TRUE
Precision (SKIP)	23.5%	53.1%	TRUE
Recall (SKIP)	9.6%	48.7%	TRUE

Table 4.9: Step 5 (Inter-Subject): Proposed Feature (AF8 delta peakfreq) vs EI. Significance tested using Wilcoxon signed-rank test against EI (two-sided, $\alpha = 0.05$, $N=25$).

4.5 Discussion

The results above do not yet justify strong causal claims about underlying brain mechanisms, but they do point to several robust interpretive directions. Most importantly, the analyses suggest that the conventional Engagement Index is not wrong in a trivial sense; rather, it appears to capture a limited and asymmetrical subset of the behaviorally relevant signal. In particular, it seems more aligned with sustained **STAY**-like states than with the more transient, decision-like **SKIP** events. This asymmetry is crucial, because it implies that the Engagement Index may reflect a broad state of ongoing

engagement, but not the full cognitive transition that leads to disengagement. From a scientific perspective, this is a more useful and humble conclusion than simply saying that the index "works" or "does not work": it works partly, in a constrained way, and its failure modes are themselves informative. Two structural explanations are available from the literature. First, the index was developed and validated for slow-changing, sustained-vigilance monitoring rather than rapid, self-paced micro-decisions [5]. Second, its denominator is not neutral under rapidly shifting attentional states: if alpha power tracks momentary cortical idling rather than a fixed physiological baseline [64], the ratio $\beta/(\alpha + \theta)$ becomes unstable precisely in paradigms where participants transition between engagement and disengagement within seconds.

A second important point is that the strongest features are not distributed uniformly across the EEG. Instead, they cluster in specific electrodes, frequency bands, and feature types, with raw broadband variability at TP10, high- γ activity at AF7, and delta peak frequency at AF8 repeatedly emerging as informative. This pattern suggests that the relevant signal may not be a single abstract "engagement" quantity, but rather a combination of lateralized, frequency-specific, and task-sensitive components. The fact that raw broadband signal sometimes outperforms band-limited features also raises the possibility that the discriminative information is not fully captured by conventional band definitions alone. At the same time, this may also indicate sensitivity to non-neural factors, including motion-related or muscle-related artifacts, which must be treated cautiously rather than ignored.

4.5.1 Neuroscience

From a neuroscientific perspective, the most consistent result is that separability increases when the signal is described more specifically, especially in the raw broadband TP10 feature and in AF7 high- γ activity. This may indicate that the decision to continue scrolling versus disengage is associated with a distributed and frequency-sensitive pattern rather than with a single canonical engagement rhythm. The repeated prominence of high- γ and delta-derived features is notable because these ranges are often linked to sharply different physiological processes; however, the present data do not allow a definitive mapping from feature to neural mechanism. What can be said more cautiously is that the observed pattern is compatible with the idea that the task engages both rapid sensorimotor dynamics and higher-order control or monitoring processes, with anterior frontal and temporal sites contributing differently to the observed class structure. The involvement of the anterior frontal sites is at least anatomically plausible: the prefrontal cortex underlying AF7 and AF8 serves as the primary cortical substrate for top-

down cognitive control, maintaining goal-relevant representations and biasing downstream processing toward task-appropriate responses [66], the class of process that a moment-to-moment appraisal of whether a video warrants continued attention would recruit.

At the same time, the broadband superiority of TP10 and the strong performance of AF7 in high-frequency ranges create an important interpretive friction point. If the signal were purely reflecting a clean cortical engagement index, one might expect a more stable advantage for the conventional composite metric. Instead, the data imply that narrower and more local signal components contain more class-relevant information. This opens at least two competing hypotheses: either the task truly recruits region-specific neurophysiological signatures that the Engagement Index averages away, or the apparent gain is partly driven by peripheral contamination, such as facial muscle activity, motion, or device-dependent noise.

This pattern is consistent with established evidence that frontal and temporal electrodes pick up EMG from the forehead and temple muscles, which persists even at rest and is not fully removed by standard ICA-based artifact correction [48, 49, 52]; because EMG measured over these muscles shows beta-like peaks as low as 13 Hz, and near a tensed muscle reaches every band, delta included [48], this ambiguity is not confined to the high-gamma and broadband features discussed above, and plausibly extends to lower-frequency features at these same electrodes, including AF8 delta peak frequency, while beta and gamma findings near scalp muscles, such as AF7 high-gamma mean, call for extreme caution and at minimum a scalp map of where each effect lies, which four electrodes cannot provide [52]. Because two independent Muse S device units were used across participants (Section 3.2), device-dependent differences in electrode contact or impedance could further modulate this susceptibility, compounding rather than resolving the ambiguity between neural and myogenic origin.

A related and separate source of ambiguity concerns the decision to report all results on un-notched data (Section 3.6.4). Because the 50 Hz powerline-noise falls inside the high-gamma band, and because a different low-cost amplifier, OpenBCI, has been shown to add extra noise peaks 14 Hz below and above the 50 Hz powerline-noise, at 36 and 64 Hz [56] (Section 2.13), any high-gamma or broadband finding reported here (e.g. AF7 high-gamma mean, TP10 raw standard deviation) cannot be fully dissociated from possible powerline-noise contamination. This is an acknowledged limitation of the present analysis rather than a validated non-issue: a targeted notch-versus-no-notch ablation on the reported top features is left to future work.

The delta-band findings of the present thesis (Section 4.4.1) are consistent with a review linking delta right after a stimulus to decision making and

attention [53], and with a review linking delta to attention and salience detection [54]. The delta responses reviewed by Güntekin and Başar follow a stimulus, whereas the SKIP windows precede a self-paced swipe (Section 2.12). Two short-form video studies report on delta, with different questions: in adolescents browsing Douyin, the change in delta during browsing predicted an addiction score [55], and the adjacent study found beta and gamma the most informative bands for telling short-form video viewing apart from other activity [22]. Neither study labels the seconds before a self-paced swipe. The cited works also study mainly the strength of delta, whereas AF8 delta peak frequency counts how often per second the strength of the delta band peaks (Section 3.7.4), so the AF8 delta peak-frequency finding is at most loosely compatible with a general role of delta in attention and salience, and is no validated TikTok-specific marker.

The electrode pattern also raises a lateralization question that cannot yet be answered decisively. AF7 and TP10 appear more informative than AF8 and TP9, which may point to a functional asymmetry in how engagement-related and action-related states are expressed during naturalistic scrolling. However, because two physical units of the headband were used and because broadband effects may partially reflect artifact structure, this should be interpreted as a hypothesis for future work rather than as a confirmed hemispheric specialization. In the next step, a targeted analysis of artifact-sensitive channels, spectral decomposition, and device-specific effects would be needed to determine whether these patterns reflect genuine neurophysiology or measurement confounds.

4.5.2 Psychology

Psychologically, the results suggest that the act of skipping a TikTok video is not simply the inverse of being engaged. The Engagement Index performs better for STAY than for SKIP, which implies that sustained attention or passive continuation may be easier to detect than the brief decision point at which a participant chooses to move on. This distinction matters because it suggests that SKIP is not merely a low-engagement state, but may represent a qualitatively different behavioral event: an active choice, a threshold crossing, or a moment of disengagement that is preceded by a narrower and more transient signal profile. The stronger performance of features such as AF8 delta peak frequency and TP10 raw standard deviation supports the idea that the decisive moment may be better captured by fine-grained temporal structure than by a broad engagement aggregate.

The burst analysis adds an additional psychological layer. Some participants behave as “fast scrollers” with repeated burst blocks, whereas others

show mostly isolated swipes. This heterogeneity implies that the psychological process underlying scrolling is not uniform across participants. Some users may evaluate content quickly and repeatedly, while others may linger, reflect, or maintain a more stable engagement rhythm. That individual variability likely explains why per-participant and cross-participant performance differ so strongly, and why the same feature may appear informative in one setting but unstable in another. This pattern is supported by prior work on both the neural and the methodological side. Even during passive naturalistic viewing, higher-order association areas, including prefrontal cortex, consistently fail to show intersubject coherence, a dissociation attributed to these regions expressing unique, individual variations rather than stereotyped stimulus-driven responses [60]. Correspondingly, the non-stationarity of EEG produces pronounced between-subject variability, making cross-subject classification a recognized challenge even under controlled laboratory conditions [16], and inter-subject variability remains a persistent open problem across the EEG-based user engagement literature [18].

The modest but consistent success of individual features over the Engagement Index also suggests that a psychologically meaningful model of scrolling may need to distinguish between sustained engagement, momentary decision pressure, and motor execution. The present results are compatible with such a layered interpretation, but they do not yet prove it. What they do show is that an aggregate engagement score is too coarse to explain the full behavioral variance, especially when the behavior is embedded in a rapid, naturalistic, and self-paced interaction environment.

4.5.3 Behaviour

From a behavioral standpoint, the strongest conclusion is that the SKIP/STAY decision is heterogeneous and temporally structured. The burst analysis shows that many swipes occur in clusters rather than as isolated events, which means that the observed behavior is not simply a sequence of independent binary decisions. Instead, it appears to reflect a mixture of cautious sampling, rapid dismissal, and occasional repeated disengagement. A partial parallel exists in habitual social media use more broadly: reward-insensitive *posting* behavior has been shown to become controlled by contextual cues rather than content appraisal [67]. That evidence concerns posting, not swiping, and comes from a different behavioral paradigm; whether the same cue-controlled, appraisal-independent pattern extends to the swipe decision studied here is not established by that work and is treated as an open question rather than assumed. This helps explain why feature-level models can outperform the Engagement Index in some cases: the neural signal may be tracking not

only whether a participant is engaged, but also where they are in a short behavioral sequence leading toward or away from a swipe.

The fact that raw TP10 variability and AF8 delta peak frequency improve SKIP detection suggests that the behavioral decision to move on may leave a measurable trace in the EEG, but only weakly and only under certain feature definitions. The most defensible interpretation is therefore not that the model “predicts choice” in a strong sense, but that it detects behavioral precursors or correlates of the SKIP decision with limited but meaningful sensitivity. That is an important distinction for the thesis: the signal is informative, but it is not deterministic.

The results also show that behavior is strongly shaped by parameter choices, especially in the burst analysis. Changing the window definition changes how events are grouped, and changing the feature definition changes which patterns emerge as separable. This dependency implies that future behavioral analyses should not rely on a single segmentation or metric, but rather test whether the observed structure is stable across multiple reasonable parameterizations.

4.5.4 Application

In applied terms, these findings are best understood as a proof of concept for more transparent and behaviorally grounded engagement modeling. Rather than demonstrating a ready-to-deploy predictive system, the results show that naturalistic scrolling behavior can be measured with sufficient structure to support interpretable feature-level analysis, and that some EEG-derived features outperform the conventional Engagement Index in ways that are meaningful for both explanation and future application.

The practical relevance of these findings lies less in the idea of a fully deployable EEG-based prediction system and more in the methodological and conceptual direction they provide. The present results show that naturalistic scrolling behavior can be measured, segmented, and partially explained using a compact set of EEG-derived features, and that these features can outperform a conventional aggregate index in ways that are interpretable at the level of signal structure. This matters because it suggests a path toward more transparent models of engagement: rather than relying on a single composite score, future systems may be built from a small number of physiologically meaningful components that are easier to interpret, debug, and adapt.

One immediate application is in the optimization of message delivery in health communication and behavior-change interventions. If certain EEG signatures are more associated with continued attention or imminent disengagement, then intervention content could be structured to better match

those moments of attentional stability or vulnerability. In principle, this could help improve the timing, length, or sequencing of health messages, public-health prompts, educational micro-content, or persuasive interventions in digital environments. For example, short-form health information could be adapted to identify when a user is likely to continue engaging versus when they are likely to skip, which could inform both content design and delivery timing – though at present this discrimination remains modest and, in the cross-participant setting, does not yet exceed the chance-equivalent F_1 reference established in Section 4.4.2.

A second application is in the development of explainable and lightweight screening tools for user engagement in naturalistic settings. Because the strongest results here emerged from a small number of features rather than a large black-box model, the present work may support the design of interpretable real-time monitors that are more feasible than complex deep-learning approaches. Such monitors could be useful in user-testing environments, educational research, human-computer interaction, or cognitive ergonomics, where the goal is not necessarily to make a diagnostic prediction but to identify when a stimulus fails to hold attention or when a participant shifts into a disengaged state.

At the same time, the present results also define the limits of immediate application. The predictive performance is not yet strong enough to justify high-stakes individualized deployment, and the possibility of artifact-related contributions means that any translational use would require substantial validation, device standardization, and robustness testing. In particular, the strong effects observed at TP10 and AF7 raise the possibility that some predictive power may arise from non-neural sources such as jaw movement, facial tension, or other task-correlated artifacts – a risk that, per Section 4.5, plausibly extends into lower-frequency features at these electrodes and not only into the high-gamma and broadband features discussed above. For this reason, the present study is best interpreted as a proof of principle for explainable signal extraction rather than as a finished applied system.

Recommendations for Policy Makers

Three concrete regulatory implications follow from the present findings without requiring any claim beyond what the data and the cited literature support.

First, **watch time is the wrong regulatory quantity**. Both platform self-reporting and emerging digital-services regulation frequently treat viewing duration as a proxy for user interest or content value. The present results, together with the behavioral literature, indicate that continuous non-

skip viewing is equally consistent with dissociative, non-encoding consumption as with genuine engagement [4, 3]; a metric that cannot distinguish the two should not be used as evidence of user benefit in transparency reports or audit frameworks. Policy makers evaluating platform disclosures should therefore require engagement measures that are validated against independent behavioral or physiological criteria rather than dwell time alone. This recommendation follows from the cited dissociation and flow literature [4, 3] rather than from the present EEG results directly, which characterize a candidate physiological signature of the pre-swipe state but do not themselves measure dissociation, flow, or recall.

Second, **friction is an evidence-backed lever**. Interface features that segment or interrupt continuous scrolling, in particular custom lists and reading-history labels, have been shown to significantly reduce self-reported dissociation without removing content [4] (a single study, on a Twitter client rather than short-form video; replication on TikTok-like platforms is needed before generalizing). Mandating or incentivizing such segmentation cues in feed-based products is a realistic, minimally invasive intervention whose mechanism is already empirically characterized, in contrast to blunt instruments such as age-gating or time caps.

Third, **auditability of engagement optimization**. The present thesis demonstrates that the neural precursors of the skip decision are partially measurable with consumer-grade hardware. Platforms plausibly have access to behavioral, and possibly physiological, signals at a scale and precision this thesis cannot match; if so, the same auditability logic applies with greater force. This thesis has no access to platform-internal data, and this claim should be read as a reasonable extrapolation rather than a demonstrated fact. Regulators should treat engagement-optimization systems as auditable algorithmic systems and require disclosure of the behavioral targets they optimize, precisely because the optimized quantity (continued viewing) is, per the above, not a wellbeing proxy.

Recommendations for Digital Global Public Health Research

For digital global public health, including the research agenda of the HPI Chair for Digital Global Public Health, whose stated priorities include disease prevention, evidence-based health communication, and precision public health, the present work suggests three tangible extensions.

First, **message pre-testing panels**: the full recording and analysis pipeline of this thesis runs on a consumer headset, a laptop, and open-source software, and is reproducible by an independent researcher (Section 4.2). This makes small- n physiological pre-testing of health messages, that is, com-

paring candidate formats of a vaccination or antimicrobial-resistance message for their pre-skip signature before campaign deployment, more accessible to public institutions than fully custom or clinical-grade EEG pipelines would be, analogous to the message pre-testing already standard in commercial advertising but at research-grade transparency – though, as Section 4.2 notes, substantial engineering investment was still required to reach this point, and this should be read as a reduction in barrier-to-entry rather than a turnkey solution.

Second, **attentional reach as a campaign endpoint**: public health communication on short-form platforms is known to underperform entertainment content in engagement metrics, with institutional accounts reaching orders-of-magnitude fewer users than individual creators [1]. Campaign evaluation should therefore report not only impressions but survival within the feed, meaning the probability of a health message being skipped within its first seconds, a quantity the present labeling framework provides an initial, currently modest-power operationalization of (Section 4.4.2).

Third, **precision public health applied to attention**: in the same way precision public health stratifies prevention by risk profile, the inter-individual heterogeneity observed here (“fast” versus “careful” scrollers, Section 4.3; the failure of cross-participant models relative to within-participant models) implies that attentional response to health content is itself a stratifiable trait, and that one-size-fits-all message design leaves systematic reach on the table.

Recommendations for Health Message Design

At the level of individual message creation, the findings translate into concrete design guidance.

(i) **The decision plausibly precedes the swipe**: the SKIP-predictive window examined here (2–5 seconds before the physical action) was adopted as a literature-motivated heuristic starting point rather than an independently validated decision boundary (Section 3.4.1), and class separability within it, while statistically distinguishable from the Engagement Index baseline, remains modest in absolute terms (Section 4.4.2). With that scoping in mind, the general pattern – that some pre-swipe neural signature precedes the physical swipe action – is broadly consistent with content needing to establish interest early rather than relying on a delayed payoff; testing where within a video’s opening seconds this signature is strongest, and whether it maps onto specific content features, is a natural extension of the present labeling framework rather than a claim established by it here.

(ii) **Design for the dissociated viewer**: because a substantial fraction

of viewing occurs in a passive, low-encoding state [4, 3], message formats should include pattern interrupts, such as changes in pacing, framing or sensory structure, intended to manufacture re-engagement points rather than rely on spontaneous recovery of attention, since Baughan et al. found such recovery is often only recognized retrospectively, if at all [4].

(iii) **Do not optimize health content for dwell time:** a message that is watched to completion in a dissociative state may be encoded worse than a shorter message that triggers active appraisal; evaluation should weight recall and behavior change over watch-through rate.

(iv) **Respect burst structure:** swipes cluster into rapid bursts for a substantial share of events, though the exact proportion is parameter-dependent – 67.4% of swipes are isolated singletons under the default ± 3.0 s window, falling to 48.9% under a wider ± 5.0 s window (Section 4.3) – and burst tendency varies sharply between participants (SD of 16–21 percentage points around these means), consistent with the individual heterogeneity noted in Section 4.5. This favors repeated, distributed placement over single high-cost placements as a population-level default, while flagging burst-aware, per-user delivery timing as a natural extension once outcome-linked validation is available; the present work establishes that swipes cluster, not that burst membership itself predicts weaker message encoding or evaluation.

Societal Scope: An Explicitly Speculative Outlook

The following paragraph is a deliberate change of register: it is a hypothesis-generating extrapolation, not a result of this thesis, and is marked as such. What the data of this thesis and the cited literature jointly establish is narrow but non-trivial: short-form feeds occupy users in a high-frequency stream of appraisal micro-decisions; a substantial share of the resulting viewing time is spent in states of reduced self-awareness and poor encoding [4, 3]; and attention is a finite, fluctuating resource whose allocation determines what is subsequently remembered [2]. From these premises, a *time-displacement hypothesis* follows naturally, though it is not tested here: hours spent in dissociative feed consumption are hours of attentional and social bandwidth not spent elsewhere, including on the effortful, in-person interactions through which relationships are formed. Whether this displacement contributes causally to widely discussed, though contested, population-level trends in youth socialization, loneliness, or, further downstream, family formation is precisely the kind of long-range causal chain the present study can neither support nor refute, and this thesis makes no such claim. What it can state is the conditional: *if* such displacement effects exist, then the mechanisms studied here, namely the moment of attentional lock-in, the dissociative default and

the decodable pre-skip state, are the proximate levers through which they operate, and the same measurement framework used here to describe them is the framework with which mitigation (segmentation cues, adaptive interruption, attention-aware content policy) would be evaluated. Quantifying the displacement chain, and testing realistic mitigations against it, is left to future research; the contribution of the present work is to make its first link measurable.

Finally, a dual-use note is warranted: any marker that helps health communicators survive the feed helps commercial and political actors equally. The appropriate response is not to withhold such research, since platforms already possess strictly superior behavioral data, but to conduct it transparently, under ethics oversight (Section 3.1), and to pair it with the auditability recommendations above.

4.6 An Updated Definition of Engagement

Everything reported so far concerned a swipe. This section does not. The preceding sections established what the data say about a specific behaviour on a specific platform; this one asks what those data, read together with the definitional literature reviewed in Section 2.4, force us to say about *engagement itself*, as a psychological construct, in any setting where something competes for a person's limited attention.

The output is one definition and the measurement requirements that follow from it. It is offered as basic research (*Grundlagenforschung*): a construct claim, deliberately stated without reference to TikTok, EEG, or any particular instrument, so that it can be reused, cited, and attacked outside the paradigm that produced it. It counts as a result of this thesis in the only sense that matters for a definition, because it is constrained by evidence and can be shown to be wrong (Section 4.6.5).

4.6.1 What the Results Teach Us

Five findings, four from this thesis and one from the literature it reviews, constrain what a definition of engagement may say. Each is stated here as an exclusion, because each removes a family of candidate definitions.

1. Staying is not engaging. The Engagement Index reached 89.2% recall for **STAY** at the inter-subject level and failed to detect **SKIP** (Section 4.4.2). It is good at noticing that nothing has happened. Independently of the physiology, extended uninterrupted viewing is as consistent with dissociative,

low-encoding consumption as with attention [4, 3], and sustained attention fluctuates trial by trial rather than accumulating with time on task [2]. Any definition under which continued presence counts as evidence of engagement is therefore excluded, because it renders the most common failure mode invisible.

2. Leaving is an event, and it has a physical signature. The seconds preceding a voluntary swipe are separable from continuous viewing, and the strongest single feature separates them with roughly ten times the standardized effect of the Engagement Index ($|d| = 0.196$; Section 4.4.1). The magnitude is small and supports no decoder. What it does support is a distinction that matters for the construct: the difference between *engagement declined* and *engagement ended at a locatable moment*. A definition must have room for a terminating event, not only for a falling value.

3. The realisation is individual; the construct is not. Within participants, the features carried structure. Across participants, models collapsed, with F_1 -test up to 45.5%, still below the chance-equivalent reference (Section 4.4.3), and burst tendency varied between individuals by 16–21 percentage points of standard deviation (Section 4.3). This is expected rather than anomalous: higher-order cortical activity is idiosyncratic even during passive naturalistic viewing [60], and cross-subject classification is a recognised open problem in this literature [16, 18]. No particular signal may therefore be written into the definition. Whatever engagement is, it is not AF7 high-gamma power, and it is not $\beta/(\alpha + \theta)$.

4. A measured quantity is undefined without a declared interval. Class separability rose monotonically as the labelled window was lengthened and as it was moved further from the swipe, and the share of swipes counted as isolated rather than bursty fell from 67.4% to 48.9% on the strength of a single threshold choice (Sections 3.4.5 and 4.3). The same behaviour, measured over different intervals, yields different numbers. A definition that does not require a timescale does not describe a measurable quantity.

5. Composites hide more than they summarise. The Engagement Index collapses three frequency bands into one scalar, and separated the two classes worse than any individual band did (Section 4.4.1). Aggregating indicators before the construct is settled destroys exactly the information needed to check whether the aggregation was justified. Definition must precede composition, not follow it.

4.6.2 The Proposed Definition

The definition below is derived from and informed by the results reported above in this chapter, and it inherits their limits: the effects constraining it are small in absolute terms, come from a single dataset of twenty-five demographically narrow participants, and carry unresolved artifact ambiguity (Section 4.5). It is therefore offered as a version 1, a definition precise enough to be used and precise enough to be corrected, rather than as a settled account. It generalises the working definition of Section 1.2 by removing the two things that made that one domain-bound: the screen, and the swipe.

Definition (Engagement). *Engagement is the volitional, effortful allocation of a person’s limited attentional capacity to a particular object, sustained over a declared interval against available alternatives, and ending in a disengagement event that is observable independently of how long the allocation lasted.*

Each clause does work. Each excludes something.

Volitional, effortful excludes dissociative absorption and pure habit. A viewer who cannot later recall what they watched [4] was present, not engaged. This is the clause that time-in-app violates, because a clock counts the dissociated viewer and the attentive one identically (Constraint 1).

Allocation of a person’s limited attentional capacity makes engagement a resource that is spent rather than a preference that is held. It is therefore zero-sum at the moment of measurement: engagement with one object is simultaneous evidence of non-engagement with everything else then available. Liking something is not engaging with it; paying for it in attention is.

To a particular object requires that engagement always be engagement *with* something, such as a message, a task, a person or a piece of work. “An engaged user” or “an engaged employee”, stated without an object, is not yet a measurable claim.

Sustained over a declared interval makes the timescale part of the measurement rather than a result of it (Constraint 4). Engagement measured over three seconds and over three weeks are different quantities, and a figure reported without its interval cannot be interpreted or compared.

Against available alternatives distinguishes engagement from captivity. Attention that has nowhere else to go, as in a mandatory meeting, a queue or a waiting room, is occupancy, not engagement. This is also why an infinite feed is an unusually clean measurement setting: the alternative is always exactly one thumb movement away.

Ending in a disengagement event observable independently of how long the allocation lasted is the clause that carries the measurement weight (Constraints 1 and 2). It asserts that the end of engagement is something that happens, not merely a level that drops, and it forbids the clock from serving as the evidence for it. The four-stage process model of O'Brien and Toms anticipated this by naming disengagement as a distinct stage, but reconstructed those stages from retrospective interviews and so could not supply a time-resolved criterion for when the stage begins [32].

What it inherits. From O'Brien and Toms, that engagement is a process with a beginning, a middle, and an end, and that disengagement is a stage of it rather than the absence of one [32]. From intervention science, that engagement is the precondition for effect and therefore worth measuring in its own right [7, 8]. From attention research, that it is a fluctuating state rather than a stable trait [2].

What it drops. Duration is removed from the definition and demoted to an indicator. Duration here covers two things: the period of time over which a person uses an intervention, which the DBCI definition of Perski et al. includes [8], and the watch time of one video. Behavioural counters of use, the most used proxy in social-media engagement research [14], are demoted together with duration. Every named instrument, whether index, scale, rating, or telemetry, is removed as well.

What it adds. Three requirements no existing definition states: a declared timescale, without which the measured quantity is undefined; an exit criterion that must be observable without reference to the clock; and the separation of the construct from whatever instrument is used to detect it.

What the definition deliberately omits is as important as what it contains. It names no indicator, no modality, and no formula (Constraints 3 and 5). It is compatible with an EEG feature, a self-report item, a linguistic marker, or a behavioural trace, and commits to none of them, which is precisely what makes it portable across the settings where different instruments are the only ones available.

4.6.3 Implications for Measurement

Three consequences follow directly, and they are the practical content of the definition.

Duration is demoted from criterion to indicator, and a weak one. It may be reported; it may not be used as the standard against which other measures are validated. Where duration and an exit-sensitive measure disagree, the definition sides with the latter.

The exit is the primary measurement target. An instrument should be judged on how well it detects the transition out of engagement, not on how well it scores the uneventful middle. Judged on that criterion, the field’s standard physiological instrument fails in this paradigm (Section 4.4.2), which is a finding about the instrument, not about the construct.

Indicators are plural and must be validated against one another. Because the construct is defined without an indicator, EEG features, self-report, and behavioural or linguistic traces are all candidate indicators of the same thing, and none of them *is* the thing. The work this creates is convergent validation, not the selection of a winner.

Operationally, these collapse into a reporting requirement. A well-formed engagement measurement states four things: the **object** engaged with, the **interval** over which engagement is claimed, the **indicator** used, and the **exit criterion**, meaning what would have counted as disengagement. A study that cannot state the fourth of these has measured occupancy rather than engagement.

4.6.4 The Importance of Field Measurement

EEG measures engagement closely, but only in a laboratory. The EEG pipeline of the present thesis needed a headband, a baseline recording for each participant, keypresses synchronised with the EEG to the millisecond, and a quiet room free of distraction. Even under these conditions, the models predicted the swipe only slightly better than chance (Sections 4.4.2 and 4.4.3). A classroom, a hospital ward, or an office offers few of these conditions, so EEG is hard to use in such places at scale.

The definition of engagement in Section 4.6.2 names no measuring instrument, and so leaves room for cheaper instruments. Fever offers an analogy. Fever is defined by body temperature, and a clinical thermometer reads body temperature precisely. A hand on the forehead also reads body temperature, cheaply and roughly, and becomes useful once the hand has been checked against the thermometer. In the analogy, EEG is the thermometer, and a short repeated questionnaire or the pattern of messages in a team chat could be the hand.

On that basis, the present thesis states the following hypothesis. The hypothesis awaits testing in follow-up research.

Where EEG cannot be used at scale, engagement as defined in Section 4.6.2 may be measurable with cheaper field measures: short self-report questionnaires repeated over time (the ask route), and behavioural or linguistic traces in everyday activity and communication (the log route). A field measure qualifies once the field measure has been checked against a precise measure such as EEG. Among other properties, the present results suggest two checks: the field measure should react at the moment a person disengages, and the field measure should do more than count time.

The present data cannot test the hypothesis. Each participant took part in a single twenty-five-minute session of one activity, alone. The session included a demographic questionnaire before viewing and a short questionnaire on keypress errors after viewing, but no self-report of engagement during viewing, no language data, no second setting, and no later outcome against which a field measure could be checked. Testing the hypothesis would need three things the present data lack. First, a setting where EEG and a field measure are recorded from the same people at the same time. Second, a moment of disengagement in that setting as clear as a swipe. Most everyday settings, such as a meeting or a lecture, lack such a clear moment, which makes the second requirement the hardest. Third, agreement between the field measure and EEG on that moment of disengagement, beyond agreement on the average level of engagement. Until the field measure and EEG agree on the moment of disengagement, the field measure has not been shown to measure engagement. Stated this concretely, the hypothesis also names the study that could prove the hypothesis wrong.

4.6.5 Limits of the Proposed Definition

Three findings would break it, and each is reachable with existing methods.

If a person-invariant continuous indicator of engagement were demonstrated, meaning one transferring across individuals without calibration, then keeping indicators out of the definition would be unnecessary caution, and a stronger, instrument-bearing definition would be preferable to this one.

If disengagement were shown to have no locatable onset, being a smooth decline whose apparent “moment” is an artefact of where an analyst places a threshold, the exit clause would fall, and with it most of the definition’s measurement value. The parameter-sensitivity result reported here (Section 3.4.5) is in fact the strongest available argument for that possibility, and is the reason the clause requires the event to be *observable* rather than instantaneous.

If dissociated, non-volitional consumption were shown to produce the same downstream outcomes as effortful attention, such as the same recall and the same behaviour change, then the volitional and effortful clauses would buy nothing, and duration could be rehabilitated as a criterion.

4.6.6 Scope of the Proposed Definition

Nothing in the definition is specific to short-form video. TikTok served here as a field site with an unusually convenient property: a clean, frequent, voluntary exit event, a few hundred observable disengagements per person per half-hour, which is what makes the exit clause testable at all. The construct is intended for settings where exits are rarer, slower, and messier, for example a learner abandoning a course, a patient dropping out of a digital behaviour-change intervention [8], or a colleague withdrawing from a shared piece of work. In such settings, the field’s habit of substituting duration for engagement is hardest to detect and most expensive to get wrong.

Because none of the three additions mentions a screen, a swipe, or an electrode, the definition holds wherever a person can attend to something and leave it. Engagement with a health message in a feed and engagement with shared work in a distributed team are, on this account, the same construct measured under different exit conditions: in the feed the exit is a thumb movement and is logged; in remote collaborative work it is slower, quieter, and visible only in what someone stops contributing to. That is precisely where substituting duration, such as hours logged, messages sent or sessions attended, for engagement is hardest to catch, and where a definition that forbids the substitution earns the most.

The contribution offered here is therefore not the effect sizes that constrain the definition, which are modest and provisional, but the definition and its four reporting requirements, which are neither. They are stated once, here, so that later work can cite them rather than re-derive them.

Chapter 5

Conclusion and Future Research

5.1 Conclusion

This thesis asked whether the moment at which a person decides to stop watching can be seen in the brain before the thumb moves, and whether the field’s conventional composite marker of engagement, the Engagement Index $\beta/(\alpha + \theta)$, is the right instrument for seeing it. Twenty-five participants browsed their own algorithmic feed, unconstrained and alone, while a four-channel consumer headband recorded continuously and every swipe was marked to the millisecond. From that recording, 236 univariate features per one-second window were derived and evaluated against a chance baseline and against the Engagement Index, under both within-participant and across-participant validation. The answers that follow are modest in magnitude and deliberately narrow in scope, but they are not ambiguous.

On separability, the answer is a qualified yes. The period preceding a voluntary skip is not statistically indistinguishable from continuous viewing: individual features separate the two classes reliably, and the strongest of them, raw-signal standard deviation at TP10 ($|d| = 0.196$), does so with roughly ten times the standardized effect of the Engagement Index. The magnitudes remain small in absolute terms, and no single feature approaches the separability that would be required for individual-level inference. What the data support is the existence of the signal, not its sufficiency.

On the benchmark comparison, the answer is more pointed, and it is the central empirical contribution of this work. The Engagement Index is not uninformative; it is *asymmetrically* informative. It detects sustained, non-skip viewing well, reaching 89.2% recall for **STAY** at the inter-subject level, and detects imminent disengagement poorly. Three features identified through effect-size ranking (TP10 raw standard deviation, AF7 high-gamma

mean, AF8 delta peak frequency) each improved SKIP detection significantly over that baseline at both levels of analysis, yet in the cross-participant setting their absolute performance (F_1 -test up to 45.5%) remained below the chance-equivalent reference. The correct reading is therefore not that a better index has been found, but that the existing one has a specific and locatable blind spot: it tracks the state of continuing, not the transition to leaving. A metric that measures the former and is used to reason about the latter will systematically mistake the absence of a decision for the presence of engagement.

On parameter sensitivity, the answer is a caution addressed to the method rather than to the phenomenon. Class separability rose monotonically as the labeled window was lengthened and as it was moved further from the swipe, and the proportion of swipes classified as isolated rather than bursty fell from 67.4% to 48.9% on the strength of a single threshold choice. Separability, in this paradigm, is not a property of the brain alone; it is a joint property of the brain and the labeling scheme imposed upon it. Any reported effect size here, including every one above, is conditional on three temporal parameters that, to our knowledge, no prior work had established for a self-paced behavioral event and were therefore fixed by argument rather than by evidence.

What this thesis cannot conclude deserves to be stated as plainly as what it can. The two features carrying the most discriminative weight are also exposed to contamination: TP10 lies at the lower edge of the scalp near jaw and neck muscles, whose EMG is present at rest above 20 Hz and not fully removable by standard correction; and the high-gamma band straddles the 50 Hz powerline-noise. Neither confound was resolved here. Two physical devices were used across the cohort, so hardware effects cannot be separated from participant effects. The sample was demographically narrow and academically skewed. The reported p -values are uncorrected across a family of comparisons and should be read as screening criteria. It is entirely possible that some of what looks like a brain signal is really jaw muscle activity. Figuring out how much of the effect is real and how much is muscle artifact is the most important thing to do next, not a nice-to-have, but a requirement before these findings can be trusted.

What does survive these caveats is a paradigm and an instrument. The experiment demonstrates that ecologically valid short-form video consumption, with a personal feed, no imposed stimulus, no fixation and no trial structure, can be recorded with millisecond behavioral synchronization on consumer hardware, and analyzed reproducibly end to end with open-source components. That means this kind of question no longer needs an expensive clinical lab: a laptop and a consumer headband are enough. And it gets there without needing any data from the platforms themselves.

One further conclusion is not about TikTok at all, and is stated in Section 4.6: the results above constrain what engagement can be taken to mean as a general construct, and that chapter sets out a definition independent of any instrument, the four things any engagement measurement must therefore report, and an explicitly hypothetical bridge from physiological measurement to the cheaper proxies that field settings would require.

That last point is why this work matters beyond its effect sizes. Billions of these micro-decisions are made every day, and the systems that shape them are optimized against watch time, a proxy that the literature reviewed here shows to be equally consistent with absorbed attention and with dissociated, unencoded consumption. This thesis provides no license to claim that the neural precursor of a swipe has been decoded. It provides something smaller and more durable: evidence that the moment of leaving has a physiological correlate distinct from the state of staying, that the field’s standard engagement metric largely misses it, and that the tools required to study it are now within reach of the public institutions whose messages must survive that same feed. Attention is finite, unevenly distributed, and, for the moment, measured almost exclusively by those with an interest in consuming it. Building an independent, transparent, and falsifiable account of how it is lost is not a neuroscientific curiosity. It is a precondition for anyone else, such as educators, clinicians and public health communicators, to compete on honest terms for the seconds that decide *what a population knows, and acts on*.

5.2 Future Research

Future research should first focus on validation and replication. The present findings are promising, but they are based on a single dataset with a relatively small sample size and two physical units of the same recording device. Replication in an independent sample is therefore essential, ideally with more uniform hardware, tighter artifact control, a broader and less academically-skewed participant pool, and multiple-comparison-corrected significance testing across the feature–baseline comparisons reported here. A key priority should be to test whether the same features – especially TP10 raw standard deviation, AF7 high- γ mean, and AF8 delta peak frequency – remain informative under repeated acquisition, different content types, and different user populations.

A second major direction is mechanistic clarification. The present results leave open whether the strongest effects reflect genuinely neural engagement processes, movement-related artifacts, or a mixture of both. Future

work should therefore include explicit artifact analyses, such as electromyographic control across the delta-through-gamma range (not only high-gamma or broadband), motion monitoring, ICA-based comparison, channel-quality assessments, and device-specific sensitivity checks. It would also be valuable to compare broadband features against strictly cleaned, source-reconstructed, or artifact-masked data to determine how much of the predictive signal survives after conservative preprocessing.

A third direction concerns theory development. The Engagement Index appears to capture a limited but real component of sustained engagement, yet it may not capture the behavioral threshold that leads to a swipe. Future studies should therefore distinguish more explicitly between sustained attention, momentary decision pressure, action readiness, and actual behavior. This could be pursued by combining EEG with behavioral timing, self-report, eye tracking, or post-hoc content ratings, and should include testing whether burst membership itself (Section 4.3) predicts weaker encoding or evaluation, rather than merely describing swipe clustering. In particular, event-aligned designs could help determine whether the relevant neural signature precedes disengagement, coincides with motor preparation, or reflects post-decisional processing. The definition proposed in Section 4.6 is intended as the target of exactly this work: its falsification conditions (Section 4.6.5) and its proxy-measurement bridge (Section 4.6.4) are stated as testable hypotheses for follow-up studies rather than as findings of the present one.

A fourth opportunity is methodological expansion. The current pipeline is deliberately simple and interpretable, but future work could test whether other transparent models improve both performance and explanatory value, such as regularized generalized linear models, hierarchical models, or hybrid approaches that preserve interpretability while capturing nonlinearity. At the same time, more elaborate feature engineering may be worthwhile if it remains conceptually grounded; examples include temporal derivatives, burst-aware features, cross-frequency interactions, or individualized feature selection. However, this should be pursued carefully, because greater complexity can easily reduce interpretability and increase the risk of overfitting.

A fifth and highly relevant direction concerns public health and behavior change. Short-form digital media is increasingly used not only for entertainment but also for public communication, education, and health promotion. If EEG-informed engagement models can eventually be made robust and explainable, they may support the design of interventions that adapt message format, timing, and length to the user's attentional state. This could matter for smoking cessation messages, vaccine communication, mental-health micro-interventions, nutrition prompts, physical-activity nudges, and other digital behavior-change tools. The challenge is not only technical but also

ethical: such systems must avoid manipulation, respect autonomy, and be transparent about how engagement is inferred and used.

Finally, future research should examine generalizability across contexts, populations, and content domains. The present sample skewed toward an academically-oriented population recruited via university mailing lists (Section 3.1), and the present task used naturalistic TikTok browsing; similar engagement dynamics may appear in other forms of media consumption, educational content, or digital decision tasks, but this would need to be tested in a broader participant pool before being assumed. Testing whether the same feature families generalize to different content domains would help determine whether the observed effects are specific to short-form video or whether they reflect a broader principle of human engagement and disengagement. If replicated, this line of work could contribute not only to neuroscience and psychology, but also to media research, user-experience design, and public-health communication.

Overall, the present study opens a broad but carefully bounded research program: one that combines explainable EEG analysis, naturalistic behavior, and applied communication research. Its value lies not in claiming a final solution, but in identifying a credible path toward more interpretable, behaviorally grounded, and potentially useful models of engagement and disengagement in real-world digital environments.

Chapter 6

Plain-Language Summary: Common Questions

The questions that dominate public debate about short-form video are not the questions an EEG study is built to answer. They are, however, the questions on which people actually act, as parents, as teachers, as voters, as regulators, and as communicators trying to reach an audience. This chapter therefore steps out of technical register and answers the most common of those questions in ordinary language, using only what this thesis and the literature it reviews can support. It deliberately revisits ground already covered in Chapters 4 and 5, restated for a reader who has not read them.

Three cautions apply to everything below. This was one study of 25 young adults, mostly students, each recorded once for 25 minutes. It measured brain activity in the seconds around a swipe; it did not measure anyone's health, mood, grades, sleep, or long-term wellbeing, and it followed nobody over time. Every answer here is therefore a *direction of evidence*, not a settled finding, and is written to be revised. Where the honest answer is "this study cannot tell you," that is stated plainly rather than filled in with plausible-sounding inference.

Is short-form video damaging my brain? This study cannot tell you, and nobody should cite it as evidence either way. It never looked for damage. It compared two ordinary moments, still watching and about to swipe away, in healthy young adults over 25 minutes. What it did find is that those two moments look measurably different in the brain, which says the decision to move on is a real physical event and not merely a habit or a figure of speech. That is a statement about how attention works, not about harm. Claims in circulation that scrolling "rewires" or "shrinks" the brain go well beyond what evidence of this kind can carry. *What to do with this:* treat confident claims

in both directions with suspicion. The strong version of the harm story and the dismissive “it’s all moral panic” reply are both running ahead of the data.

If I watch something for a long time, does that mean I was interested in it? Probably less often than you would assume, and this is the clearest practical message of the thesis. Platforms, advertisers and much public commentary treat time spent watching as if it measured interest. But long, uninterrupted viewing is just as consistent with being absorbed as with being zoned out and later remembering almost nothing [4, 3]. The measurement side of this thesis points the same way: the standard laboratory formula for “engagement” (Section 4.4.2) was far worse at telling these two moments apart than several plain, single measurements of the raw signal, the best of which separated them by close to ten times as much. If watch time reliably captured engagement, that gap should not exist. *What to do with this:* stop reading watch time as proof that something worked. For anyone reporting reach, health campaigns especially, a high view count is a statement about distribution, not about whether anything landed. Ask separately whether people acted, returned, or could recall the message.

Can a cheap headband read my mind, or tell what I am about to do? No, not in any sense that should worry you today, and the failures are as informative as the successes. Using a consumer headband costing a few hundred euros, the signal that distinguished “about to swipe” from “still watching” was real but weak, and it worked only when the system was tuned to one specific person. Trained on a group and applied to a new individual, performance collapsed to roughly guessing (Section 4.4.3). In plain terms: there is a faint, person-specific fingerprint, not a readable thought. *What to do with this:* do not buy claims that wearable brain sensors can currently detect engagement, attention or intent in a general population. Equally, note the direction of travel: this was four cheap sensors, one short session, one small group. Organisations with far more data and far longer observation are not bound by those limits, which is the real reason to ask now who is permitted to measure attention and under what oversight.

Why is it so hard to stop scrolling once I start? This study offers one small, concrete piece of the picture. Swipes do not arrive evenly spaced. They cluster into short bursts, meaning runs of rapid skipping with calmer stretches in between (Section 4.3). About two-thirds of swipes were isolated, so most of the time people were not in a burst at all. But how burst-prone someone was varied enormously between individuals, far more than any average conveys.

The pattern is real; the “it” that happens is not the same for everyone. *What to do with this:* if you want to interrupt your own scrolling, the useful target is the fast run rather than the total minutes, since the two are not the same behaviour. And be sceptical of advice derived from an average user, because on this measure the average user barely exists.

Should phones be banned in schools, or social media restricted by age? This thesis has nothing to say about either policy, and it would be misleading to enlist it on either side. It tested no intervention, involved no minors, and measured no outcome that such a policy would target. One indirect observation is worth carrying into that debate, however: the behaviour measured here differed so much from person to person that a single rule calibrated to the average would fit very few individuals well. That is a caution about uniform rules generally, not an argument for or against any particular one. *What to do with this:* demand that whoever proposes a rule states which outcome it is supposed to improve and how that will be checked afterwards. This study cannot supply that evidence, and neither can most of what is currently cited in the debate.

What should regulators actually require, if not bans? The most defensible ask suggested by this work is disclosure rather than prohibition. Feeds are tuned to maximise something, most commonly continued watching, which on the evidence reviewed here is not a reliable indicator that the viewer is attending, benefiting, or even remembering. Optimising it is therefore not the same as serving the user, and the gap between the two is currently invisible from the outside. A further practical point: this study was run on off-the-shelf hardware by a single researcher, which means independent measurement of these questions is now cheap enough that no regulator need rely solely on platform-supplied figures. *What to do with this:* require platforms to disclose what their systems optimise for and to open those targets to outside audit; fund small independent replication rather than waiting for internal data to be volunteered.

I make health or education content. How do I hold attention? Honestly: this thesis does not yet give you a tool. It shows that the moment of leaving has a physical signature, which is a precondition for one day detecting disengagement as it begins, but the signal is currently too weak, and too person-specific, to build anything you could deploy. What it does give you is a warning about your own dashboard. If you are judging content by average watch time, you are optimising against a number that cannot distinguish an

absorbed viewer from a dissociated one, and you may be steadily selecting for content that holds eyes without landing a message. *What to do with this:* keep watch time as a distribution check, but judge success on something downstream that you actually care about, such as recall, clicks, appointments or behaviour change, and accept that this is harder and slower to measure. That difficulty is not a reason to fall back on the easier number.

So what is the single most useful thing to take away? That the industry-standard way of measuring engagement is weaker than its ubiquity implies, and that better measurement is now within reach of people outside the platforms. This thesis does not show that anyone has decoded the decision to swipe; it shows that the moment of leaving is physically distinguishable from the state of staying, that the conventional engagement metric largely misses it, and that the equipment required to investigate this costs a few hundred euros rather than a laboratory budget. Whether that capability is used to make feeds more compelling or to hold them accountable is not a scientific question, and will not be settled by studies like this one.

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